

Uncertainty-Aware Attention, Geometry and State-Space Modelling for Streaming Motion Forecasting

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September 2026

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WORD COUNT

Number of Pages: 86

Number of Words: 21,115

(Chapters 1–5: Introduction through Conclusions and Future Work).

ABSTRACT

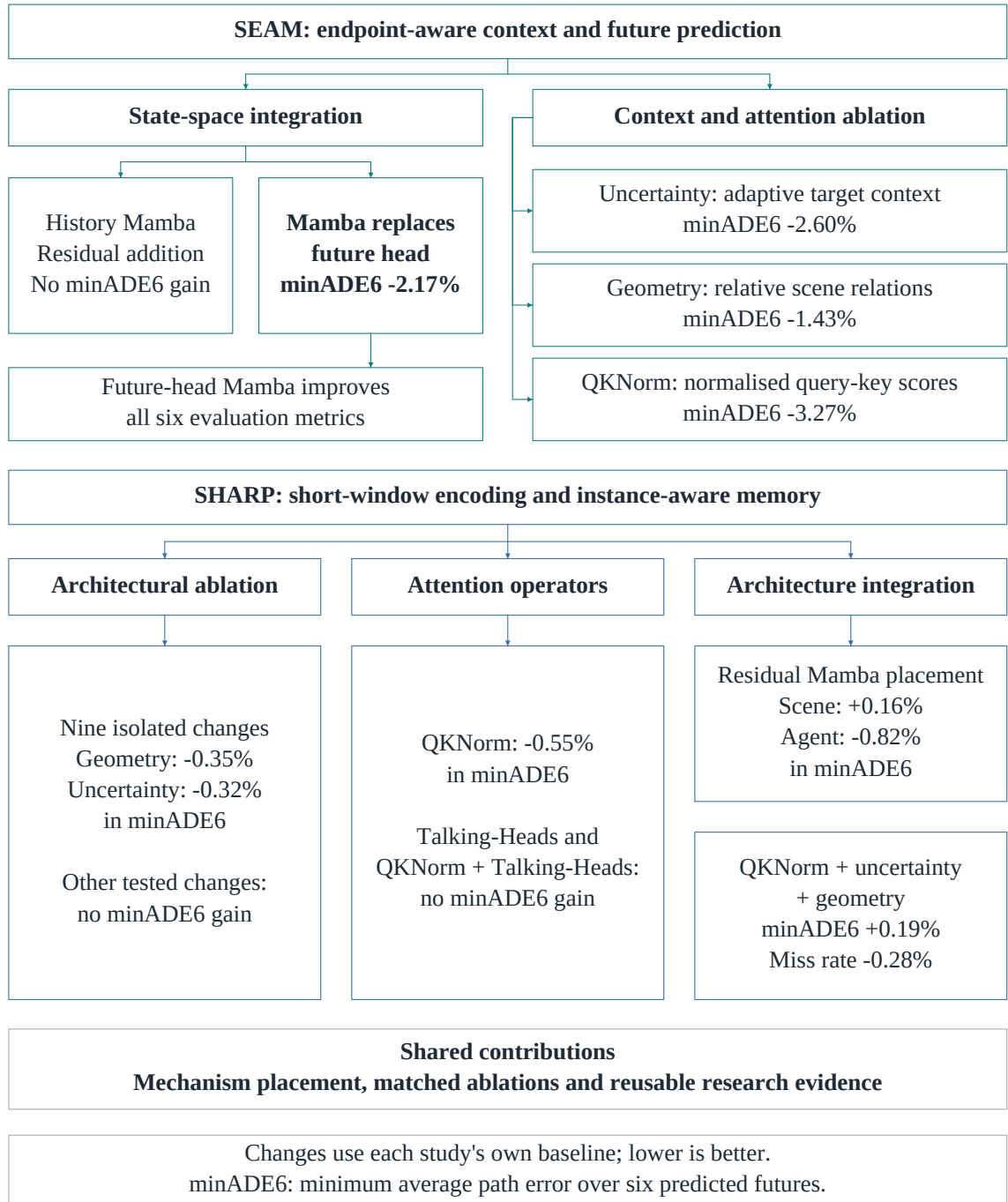
Streaming motion forecasting updates plausible road-user trajectories as observations arrive. Its accuracy depends on how current evidence interacts with retained context. This dissertation investigates whether uncertainty-aware retrieval, relative geometry, attention normalisation and selective state-space processing improve that interaction while preserving the streaming mechanisms of SEAM and SHARP. Five protocol-matched experimental families on Argoverse 2 distinguish component effects, recurrent placement and mechanism composition. Six-mode minimum average displacement error (minADE6), measured in metres, provides the primary comparison criterion; endpoint, miss-rate and confidence-sensitive metrics provide complementary evidence.

For SEAM, Mamba in the future-coordinate head improves all six final-epoch metrics and reduces selected minADE6 from 0.662859 to 0.648480, a 2.17% reduction. Its separate context and attention ablation finds improvements from uncertainty-aware target context, relative-geometry bias and Query–Key Normalisation (QKNorm). QKNorm produces the largest selected minADE6 reduction, from 0.727668 to 0.703845 (3.27%). These findings support interventions matched to endpoint retrieval, physical scene relations and chronological prediction.

For SHARP, uncertainty and geometry improve minADE6 in the architectural ablation; QKNorm reduces it by 0.55% in the independent attention study. Residual Mamba over agent history reduces selected minADE6 by 0.82% against its matched control, whereas replacing temporal attention does not improve it. Joint QKNorm, uncertainty and geometry improve four secondary metrics but increase minADE6 by 0.19%, demonstrating that individually useful mechanisms need not yield additive gains.

The contribution is a controlled account of how representation, operator placement and mechanism interaction shape streaming prediction. Separate training-time comparisons and traceable evaluation artefacts support interpretation. The single-seed evidence identifies promising architectural choices without establishing universal superiority, providing a basis for targeted confirmation and further model development.

**Graphical abstract: streaming motion forecasting
Argoverse 2; original streaming architecture preserved**



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ABBREVIATIONS

The following specialist abbreviations, metric labels and method names support the interpretation of motion forecasting, model architectures and training procedures. Mathematical symbols are defined alongside their equations. Metric suffixes 1 and 6 denote the number of trajectory modes considered.

Abbreviation	Meaning
AdamW	Adaptive moment estimation with decoupled weight decay
ADE	Average displacement error
AV1	Argoverse 1
AV2	Argoverse 2
BatchNorm	Batch normalisation
b-minFDE6	Brier-adjusted minimum final displacement error over six modes
CA	Cross-attention
CUDA	Compute Unified Device Architecture, NVIDIA's GPU computing platform
cuDNN	CUDA Deep Neural Network library
DDP	Distributed data parallel training
DeMo	Decoupling Motion Forecasting into Directional Intentions and Dynamic States
DenseTNT	Dense target-driven trajectory prediction
DESIRE	Distant Future Prediction in Dynamic Scenes with Interacting Agents
FDE	Final displacement error
FP32	32-bit floating-point precision
GAN	Generative adversarial network
GELU	Gaussian error linear unit
HiVT	Hierarchical Vector Transformer
L1	Sum of absolute component magnitudes, as used in regression losses
L2	Euclidean norm
LaneGCN	Lane graph convolutional network
LN	Layer normalisation
LR	Learning rate
LSTM	Long short-term memory
MHA	Multi-head attention
minADE1	Minimum average displacement error for the highest-confidence trajectory
minADE6	Minimum average displacement error over six trajectory modes
minFDE1	Minimum final displacement error for the highest-confidence trajectory
minFDE6	Minimum final displacement error over six trajectory modes
MLP	Multilayer perceptron
MR	Miss rate
MTR	Motion Transformer
QCNNet	Query-centric trajectory-prediction network
QK+TH	Query-Key Normalisation combined with Talking-Heads attention
QKNorm	Query-Key Normalisation
Q+U+G	QKNorm, uncertainty-aware target context and relative-geometry attention bias
ReLU	Rectified linear unit
S4	Structured state-space sequence model

Abbreviation	Meaning
S5	Simplified state-space layers for sequence modelling
SA	Self-attention
SEAM	Streaming trajectory-prediction model using endpoint-aware modelling
SHARP	Short-Window Streaming for Accurate and Robust Prediction in Motion Forecasting
SSM	State-space model
SyncBatchNorm	Synchronised batch normalisation
TH	Talking-Heads attention
TNT	Target-driven trajectory prediction

1 INTRODUCTION

1.1 Motion forecasting as a streaming problem

An autonomous vehicle must anticipate how surrounding road users may move before selecting its own actions. These futures are often *multimodal*: several distinct manoeuvres, such as continuing through a junction or turning, can remain consistent with the same observed motion. Forecasting therefore requires a set of plausible trajectories rather than a single extrapolated path. Models associate each trajectory hypothesis with a confidence score, allowing different possible outcomes to be represented explicitly [1, 2].

The relevant evidence extends beyond an individual agent’s coordinates. Here, an *agent* denotes a road user represented by a tracked motion history. Nearby agents influence interaction, while lane geometry constrains where motion is physically plausible. VectorNet and LaneGCN demonstrate how structured representations of tracks and maps support this reasoning without converting the complete scene into an image [3, 4]. Context-sensitive planning and manoeuvre-anticipation research further motivates selecting surrounding evidence according to its relevance to future behaviour [5, 6].

Repeated prediction introduces a further requirement. A vehicle receives new observations continuously, whereas a conventional forecasting input contains a fixed history and a fixed prediction horizon. Re-encoding overlapping histories at every update duplicates computation. Discarding previous predictions also loses information about how a plausible future has evolved. A streaming predictor instead carries useful information between updates. This retained information must remain associated with the correct agent and coordinate frame, and it must be revised when new observations change the interpretation of the scene. RealMotion formalises complementary streams for scene context and previous trajectories [7].

This dissertation investigates how the information-processing operations inside such predictors can be improved without removing their streaming mechanisms. Argoverse 2 (AV2) supplies the common dataset: its motion-forecasting benchmark contains 250,000 scenarios from six cities, with agent tracks and high-definition maps [8]. The investigation concerns prediction from recorded observations, rather than the behaviour of a vehicle executing the predicted trajectories. Within this setting, the central problem is to identify which architectural changes improve forecasting accuracy and which representations make those changes useful.

1.2 Reference architectures and research gap

SEAM and SHARP provide complementary ways of organising streaming evidence. SEAM uses previous trajectory endpoints to identify regions of the scene likely to matter to each forecast [9]. An endpoint is the predicted position at the end of a trajectory hypothesis. SEAM combines broad scene information with local evidence retrieved around these endpoints through its dual-context decoder, the component that converts encoded information into predicted trajectories. This organisation makes endpoint retrieval, scene interaction and future-coordinate generation distinct sites for intervention.

SHARP processes non-overlapping short observation windows and transfers latent context between instances associated across windows [10]. A latent context is a learned feature representation, rather than a stored copy of the complete observation history. Its instance-aware transfer preserves information about an agent while incorporating the next short interval of motion. Temporal encoding, scene interaction and endpoint-conditioned retrieval consequently have separate roles within the predictor. These roles allow local modifications to be tested while retaining SHARP’s incremental processing and identity-associated memory.

The shared question is how to select, compare and accumulate evidence. Attention weights scene elements by learned relevance; recurrence updates a state along a sequence. Neither inherently supplies physical relations or adapts endpoint neighbourhoods to predictive ambiguity, motivating targeted rather than whole-architecture changes.

The reference models and their antecedents establish the value of streaming representations; they do not establish the effect of the particular interventions examined here [7, 9, 10]. The gap addressed is therefore empirical and architectural: how do uncertainty-sensitive retrieval, explicit geometry, attention normalisation and selective recurrence behave inside otherwise controlled streaming predictors? A second question follows from the first. Even when individual components improve their own matched controls, their combination need not preserve those effects. Interactions between components must be measured rather than inferred by adding their separate gains.

1.3 Architectural hypotheses

Four mechanisms introduce different *inductive biases*: structural assumptions that make particular relationships easier to represent. Their relevance depends on the operation and the information it processes.

Uncertainty-aware target context: Dispersed mode probabilities indicate weaker support for a single future. The uncertainty-aware pathway uses this ambiguity to adjust endpoint-centred retrieval or feature weighting. Broader context may reveal competing futures; narrower context may exclude distractors for confident predictions. Confidence-conditioned context is therefore hypothesised to improve relevance. These model-derived scores are not necessarily calibrated: their values need not match observed event frequencies.

Relative-geometry attention bias: Attention ordinarily learns relationships from feature vectors. Adding a learned contribution derived from relative displacement, distance and heading makes physical relations directly available to its scores. This adapts relation-aware attention [11] to interactions between road-scene elements. The hypothesis is that explicit geometry helps distinguish physically different relationships that could otherwise have similar feature-based scores. It is applied where tokens, the feature vectors processed by attention, correspond to elements with comparable spatial coordinates.

Query–Key Normalisation: Multi-head attention (MHA) compares query and key vectors to determine how strongly one representation attends to another [12]. Query–Key Normalisation (QKNorm) normalises these vectors before comparison and learns a similarity scale [13]. It may improve the distribution of attention weights by reducing sensitivity to vector magnitude. This is distinct from uncertainty-aware retrieval: QKNorm changes the scoring operation, whereas the uncertainty mechanism changes the support and contribution of target context. Talking-Heads attention, which mixes information across attention heads, supplies an additional operator comparison in SHARP [14].

Selective state-space processing: A state-space model represents a sequence through an evolving hidden state. Mamba makes parts of that state update depend on the current input, enabling information to be retained or suppressed selectively [15]. DeMo provides a motion-forecasting precedent for combining attention with Mamba on history and future-state sequences [16]. Chronological observations and successive future positions have a meaningful order; a collection of agents and lane segments does not necessarily have the same property. The distinction between ordered sequences and permutation-invariant sets is therefore relevant to placement [17]. The hypothesis is that recurrence is more useful when its scan follows a meaningful progression.

Adding a residual pathway, whose output supplements the existing representation, and replacing an existing prediction component test different versions of this hypothesis.

These hypotheses require separate placement and combination tests; no operator is assumed to improve every location.

1.4 Research aim and questions

The aim is to determine how uncertainty-aware context retrieval, attention normalisation, relative geometry and selective state-space processing affect multimodal forecasting accuracy in SEAM and SHARP while preserving their defining streaming processes. Five questions organise the investigation:

1. **RQ1 – Mamba placement in SEAM:** How do agent-history addition and future-head replacement compare under a matched control?
2. **RQ2 – Context and attention mechanisms in SEAM:** How do uncertainty-aware context, relative geometry and QKNorm individually affect accuracy?
3. **RQ3 – Mamba placement in SHARP:** How do residual scene and agent-history additions compare with temporal-attention replacement within their respective studies?
4. **RQ4 – Context and attention mechanisms in SHARP:** How do uncertainty-aware context, relative geometry and QKNorm affect accuracy individually and as a combined model?
5. **RQ5 – Architecture-dependent mechanism transfer:** Which improvements recur across the architectures, and how do placement and composition qualify those effects?

1.4.1 Measurable objectives

The following outcomes provide criteria for assessing the aim:

1. **Reference specification:** documented architecture, optimisation and evaluation settings corresponding to each selected reference protocol, with measured local controls.
2. **SEAM intervention evidence:** matched state-space placement and independent context and attention comparisons addressing RQ1 and RQ2.
3. **SHARP intervention evidence:** matched residual-placement, architectural-ablation, attention-operator and mechanism-composition comparisons addressing RQ3 and RQ4.

4. **Architectural explanation:** each intervention’s representation, location, operation and retained pathways linked to its rationale and measured effect, supporting RQ5.
5. **Reproducible comparison:** linked source, configurations, checkpoint-selection records, metric tables, plots and within-study training-time comparisons supporting the numerical conclusions.

1.5 Experimental approach and contributions

Five families organise the programme: the SEAM state-space integration study; SEAM context and attention ablation study; SHARP architecture integration study; SHARP architectural ablation study; and SHARP attention-operator study. SHARP integration contains separate reference, placement and composition contrasts. Integration and attention studies use 80 epochs; ablations use 20. An epoch denotes one training-data pass. Each comparison retains its own control.

Each comparison fixes data splits, seed, batch construction, optimisation and evaluation. SEAM integration follows its selected implementation protocol; SHARP integration follows the publication and supplement. Ablations have separate schedule-matched controls.

The primary selection criterion is minADE6: the minimum average positional error, in metres, among six predicted trajectories relative to the observed future. It measures the quality of the closest predicted trajectory over the complete prediction horizon rather than only at its endpoint. Lower values are better. Endpoint error, miss rate and confidence-sensitive endpoint error provide complementary evidence. The full metric definitions and the distinction between checkpoint selection and complete evaluation vectors are developed in Chapter 2. Small differences from single-seed comparisons are interpreted as local experimental evidence rather than estimates of cross-seed superiority [18].

The dissertation makes three connected contributions. First, it implements and evaluates a representation-specific intervention map: uncertainty-sensitive endpoint retrieval, geometry-informed scene interaction, query-key normalised attention and selective recurrence over histories or future positions. The contribution concerns their adaptation and controlled placement within both architectures.

Second, the experiments identify useful mechanisms within each model. In SEAM, future-head Mamba reduces selected minADE6 from 0.662859 to 0.648480 m, a 2.17% reduction, and improves all six metrics in the common full-vector evaluation. The history addition does not improve the primary criterion. In the separate SEAM context and attention ablation study, uncer-

tainty, geometry and QKNorm reduce selected minADE6 by 2.60%, 1.43% and 3.27%, respectively, and all three improve the six final-epoch metrics. These findings support both the relevance of the three individual mechanisms and the importance of Mamba’s functional placement.

SHARP supplies complementary evidence. Uncertainty and geometry improve the architectural ablation control, QKNorm improves the attention-study control and residual temporal-agent Mamba improves selected minADE6 within its state-space placement comparison. The final combined QKNorm–uncertainty–geometry model improves endpoint and miss-rate measures, but minADE6 changes from 0.679282 to 0.680589 m. The combined experiment therefore distinguishes improved secondary outcomes from a primary-metric improvement. Taken together, the studies support targeted modifications while showing their isolated gains.

Third, the dissertation connects these observations to inspectable experimental artefacts. The project repository, <https://github.com/madvidd/Salehi-MSc-Thesis>, organises source, configuration records and original evaluations by model and scientific study under `Studies/`, with comparison summaries and numerical tables under `Results/`. Recorded model stores retain the associated large checkpoints and raw logs. This evidence supports reproduction and further analysis in the sense advocated by established machine-learning reproducibility guidance [19]. Detailed metric contrasts, training-time comparisons and model-specific interpretation are consolidated in Chapter 4 rather than repeated here.

1.6 Scope and dissertation structure

Open-loop AV2 evaluation compares forecasts with recorded futures without influencing later observations. It does not measure closed-loop planning or inference latency, the time needed to produce a forecast. SEAM and SHARP are experimental controls; DeMo is a literature precedent.

Chapters 2, 3, 4 and 5 develop the theory, methods, results and conclusions, respectively. The Appendix indexes supporting evidence.

2 BACKGROUND AND RELATED WORK

The questions in Chapter 1 concern how ordered observations, interacting scene elements and alternative futures are represented, retrieved and retained. Task and metric definitions lead into the forecasting literature, attention and state-space theory, then the SEAM and SHARP architectures. The synthesis establishes the model-specific questions addressed by Chapter 3.

2.1 Motion-forecasting formulation

Motion forecasting estimates the future states of road users from their observed motion and surrounding scene context. Consider a scene sampled at discrete times $t = -T_{\text{obs}} + 1, \dots, 0$. For agent i , an observed state may contain planar position $\mathbf{p}_i^t \in \mathbb{R}^2$, velocity $\mathbf{v}_i^t \in \mathbb{R}^2$, heading θ_i^t , semantic class c_i and an availability indicator. Its history is denoted

$$\mathbf{X}_i = \{\mathbf{x}_i^t\}_{t=-T_{\text{obs}}+1}^0, \quad \mathbf{x}_i^t = [\mathbf{p}_i^t, \mathbf{v}_i^t, \theta_i^t, c_i]. \quad (2.1)$$

The full input also contains histories $\mathcal{X} = \{\mathbf{X}_i\}_{i=1}^{N_a}$ for neighbouring agents and a vector map $\mathcal{M}$ containing lane centre-lines, boundaries and semantic attributes. Coordinates are commonly transformed into an agent-centred or query-centred frame to reduce irrelevant global translation and rotation variation [20, 21].

Here T_{obs} and N_a count observed samples and retained agents. Bold symbols denote vectors or matrices, $\mathbb{R}^2$ denotes two real-valued coordinates and brackets concatenate features. Availability masks distinguish missing observations from physical zeros. Semantic classes are encoded before projection into the latent feature space.

A latent feature is a learned numerical description, not a physical measurement. Linear projections mix features through learned weights; embeddings place observations or categories in a shared feature space. A multilayer perceptron (MLP) adds nonlinear activations between projections. Each token is a feature vector with components called channels. Pooling summarises tokens, for example through each channel’s maximum over valid observations.

For a target agent, a multimodal predictor returns K future trajectories

$$\hat{\mathbf{Y}}_{i,k} = \{\hat{\mathbf{y}}_{i,k}^t\}_{t=1}^{T_{\text{pred}}}, \quad k \in \{1, \dots, K\}, \quad (2.2)$$

and confidence values $p_{i,k} \geq 0$ satisfying $\sum_{k=1}^K p_{i,k} = 1$. Here T_{pred} counts future samples, and a hat distinguishes a prediction from an observed value. Each mode represents one plausible future, such as proceeding through a junction, yielding or turning. Only one future is observed in a recorded scenario, even though several futures may have been reasonable at prediction time. The learning problem is consequently one-to-many.

Two forms of uncertainty should be distinguished. *Aleatoric uncertainty* arises from ambiguity intrinsic to the scene, including unobserved intention and stochastic human behaviour. It cannot be removed simply by collecting more examples of the same situation. *Epistemic uncertainty* arises from limited data or model knowledge and can, in principle, be reduced through improved coverage or modelling [22]. A finite set of K trajectories is primarily an approximation to the aleatoric distribution. Its confidence scores are useful for ranking modes, but they are not automatically calibrated probabilities; modern neural networks can remain overconfident even when their ranking accuracy is high [23]. Deep ensembles provide one practical way to expose epistemic variation, although their repeated training and inference cost is substantial for the multi-day experiments in this dissertation [24].

2.1.1 Snapshot and streaming inference

In the conventional snapshot formulation, one forward pass maps a complete history and map to a set of futures:

$$(\hat{\mathcal{Y}}, \mathbf{p}) = f_{\theta}(\mathcal{X}, \mathcal{M}). \quad (2.3)$$

Here f_{θ} is the network with learned parameters θ ; a forward pass evaluates it on its inputs. Re-encoding the scene whenever the window advances serves isolated benchmark samples but does not exploit overlapping observations during continuous driving.

A streaming model additionally carries state between invocations. For observation window w , a general form is

$$(\hat{\mathcal{Y}}^w, \mathbf{p}^w, \mathcal{S}^w) = f_{\theta}(\mathcal{X}^w, \mathcal{M}^w, \mathcal{S}^{w-1}, \hat{\mathcal{Y}}^{w-1}), \quad (2.4)$$

where $\mathcal{S}^{w-1}$ denotes retained latent context and $\hat{\mathcal{Y}}^{w-1}$ denotes previous forecasts when the method uses them. Not every streaming method uses both terms. Some cache scene features, some relay trajectories and others maintain state for individual tracked instances. Equation 2.4 provides a common language for comparing these choices.

Streaming state must remain aligned as the ego vehicle moves and tracked agents enter or disappear. Earlier prediction errors can contaminate later context, so current observations must be able to correct stale memory.

Equation 2.4 also separates three functions that should not be conflated: encoding evidence inside the current observation window, associating retained state with entities across windows and using previous trajectory hypotheses to retrieve future-relevant context. A local operator can change the first function without redesigning the other two. This distinction is central to the dissertation: attention, geometry and Mamba interventions modify defined representation paths while short-window execution, identity-aware state association and trajectory relay remain invariant. The resulting comparisons therefore concern operator and context design inside a streaming model, rather than an uncontrolled change from streaming to snapshot prediction.

2.2 Evaluation metrics and reporting conventions

Let $\mathbf{Y}_i = \{\mathbf{y}_i^t\}_{t=1}^{T_{\text{pred}}}$ be the ground-truth future. For mode k , average displacement error (ADE) and final displacement error (FDE) are

$$e_{i,k}^{\text{ADE}} = \frac{1}{T_{\text{pred}}} \sum_{t=1}^{T_{\text{pred}}} \|\hat{\mathbf{y}}_{i,k}^t - \mathbf{y}_i^t\|_2, \quad (2.5)$$

$$e_{i,k}^{\text{FDE}} = \|\hat{\mathbf{y}}_{i,k}^{T_{\text{pred}}} - \mathbf{y}_i^{T_{\text{pred}}}\|_2. \quad (2.6)$$

The Euclidean norm $\|\cdot\|_2$ gives the straight-line distance between predicted and observed coordinates. ADE averages this distance across the complete future, whereas FDE measures the endpoint only. Their units are metres.

For N evaluated agents, the local SHARP metric implementation used in the experiments computes

$$\text{minADE}_K = \frac{1}{N} \sum_{i=1}^N \min_{1 \leq k \leq K} e_{i,k}^{\text{ADE}}, \quad (2.7)$$

$$\text{minFDE}_K = \frac{1}{N} \sum_{i=1}^N \min_{1 \leq k \leq K} e_{i,k}^{\text{FDE}}. \quad (2.8)$$

The two minima are selected independently in that implementation. This detail matters because the official Argoverse 1 forecasting evaluator selects the mode with minimum FDE and then reports both its ADE and FDE [25]. Metric names alone are therefore insufficient evidence of exact comparability; the evaluation code and protocol must also agree.

The miss rate is the proportion of samples for which all K endpoints lie beyond a threshold τ :

$$\text{MR}_K = \frac{1}{N} \sum_{i=1}^N \mathbb{I} \left[\min_{1 \leq k \leq K} e_{i,k}^{\text{FDE}} > \tau \right]. \quad (2.9)$$

The AV2 forecasting implementation uses $\tau = 2$ metres [26]. MR is less sensitive than FDE to small endpoint changes but exposes whether the model fails to place any mode near the true destination.

The indicator $\mathbb{I}[\cdot]$ equals one when its condition holds and zero otherwise; its average is therefore the fraction of agents missing the threshold.

The Brier-adjusted endpoint metric combines endpoint error and confidence:

$$\text{b-minFDE}_K = \frac{1}{N} \sum_{i=1}^N \min_{1 \leq k \leq K} [e_{i,k}^{\text{FDE}} + (1 - p_{i,k})^2]. \quad (2.10)$$

The penalty discourages low confidence in geometrically accurate modes. It supplements oracle coverage with confidence information without providing a complete calibration analysis.

2.2.1 Trajectory coverage, ranking and confidence

The distinction between an oracle distance and a probabilistic score is fundamental. A scoring rule is *proper* when its expected value is optimised by reporting the forecaster's true predictive distribution rather than a strategically distorted one [27]. minADE and minFDE are not proper scoring rules: the model can improve them by spreading hypotheses without assigning meaningful probability. The Brier term introduces a confidence penalty, but b-minFDE still applies a minimum over a finite mode set and therefore does not constitute a complete distributional assessment. Calibration curves, negative log-likelihood and expected calibration error would test different properties. The dissertation consequently treats b-minFDE as a useful confidence-sensitive benchmark metric, not as proof that the complete forecast distribution is calibrated.

All reported forecasting metrics are lower-is-better. The $K = 1$ metrics assess the highest-confidence trajectory; minimum metrics with $K = 6$ select among six hypotheses after the future is known. They are therefore termed *oracle* metrics: they measure coverage, not necessarily the prediction chosen online.

A calibration curve compares stated probabilities with observed frequencies across confidence ranges; expected calibration error summarises their discrepancies. Negative log-likelihood pe-

nalises low probability assigned to the observation. These diagnostics require a defined event or probability-density model, not merely candidate paths and displacement errors.

minADE6 measures whole-path coverage; minFDE6 measures destination coverage, MR thresholds endpoint failure and b-minFDE6 incorporates confidence. When improvement on one measure accompanies deterioration on another, neither model dominates all criteria: this is a Pareto trade-off. The full metric vector is therefore retained alongside the primary criterion.

2.2.2 Single-agent and multi-agent evaluation

Single-agent evaluation scores a designated focal actor; multi-agent evaluation scores several, often within coherent scene-level “worlds”. Its actor-aggregated avgMinADE, avgMinFDE and actorMR are distinct from focal-agent metrics. Comparisons therefore identify the split, mode count, actor protocol and checkpoint.

Validation and challenge-test results are distinct, as are selected-checkpoint and final-epoch measurements. Checkpoint differences can be comparable to the architectural effects being studied.

2.2.3 Matched protocols and attribution

Training adjusts parameters to reduce a loss, a penalty for prediction error, by using backpropagation to compute its gradient and an optimiser to update the parameters at a scale set by the learning rate. During this process, each global batch supplies one update, while an epoch traverses the training data. Checkpoints preserve the resulting model state for selection or evaluation; within an ablation, these evaluations test a specified component against a matched control.

Reproducible comparison requires more than identical metric names. For study family s , let the selected protocol be represented by

$$\mathcal{P}_s = (\mathcal{D}_s, E_s, B_s, O_s, \eta_s, \mathcal{L}_s, C_s, Q_s), \quad (2.11)$$

where $\mathcal{D}_s$ denotes the dataset split and preprocessing, E_s the epoch budget, B_s the global batch, O_s the optimiser, η_s the learning-rate schedule, $\mathcal{L}_s$ the training objective and regularisation, C_s the checkpoint-selection rule and Q_s the evaluation procedure. A variant v and control c support an attributable metric difference only when both are evaluated under the same $\mathcal{P}_s$:

$$\Delta m_{s,v,j} = m_j(v; \mathcal{P}_s) - m_j(c; \mathcal{P}_s), \quad (2.12)$$

where m_j is a reported metric and negative differences indicate improvement. Comparing different $\mathcal{P}_s$ would confound architectural and protocol effects rather than isolate an intervention.

One configuration source is selected for each model family before evaluation. The SHARP studies use the publication and supplementary protocol [10]; the SEAM state-space integration study uses the configuration supplied with its reference implementation [9]. The experimental configurations reproduce the selected source parameters, while alternative schedules are excluded from the corresponding comparison. Hardware and library details remain execution metadata and are reported separately because they can influence numerical reproducibility even when the training tuple in Equation 2.11 is aligned [19, 18].

Each SEAM and SHARP study therefore retains a schedule- and batch-matched baseline rather than entering a common accuracy or runtime ranking.

2.2.4 Training duration and computational complexity

Forecast quality and computational cost answer different questions and are therefore reported in separate tables. For study family s and variant v , retained wall time is defined over the contiguous execution intervals assigned to that variant:

$$T_{s,v} = \sum_{a=1}^{A_{s,v}} \left(t_{s,v,a}^{\text{end}} - t_{s,v,a}^{\text{start}} \right), \quad (2.13)$$

where $A_{s,v}$ is the number of retained active intervals. Data loading, training, validation and checkpointing contribute when they occur inside those intervals. A separately timed final checkpoint evaluation is not added to a training-attempt duration; this distinction applies to the SHARP mechanism composition records. Inactive calendar gaps are omitted rather than assigned to the architecture.

Two derived quantities make comparisons within a study easier to interpret. The average retained time per epoch and the relative difference from the matched control c are

$$\bar{T}_{s,v} = \frac{T_{s,v}}{E_s}, \quad \Delta T_{s,v}^{\%} = 100 \frac{T_{s,v} - T_{s,c}}{T_{s,c}}. \quad (2.14)$$

These quantities compare equal epoch budgets, not forward-pass latency. Isolating operator cost from data processing, validation, storage and communication would require repeated timed steps under fixed conditions.

Asymptotic complexity does not determine these measurements by itself. Attention constructs an $L \times L$ score matrix and therefore scales quadratically with token length, whereas selective state-space recurrence scales linearly in L in its intended implementation [12, 15]. For the short sequences used by SHARP and SEAM, fixed kernel overhead, extra projections, recurrent compaction and distributed communication can dominate the asymptotic term. The empirical runtime tables consequently test realised cost within each of the five protocol-matched families rather than assuming that the theoretically lower-complexity operator must be faster.

2.3 Benchmark datasets

2.3.1 Argoverse 2

AV2 is the primary dataset in this dissertation. It contains approximately 250,000 eleven-second motion-forecasting scenarios collected in six cities and multiple seasons [8]. Each scenario contains timestamped tracks, object position, heading, velocity, semantic category and a local vector map. Tracks are categorised as focal, scored, unscored or fragments, which determines whether they are evaluated or used only as context. The standard temporal division is five seconds of observation followed by six seconds of future at 10 Hz. This corrects the shorter AV1 timing sometimes carried over incorrectly when describing AV2.

SEAM reaches the final AV2 prediction origin through overlapping three-second histories ending at three, four and five seconds. Previous endpoints guide retrieval at the next update, while current observations refresh the encoded scene [9]. The forecasting horizon and the history processed by an individual invocation are therefore different quantities.

The long observed segment is particularly relevant to SHARP. Rather than requiring all five seconds before producing a useful forecast, SHARP can process one-second windows and accumulate context as additional windows arrive [10]. Its evolving-scene analysis evaluates different prediction times and cumulative history lengths; the standard AV2 endpoint remains the five-second-history, six-second-future setting. The controlled local ablations use the SHARP-processed AV2 train and validation data. Dataset preprocessing, token limits and exact split usage are specified in Chapter 3 rather than assumed from the benchmark name.

AV2’s dense interactions and varied agent types support tests of agent–agent and agent–map context. Focal-agent accuracy nevertheless does not establish joint consistency or the dynamic feasibility of every predicted waypoint.

2.3.2 Argoverse 1 and nuScenes

Argoverse 1 (AV1) uses two seconds of history to predict three seconds of future motion and introduced a large urban forecasting benchmark with rich vector maps [28]. nuScenes provides sensor-rich autonomous-driving data in Boston and Singapore; its prediction task commonly uses two seconds of history and six seconds of future sampled at 2 Hz [29]. Both datasets appear in the SHARP article and provide context for how forecasting protocols vary across benchmarks.

Different sampling rates, horizons, scene definitions and scored objects prevent pooling these benchmarks. The experimental comparisons therefore remain centred on AV2.

2.4 Evolution of learned motion forecasting

Figure 2.1 summarises the modelling progression most relevant to this dissertation. The categories overlap rather than forming a strict chronology: modern systems often combine recurrent, graph, attention and query-based components.

The literature is organised by function: temporal encoding, interaction, map structure, multi-modal generation and confidence. This exposes which contributions transfer to streaming prediction and which requirements remain unresolved.

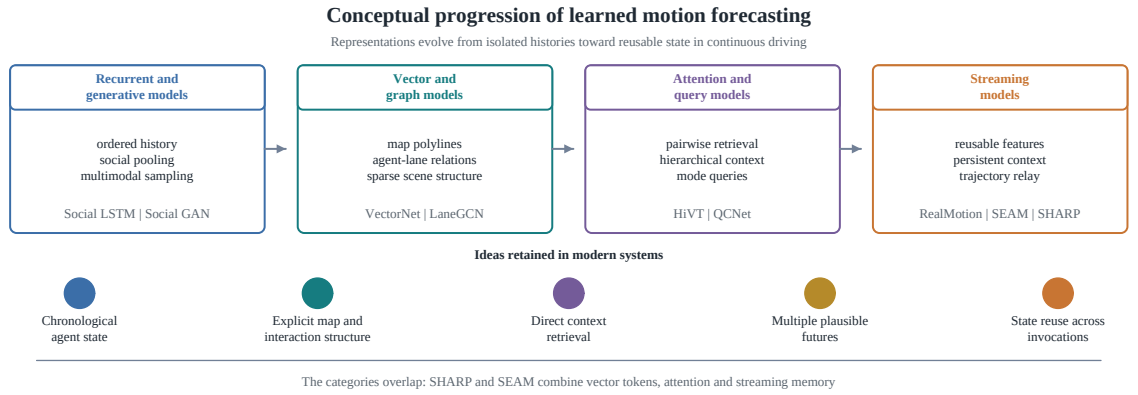


Figure 2.1: Conceptual progression of learned motion forecasting. Each stage retains useful ideas from earlier work while changing the representation or the way context is reused.

2.4.1 Recurrent and generative interaction models

Early learned predictors encoded each trajectory with a recurrent neural network. Social LSTM assigned a recurrent state to each pedestrian and pooled neighbouring hidden states to represent interaction [30]. It established that the temporal state of one agent should not be forecast independently of surrounding agents. Social GAN combined recurrent generation with adversarial learning and a variety loss, encouraging multiple socially plausible samples rather than one aver-

aged path [1]. Trajectron++ represented heterogeneous agents as a dynamic graph, introduced a conditional latent-variable model and optionally enforced dynamic feasibility [2].

DESIRE provides an earlier explicit separation between diversity generation and contextual ranking. Its conditional variational recurrent generator produces multiple hypothetical futures, after which a scene-aware scoring and refinement module reasons about interactions and long-term rewards [31]. This sample–rank–refine pattern remains relevant even in later deterministic-query architectures: producing different modes and deciding which mode deserves confidence are distinct computational problems.

These approaches distinguish temporal memory, diversity and interaction. A recurrent network updates hidden state sequentially; an LSTM gates its memory. Latent variables represent unobserved variation such as intention. Adversarial learning trains a generator against a discriminator of generated and recorded examples. Graph message passing updates entities from their neighbours’ features. Observation order informs motion, but generating multiple paths does not itself make them responsive to agents or maps.

A fixed recurrent state can limit exact retrieval, while neighbourhood pooling can obscure individual influences. Sampled or adversarial objectives also complicate confidence interpretation. Recurrence is consequently investigated alongside, rather than as a universal substitute for, explicit scene retrieval.

2.4.2 Vectorised maps and graph interaction

Raster methods render trajectories and map elements into a bird’s-eye-view image and process the result with a convolutional network. This representation is convenient, but it can introduce resolution loss and expensive dense computation. VectorNet instead encodes agent trajectories and map polylines in their native vector form, first modelling each local polyline and then high-order interactions among scene elements [3]. LaneGCN makes lane topology explicit and performs actor-to-lane, lane-to-lane, lane-to-actor and actor-to-actor message passing [4].

SEAM and SHARP use vectorised agent and lane tokens, learned features representing individual elements. Native coordinates support geometry, variable scene sizes and reuse; interaction still requires deciding which tokens exchange information.

A fixed graph may omit unusual long-range interactions, whereas unrestricted attention must learn which relations matter. Geometry bias combines explicit physical relations with learned retrieval without replacing the scene encoder with a separate road graph.

2.4.3 Hierarchical attention and query-centric prediction

Attention-based models address this interaction problem by allowing tokens to retrieve relevant context directly. HiVT uses local and global Transformer stages while exploiting translation and rotation invariance [20]. QCNet places scene elements in local spacetime coordinate systems and uses query-centric representations that can be reused when the observation window advances [21]. Its recurrent proposal decoder and anchor-based refinement also separate initial multimodal proposals from subsequent trajectory refinement.

Online efficiency depends on coordinate frames, interaction hierarchy and query design as well as the decoder. Direct attention retrieval does not determine what survives an update; SEAM and SHARP explicitly organise reuse through aligned context and prediction transfer.

2.5 Representing multimodal futures

2.5.1 Sampling, anchors and endpoint-conditioned trajectories

There is no unique neural representation of a multimodal future. Sampling-based latent-variable or adversarial systems draw several trajectories from an implicit distribution. They can represent flexible shapes, but finite sampling may miss a low-probability yet safety-critical manoeuvre and may not provide calibrated likelihoods for the retained samples. Anchor-based systems instead begin with a finite vocabulary of motion prototypes. MultiPath, for example, predicts probabilities over anchor trajectories, then regresses offsets and uncertainty around each anchor in one forward pass [32]. The representation is compact and interpretable, but its coverage depends on how well the anchor set spans the deployment distribution.

Goal-conditioned systems exploit the observation that destination uncertainty accounts for a large part of long-horizon trajectory uncertainty. TNT first scores target states, completes a trajectory conditioned on each target and finally estimates likelihoods for a compact selected set [33]. DenseTNT replaces sparse, hand-selected anchors and rule-based non-maximum suppression with dense goals and a learned set-selection stage [34]. These methods explain why endpoints are not merely evaluation coordinates: a destination can function as an intention variable that guides retrieval and decoding. SEAM directly uses prior endpoints as spatial context centres, while SHARP relays previous trajectory modes into later windows.

Learned intention queries avoid committing to a fixed geometric anchor vocabulary. MTR assigns pairs of learnable motion queries to global intention localisation and local trajectory refinement, allowing each query to specialise in a mode while sharing the scene encoder [35]. QCNet and DeMo likewise distinguish mode-level intention from time-varying trajectory state. The dis-

sertation’s six SHARP modes are therefore best understood as a compact set prediction problem: each mode needs a distinct role, a plausible path and a confidence that is comparable with the other modes.

2.5.2 Context fusion for multimodal decoding

Wayformer demonstrates that heterogeneous agents, maps and traffic signals can be fused with a relatively homogeneous attention encoder–decoder. It compares early, late and hierarchical fusion, as well as factorised and latent-query attention for controlling cost [36]. This finding directly informs the SHARP attention ablations: a common attention operator can process heterogeneous agent, lane and target-context tokens when their tokenisation and relation features are explicit, so an operator change can be tested without introducing separate modality-specific branches.

These formulations generate, condition and rank candidate futures. The interventions chiefly alter conditioning and retrieval, retaining six modes and the reference objective except in named auxiliary-loss tests. Improved oracle coverage does not establish calibration or exhaustive future coverage.

2.6 Sets, equivariance and joint scene futures

Observations and future coordinates have temporal order; agent and lane lists usually have preprocessing-defined indices. A model should not treat those indices as unintended motion cues.

Agent and lane collections are sets rather than natural one-dimensional sequences. A permutation-invariant set function produces the same global output after its inputs are reordered, while a permutation-equivariant function reorders per-element outputs in the same way as its inputs. Deep Sets characterises a broad invariant form as

$$f(\{\mathbf{x}_i\}_{i=1}^N) = \rho\left(\sum_{i=1}^N \phi(\mathbf{x}_i)\right), \quad (2.15)$$

where ϕ embeds each element and ρ maps the pooled representation to an output [17]. Set Transformer replaces simple pooling with attention-based interaction while retaining order-appropriate set behaviour [37]. These principles explain why a recurrent scan over arbitrarily assembled scene tokens is questionable: recurrence is order-sensitive, whereas the represented collection should normally be equivariant to token permutation unless geometry or topology defines the order.

Road scenes also require relations that a plain set abstraction does not encode. Graphormer shows that Transformer attention becomes substantially more effective on graph data when structural relations are supplied explicitly in the attention mechanism [38]. Lane connectivity, relative

displacement and heading are the corresponding traffic-scene relations. The dissertation’s relative-geometry bias follows this principle by preserving content attention while making a physically meaningful pairwise relation directly available to each head.

Marginal prediction generates a distribution for each actor independently, even if all actors share a scene encoder. Joint prediction instead generates coherent scene-level futures. Scene Transformer uses agent-permutation-equivariant attention and masking to support marginal, joint and conditionally constrained forecasting with one architecture [39]. AutoBots interleaves temporal and social set attention and uses latent variables to generate scene-consistent multi-agent futures [40]. The local SHARP study evaluates the designated focal actor, so it does not claim joint-world consistency; these works define the additional requirement that would be necessary before deploying the modified model as a complete multi-agent prediction distribution.

2.7 Attention mechanisms and transfer hypotheses

Attention supports temporal encoding, scene interaction and mode-conditioned decoding. Its role depends on the queries and context, so the ablations distinguish scoring rule from placement.

For an input token matrix $\mathbf{X} \in \mathbb{R}^{L \times d}$, a Transformer head forms queries, keys and values:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_V. \quad (2.16)$$

Scaled dot-product attention is

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_h}} + \mathbf{B} + \mathbf{M} \right) \mathbf{V}, \quad (2.17)$$

where d_h is the head dimension, $\mathbf{M}$ is a padding or causal mask and $\mathbf{B}$ is an optional relation bias. Multi-head attention (MHA) performs the calculation in several learned subspaces and concatenates their outputs [12]. In motion forecasting, different heads may specialise in temporal continuity, neighbouring vehicles, lane direction or possible conflict points.

Here L is the number of input tokens, d their feature width and $\mathbf{W}_Q$, $\mathbf{W}_K$ and $\mathbf{W}_V$ are learned projections. For one head with equal query, key and value widths, each projection lies in $\mathbb{R}^{d \times d_h}$. A query describes the information sought, keys provide matching descriptors and values supply the information to aggregate. The transpose $\top$ permits all query–key dot products to be formed as a matrix multiplication. Its raw scores are termed logits. Softmax exponentiates these scores and divides each by their sum over available keys, producing non-negative weights that sum to one for each query.

Self-attention obtains queries, keys and values from the same token set. Cross-attention obtains queries from one representation and keys and values from another, as when a mode query retrieves scene context. An excluded key receives mask value $-\infty$, while an allowed key receives zero. A causal mask additionally excludes information unavailable at the represented time. Soft-max saturation describes concentration near a single choice, leaving other keys with negligible weights.

Attention directly connects queries to available keys through $O(L^2)$ scores at fixed width. With thirty history samples in SEAM and ten in SHARP, the temporal tests primarily concern representation and optimisation, not long-sequence savings. Scene attention instead relates entities and has a different token count.

The standard MHA operator in Equation 2.17 is the control for the independent attention study. QKNorm, Talking-Heads attention and their combination are evaluated only against that control under the attention study’s common protocol. The SHARP architecture integration and architectural ablation baselines, together with both SEAM baselines, belong to separate experimental families and do not enter this operator comparison.

2.7.1 Query–Key Normalisation

The dot product in Equation 2.17 depends on both angular alignment and vector magnitude. Large norms can produce large logit differences and an arbitrarily saturated softmax. QKNorm applies ℓ_2 normalisation along each head dimension:

$$\hat{\mathbf{q}} = \frac{\mathbf{q}}{\max(\|\mathbf{q}\|_2, \epsilon)}, \quad \hat{\mathbf{k}} = \frac{\mathbf{k}}{\max(\|\mathbf{k}\|_2, \epsilon)}, \quad (2.18)$$

then uses a learned scale g_h for head h :

$$\ell_{ij}^h = g_h \hat{\mathbf{q}}_i^\top \hat{\mathbf{k}}_j. \quad (2.19)$$

The score becomes a scaled cosine similarity, and the head learns its own effective softmax temperature [13]. QKNorm was introduced and evaluated for low-resource translation, not motion forecasting. Its relevance here is therefore a transfer hypothesis: stabilising the attention geometry may help temporal and scene interaction, but the original paper does not establish that it will improve trajectories.

The positive ϵ prevents division by zero; i and j index queries and keys. Larger per-head scales g_h concentrate weights on aligned keys; smaller scales broaden them. Normalisation controls norm-driven variation without preselecting tokens.

2.7.2 Talking-Heads attention

In standard MHA, heads do not communicate until their value outputs are concatenated. Talking-Heads attention introduces learned projections across the head dimension immediately before and after the softmax [14]. If $\mathbf{L}$ stacks the logits, a simplified form is

$$\mathbf{A} = \mathbf{P}_{\text{post}} \text{softmax}(\mathbf{P}_{\text{pre}} \mathbf{L}). \quad (2.20)$$

The pre-softmax projection can combine evidence used to decide where to attend, while the post-softmax projection can combine the resulting attention maps. This adds relatively few parameters, but its original evidence also comes from language modelling. In a compact forecasting model with eight heads, cross-head mixing may be useful, redundant or harder to optimise; only a controlled ablation can distinguish these outcomes.

$\mathbf{P}_{\text{pre}}$ and $\mathbf{P}_{\text{post}}$ act on the head axis of the logits and weights; softmax acts over keys. The projections mix evidence for each query–key pair without mixing spatial indices.

2.7.3 Uncertainty-aware target-context attention

Entropy measures dispersion of mode probabilities: it is low when one mode dominates and high when confidence is evenly shared. Division by the maximum for K modes standardises its scale. It measures ambiguity among predicted alternatives, not trajectory error or knowledge of the true future.

Endpoint-conditioned forecasting associates each mode k with a confidence p_k and a prospective spatial region. These confidences rank the finite trajectory set but should not automatically be interpreted as calibrated probabilities [23]. They can nevertheless provide an operational ambiguity signal: a low-confidence mode or a high-entropy mode distribution indicates that narrowly localised context may be insufficient. Goal-conditioned methods establish the value of using prospective endpoints to organise decoding and context selection [33, 34].

In this dissertation, *uncertainty-aware attention* has a restricted architectural meaning. Mode confidence and normalised mode entropy control the support of endpoint-centred context retrieval and gate the retrieved target feature. The uncertainty signal therefore determines which scene tokens are made available to the target-context pathway and how strongly that context contributes;

it does not redefine every attention layer or assert that the complete predictive distribution is calibrated. The query–key score inside the pathway remains the selected attention operator, while QKNorm and relative-geometry bias are separately defined changes to score normalisation and pairwise logits. This separation is necessary for attributing each intervention under Equation 2.12.

2.7.4 Relative-geometry bias

Feature similarity need not imply physical relevance: distant agents may have similar features, while a semantically different crossing lane may matter. Relation bias supplements content similarity with pairwise geometry.

Attention without positional or relational features is permutation-equivariant: reordering tokens reorders the output but does not tell the model what the order means. Relative-position attention augments each pairwise score with a representation of the relation between two elements [11]. For motion forecasting, a relation feature can be defined as

$$\mathbf{r}_{ij} = [\Delta x_{ij}, \Delta y_{ij}, \|\Delta \mathbf{p}_{ij}\|_2, \cos(\Delta \theta_{ij}), \sin(\Delta \theta_{ij})], \quad (2.21)$$

with an attention-head-specific bias $b_{ij}^h = \phi_h(\mathbf{r}_{ij})$. The score becomes

$$\ell_{ij}^h = \frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d_h}} + b_{ij}^h + m_{ij}. \quad (2.22)$$

This inductive bias exposes displacement, range and relative heading directly. It does not replace learned token content; it supplements content-based similarity with a relation that has immediate physical meaning in a traffic scene.

The offsets Δx_{ij} and Δy_{ij} locate token j relative to i in a common frame; $\Delta \theta_{ij}$ is their heading difference. Sine and cosine avoid angle-wrap discontinuities. The learned ϕ_h produces one scalar bias per head, while m_{ij} retains the exclusion mask.

Relative geometry and QKNorm act on different parts of the same logit. QKNorm controls the scale and angular interpretation of content similarity, whereas b_{ij}^h injects a head-specific physical relation. Their combination can therefore be written as

$$\ell_{ij}^h = g_h \frac{\mathbf{q}_i^\top \mathbf{k}_j}{\max(\|\mathbf{q}_i\|_2, \epsilon) \max(\|\mathbf{k}_j\|_2, \epsilon)} + \phi_h(\mathbf{r}_{ij}) + m_{ij}. \quad (2.23)$$

Content alignment and physical relation contribute separate score terms, motivating SHARP’s composition study. Their joint effect still requires measurement because individually useful biases can interfere during optimisation.

2.8 State-space models and Mamba

2.8.1 From linear dynamics to selective recurrence

A continuous linear state-space model represents a latent state $\mathbf{h}(t)$ driven by input $\mathbf{x}(t)$:

$$\frac{d\mathbf{h}(t)}{dt} = \mathbf{A}\mathbf{h}(t) + \mathbf{B}\mathbf{x}(t), \quad \mathbf{y}(t) = \mathbf{C}\mathbf{h}(t) + \mathbf{D}\mathbf{x}(t). \quad (2.24)$$

After discretisation, the recurrence can be written as

$$\mathbf{h}_t = \bar{\mathbf{A}}_t \mathbf{h}_{t-1} + \bar{\mathbf{B}}_t \mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C}_t \mathbf{h}_t + \mathbf{D} \mathbf{x}_t. \quad (2.25)$$

The latent state is a compressed summary of the processed prefix. Unlike all-pairs attention, the update follows the sequence order.

$\mathbf{A}$ propagates state, $\mathbf{B}$ injects input, $\mathbf{C}$ reads out state and $\mathbf{D}$ provides a direct input-to-output path. For an n -dimensional state and scalar input/output, their shapes are $n \times n$, $n \times 1$, $1 \times n$ and a scalar. Bars denote discrete-time transitions; token-dependent parameters allow selective responses along the sequence.

For a step interval Δ , zero-order-hold discretisation assumes that the input remains constant within the interval and gives

$$\bar{\mathbf{A}} = \exp(\Delta \mathbf{A}), \quad \bar{\mathbf{B}} = (\Delta \mathbf{A})^{-1} (\exp(\Delta \mathbf{A}) - \mathbf{I}) \Delta \mathbf{B}, \quad (2.26)$$

when the inverse exists, with stable alternatives used in implementation. Structured state-space models such as S4 parameterise the transition so that long convolution kernels can be evaluated efficiently and stably [41]. S5 replaces a bank of single-input, single-output systems with a multi-input, multi-output state and uses a parallel scan [42]. These developments establish the mathematical lineage from linear dynamical systems to modern sequence layers; Mamba’s distinctive contribution is to make the effective step, input and output projections depend on the current token.

Mamba makes selected SSM parameters input-dependent, allowing the recurrence to propagate or forget information according to the current token. A hardware-aware selective-scan im-

plementation preserves linear scaling with sequence length [15]. This provides a form of content-dependent memory without constructing an $L \times L$ attention matrix.

Here $\mathbf{I}$ is the identity matrix. The matrix exponential $\exp$ propagates dynamics over Δ , rather than exponentiating entries independently. Mamba’s learned step controls retention, not the AV2 sampling interval. Recurrence within a forward pass also differs from cross-update state reuse, which requires support from the streaming architecture.

State-space duality further shows that attention and selective state-space layers are related through structured semiseparable operators rather than belonging to unrelated model families [43]. This theoretical connection supports hybrid architectures: the practical decision is not a binary choice between “Transformer” and “state space”, but which information path should be explicitly addressable and which should be compressed into recurrent state.

In the causal formulation, semiseparability bounds the rank of matrix blocks contained on or below the diagonal. Rank counts independent components: a bounded rank lets interactions between sequence segments be represented compactly rather than by unrestricted pairwise coefficients [43]. This connection concerns structured forms of attention, not equivalence between ordinary softmax attention and every Mamba layer.

2.8.2 Operator–representation compatibility

Placement determines what is compressed and along which axis. History has a past-to-present order; pooled agents and lanes may have only an indexing order. Future-head positions represent predicted times, not further observations. Bidirectional processing can use a full observed window or future latent queries without ground-truth future coordinates: its causal interpretation depends on inputs, not scan direction alone.

Bidirectionality is valuable when the complete input sequence is available, because forward and reverse scans provide context from both directions. Vision Mamba applies this idea to position-encoded image token sequences and demonstrates that an SSM can serve as a bidirectional visual backbone [44]. The analogy is limited: pixels have a fixed spatial layout, while SHARP’s concatenated scene tokens are assembled from variable agents and lanes. The temporal-agent history has the clearer ordered structure, which is why the residual Mamba experiment processes forward and reversed valid observations rather than scanning the scene-token list.

Fixed-size recurrent state also imposes an information bottleneck. Theoretical and empirical copying studies show tasks on which Transformers retrieve detailed context more effectively than

generalised state-space models with a bounded latent state [45]. Motion forecasting is not string copying, but exact retrieval of a particular agent or lane relation is closer to attention’s explicit addressability than to indiscriminate compression. This limitation supplies a second reason not to replace all SHARP attention: pairwise interaction and target retrieval remain explicit, while Mamba is tested only where a compact chronological summary is plausible.

DeMo provides the most directly relevant prior use of Mamba in motion forecasting. It decouples mode queries representing directional intention from state queries representing time-varying future states, combines attention for global aggregation with Mamba for state-sequence modelling and applies different directional choices according to the sequence role [16]. Its result supports a hybrid design principle: attention and selective recurrence can be complementary. It does not imply that either operator can replace every attention layer in a streaming model without loss.

Figure 2.2 contrasts the information paths that motivate the dissertation’s attention and Mamba ablations. The comparison is conceptual; actual runtime also depends on kernel implementation, hidden width, sequence length and hardware.

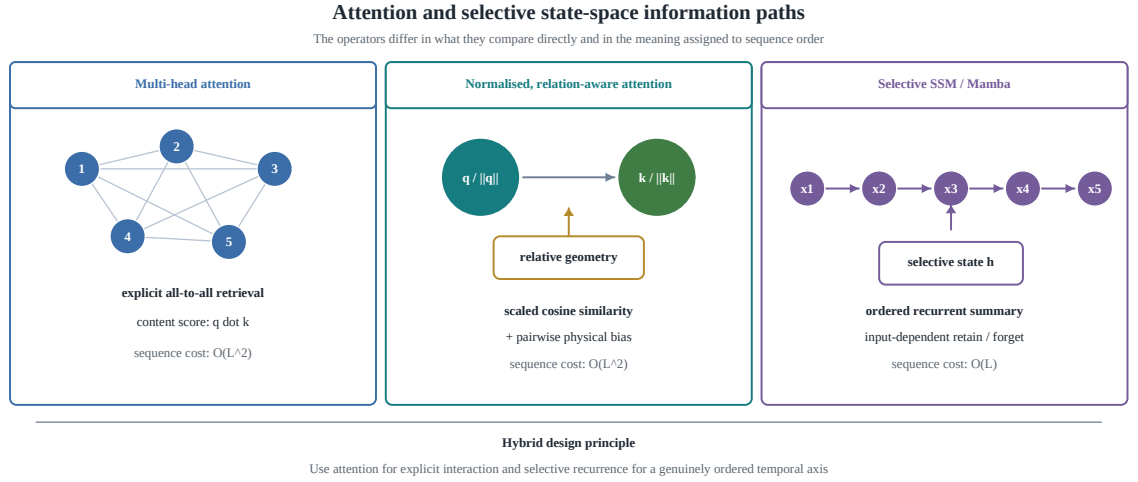


Figure 2.2: Inductive biases of the principal operators. Attention performs explicit pairwise retrieval; QK Norm and geometry bias modify its score; Mamba propagates a selective recurrent state along an ordered sequence.

2.9 Streaming motion forecasting

2.9.1 Reusable scene representations

QCNet identifies a major source of redundant online computation: agent-centred coordinates tied to the latest state require scene elements to be re-normalised and re-encoded whenever the observation window moves. Its query-centric local spacetime representations are independent of the global frame and can therefore be cached and reused [21]. This is an important form of streaming efficiency, although it does not by itself use previous trajectory hypotheses as target-region priors.

Segment-level recurrence has a longer history in sequence modelling. Transformer-XL carries hidden representations from one segment to the next while preventing gradients from propagating indefinitely through the cached memory [46]. SHARP differs because its memory is associated with tracked road instances, transformed between moving coordinate frames and combined with previous trajectories. The comparison is nevertheless useful: both architectures avoid treating every fixed-length segment as independent and must prevent cached context from becoming a stale substitute for current evidence.

2.9.2 RealMotion

RealMotion explicitly reorganises benchmark scenes into continuous observations and introduces two scene-level streams [7]. A context stream progressively accumulates historical scene information and interaction, while a trajectory stream relays past predictions to improve the current forecast. The design demonstrates that previous inference is not merely redundant output: it can become input evidence for the next invocation. It also exposes a risk common to streaming methods, because an earlier error may be propagated unless current observations can correct it.

2.9.3 SEAM

SEAM extends streaming reuse by treating the endpoints of previous trajectories as spatial priors [9]. Let $\mathbf{A}^t$ and $\mathbf{L}^t$ denote the agent-history and lane tensors available at update t , let $\mathbf{S}^{t-1}$ denote previously encoded scene context and let $\mathbf{F}^{t-1}$ denote the previous K trajectory hypotheses. The streaming prediction is represented as

$$(\mathbf{F}^t, \mathbf{P}^t) = f_{\text{SEAM}}(\mathbf{A}^t, \mathbf{L}^t, \mathbf{S}^{t-1}, \mathbf{F}^{t-1}), \quad (2.27)$$

where $\mathbf{F}^t \in \mathbb{R}^{K \times T_f \times 2}$ contains future coordinates and $\mathbf{P}^t \in \mathbb{R}^K$ contains their confidence logits. This formulation makes the retained information explicit: SEAM reuses a scene representation and a set of mode-indexed trajectories rather than concatenating every earlier observation into the current input.

Here t indexes streaming updates, not trajectory samples, and T_f counts predicted positions. Softmax converts $\mathbf{P}^t$ into mode probabilities for ranking and confidence-conditioned retrieval.

Temporal self-attention and masked pooling map each agent history to a D -dimensional token. Separately encoded lane tokens join the agent tokens for scene self-attention, producing $\mathbf{S}^t \in \mathbb{R}^{(N_a + N_l) \times D}$ for N_a agents and N_l lanes. Type embeddings distinguish categories; position and heading embeddings restore relations removed by each element’s local-frame normalisation.

Endpoint-aware modelling introduces a second, mode-conditioned representation. The endpoint of previous mode k is

$$\mathbf{e}_k^{t-1} = \mathbf{F}_{k,T_f}^{t-1}, \quad (2.28)$$

and the associated target region selects current scene elements whose centres $\mathbf{c}_i^t$ satisfy

$$\mathcal{I}_k^t(r_k) = \{i : \|\mathbf{c}_i^t - \mathbf{e}_k^{t-1}\|_2 \leq r_k\}. \quad (2.29)$$

Selected elements are re-expressed in the previous endpoint's local coordinate frame. A positional embedding preserves its relation to the focal-agent frame. Two shallow attention blocks encode $\mathbf{T}_k^t \in \mathbb{R}^{N'_k \times D}$; N'_k is normally much smaller than the complete scene-token count. This concentrates computation on map and road-user evidence likely to affect mode k at long range.

Distance evaluation first aligns the previous endpoint and current scene centres. Radius r_k defines retrieval support, the tokens available to each mode; attention assigns weights. The endpoint guides retrieval without fixing the new trajectory's destination.

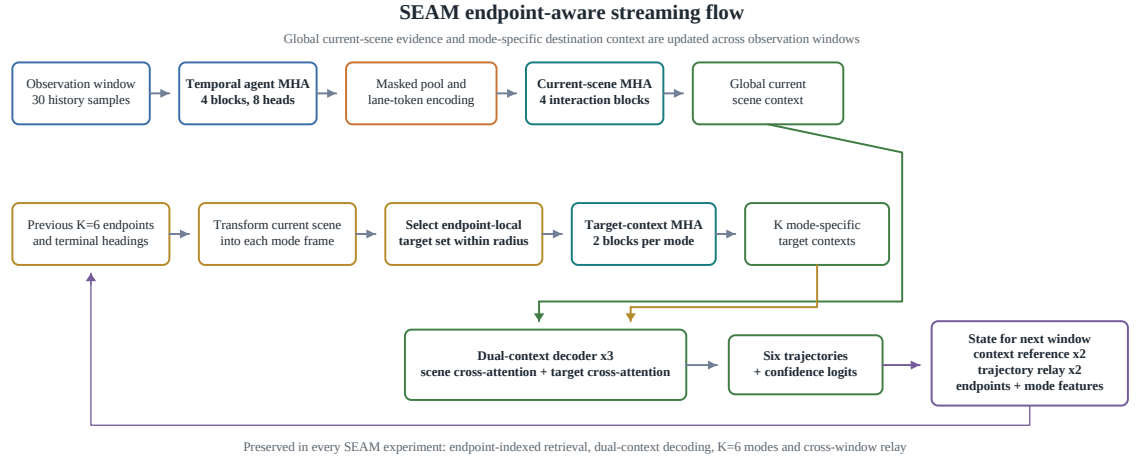


Figure 2.3: SEAM endpoint-aware streaming. Previous mode endpoints define target-centred regions in the current scene; agent-centric and mode-specific target context are then integrated by the dual-context decoder before scene features and trajectories are relayed to the next update.

The dual-context decoder uses one learned query $\mathbf{q}_k^{(0)}$ per mode. At decoder stage ℓ , the query first retrieves globally available scene evidence and then retrieves only the target set associated with the same mode:

$$\tilde{\mathbf{q}}_k^{(\ell)} = \text{CA}_S(\mathbf{q}_k^{(\ell)}, \mathbf{S}^t, \mathbf{S}^t), \quad (2.30)$$

$$\mathbf{q}_k^{(\ell+1)} = \text{CA}_T(\tilde{\mathbf{q}}_k^{(\ell)}, \mathbf{T}_k^t, \mathbf{T}_k^t), \quad (2.31)$$

where CA denotes residual cross-attention with normalisation and a feed-forward sublayer. Three stages produce mode features, then coordinates and confidence scores. The first attention selects current scene evidence; the second selects context near each hypothesised destination. SEAM thus uses the previous prediction to organise one decoding pass, rather than repeatedly refining a trajectory within the same update.

Two further pathways maintain continuity. Context referencing aligns $\mathbf{S}^{t-1}$ using relative translation, heading and elapsed time, then cross-attends current agent and lane tokens to it. Trajectory relay embeds $\mathbf{F}^t$ and $\mathbf{F}^{t-1}$, aligns the previous trajectory to the current frame and predicts a residual coordinate update:

$$\mathbf{S}^t = g_S(\mathbf{C}^t, \mathcal{T}_{t-1 \rightarrow t}(\mathbf{S}^{t-1})), \quad \mathbf{F}^t \leftarrow \mathbf{F}^t + g_F(\mathbf{F}^t, \mathcal{T}_{t-1 \rightarrow t}(\mathbf{F}^{t-1})), \quad (2.32)$$

Here $\mathbf{C}^t$ is the scene encoding before memory integration, g_S updates context, g_F predicts a trajectory correction and $\mathcal{T}_{t-1 \rightarrow t}$ transforms coordinate frames. The arrow adds the correction to the forecast. SEAM reuses spatial context, mode features and trajectories; endpoint-aware target sets link each mode to a prospective region.

SEAM is trained with winner-takes-all multimodal supervision. The mode with the lowest average displacement to ground truth receives the regression target, cross-entropy trains the confidence assigned to that mode and a dense auxiliary head predicts the futures of non-focal agents. The objective is

$$\mathcal{L}_{\text{SEAM}} = \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{aux}}, \quad (2.33)$$

with Smooth L1 regression for the selected mode and auxiliary actors [9]. Because only one mode receives the direct trajectory regression target, the confidence and endpoint pathways influence which alternative becomes useful as well as where that alternative terminates.

$\mathcal{L}_{\text{reg}}$, $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{aux}}$ penalise focal-trajectory, mode-confidence and non-focal-trajectory errors. Smooth L1 is quadratic near zero and linear beyond its transition, limiting large residuals' influence. Cross-entropy penalises low selected-mode probability; auxiliary supervision leaves the focal forecast unchanged.

This design motivates confidence-conditioned target retrieval or gain, geometry-biased scene scores and all-site QKNorm: distinct changes to context, physical relations and similarity scale.

SEAM’s endpoint-indexed dual-context architecture requires direct tests rather than gains inferred from SHARP.

SEAM’s two ordered Mamba sites serve different functions: residual history processing refines evidence before pooling, whereas future-head replacement changes trajectory generation. Their comparison tests placement as well as supplementation versus substitution. The baseline is SEAM itself, without a RealMotion checkpoint, prediction output or fusion branch.

2.9.4 SHARP

SHARP focuses on heterogeneous observation lengths in evolving scenes [10]. It encodes short, non-overlapping observation windows and uses instance-aware context streaming to transfer latent features for matched agents. A dual training objective supervises both streaming and individual-window prediction, encouraging one network to remain useful when only a short history is available and when several windows have accumulated. Previous trajectories and target-centred context support later predictions.

SHARP updates instance-associated state across windows, accommodating new agents and missing observations. Short-window processing and identity-aware memory define this streaming system; changing either would alter the research problem.

The information path begins with separate encoders for agent histories and lane polylines. Temporal attention and pooling produce one token per agent; lane geometry supplies map tokens. Position and type embeddings place these local representations in a shared focal-agent context. Before scene interaction, the streaming module aligns previous context and uses a learned bias to favour corresponding tracked instances. Subsequent self-attention integrates the current agents and map, while endpoint-centred target context and trajectory relay connect that scene representation to the multimodal decoder [10].

Residual history processing changes evidence before pooling; geometry bias changes scene interaction; uncertainty changes endpoint-local retrieval. QKNorm can alter scores throughout these pathways without changing their roles. Instance correspondence and dual training remain properties of SHARP, not of the inserted operator.

Table 2.1 compares the methods that most directly motivate this dissertation.

Reuse differs by indexing: QCNet caches coordinate-consistent encodings; RealMotion propagates scene context and trajectories; SEAM retrieves around mode-indexed endpoints; SHARP transfers instance-associated context and trajectories. These are not interchangeable memories:

time-indexed state cannot automatically replace identity-indexed context, nor a generic cache a mode-specific endpoint prior. The experiments preserve these distinctions.

Table 2.1: Methods relevant to reusable and stateful motion forecasting.

Method	Reuse	Mechanism	Study role
QCNet	Cached scene encodings	Query-centric local spacetime frames and reusable keys/values	Shows that coordinate design determines whether online computation can be reused
RealMotion	Scene context and past trajectories	Context stream plus trajectory relay across successive scenes	Establishes explicit continuous forecasting and dual information streams
SEAM	Previous endpoints and streaming context	Endpoint-centred target regions with a dual-context decoder	Supports the Mamba placement study and controlled context-attention study
SHARP	Per-instance latent context and past predictions	Short-window processing, identity-aware memory and dual training	Supports the placement, screening, attention and confirmation studies

Table 2.1: The method labels refer to QCNet [21], RealMotion [7], SEAM [9] and SHARP [10]. Reuse identifies the information carried between predictions; mechanism identifies how it is carried; and study role states how the method informs the experimental design.

2.10 Synthesis and research gap

The gap concerns how ordered propagation, physical relations and direct retrieval interact in streaming SEAM and SHARP. Evidence from other tasks determines neither placement nor combined forecasting effects.

Table 2.2 makes the inferential step from prior evidence to the controlled studies explicit. The middle column is important: it records what cannot be concluded directly from the cited source and therefore prevents a mechanism developed for another task from being presented as an established forecasting improvement.

SEAM tests endpoint-indexed context, scene interaction and ordered history or future queries; SHARP additionally uses identity-associated memory and short-window supervision. Their distinct paths require separate controls.

The research questions connect these concepts to specific comparisons. RQ1 tests Mamba placement in SEAM, and RQ2 isolates uncertainty, geometry and QKNorm within its context and attention ablation study. RQ3 examines residual Mamba placement and temporal-attention replacement in SHARP under their respective controls. RQ4 connects SHARP’s architectural screening and attention-operator selection to direct evaluation of the combined mechanisms. RQ5 synthesises the resulting evidence about representation, placement and composition across the two architectures. Reference reproduction establishes the local controls needed for attribution.

This mapping links the architectural hypotheses to Chapter 3 while preserving the within-study comparisons defined by Equation 2.12.

Table 2.2: Literature-derived rationale and unresolved transfer questions.

Evidence	Open question	Experiment
Attention scores	Both mechanisms were established outside motion forecasting, and neither proves that changing all attention sites improves trajectory geometry	Compare MHA, QKNorm, Talking-Heads and their combination under one full-schedule operator study
Geometry bias	Token-index relations do not directly encode traffic displacement, range or heading, and a useful bias at one interaction stage may be redundant at another	Apply road-relative geometry only to the scene-interaction logits and test it as an isolated intervention
Target context	A confidence score is not automatically calibrated uncertainty, so expanding a retrieval radius may add relevant evidence or distracting context	Test uncertainty-aware target retrieval independently before combining it with another operator
State-space models	SHARP contains chronological histories, unordered scene collections and mode-conditioned future states; these axes are not equivalent recurrence domains	Separate residual history processing, scene-token scanning and complete temporal-attention replacement
Streaming state	An improvement is not attributable to a local operator if the cross-window state or prediction cadence also changes	Hold the defining streaming pathways fixed throughout each controlled comparison
Reproducibility	A selected reference protocol and its local execution need not produce numerically identical outcomes	Establish one traceable reference per study and admit only checkpoint-backed, protocol-identified evidence

Table 2.2: Evidence summarises the relevant published finding; open question identifies what that finding does not establish for streaming motion forecasting; and experiment gives the corresponding test. Attention scores denotes QKNorm’s control of norm-driven softmax saturation and Talking-Heads’ cross-head mixing [13, 14]. Geometry bias denotes evidence from relative attention and graph-aware Transformers [11, 38]. Target context denotes goal-conditioned forecasting and SEAM’s endpoint-organised prediction and retrieval [33, 34, 9]. State-space models denotes Mamba’s selective compression of ordered prefixes and DeMo’s combination of attention and Mamba on semantically defined sequences [15, 16]. Streaming state denotes reuse of coordinate-stable features, scene state, trajectories, endpoints or identity-associated memory [21, 7, 9, 10]. Reproducibility denotes evidence that implementation and training variation can rival small model differences [18, 19].

Composition is a distinct hypothesis because the interventions are not independent filters. Query–key normalisation changes the scale of content similarity relative to an additive relation bias, while uncertainty-controlled retrieval changes the set and strength of contextual evidence. Joint optimisation can therefore reinforce, attenuate or redistribute their effects. The final SHARP contrast in Table 4.13 tests the package as a whole; it does not infer interaction coefficients from single-component results measured under different protocols. This distinction is necessary even when each component has an established architectural rationale [13, 11].

3 METHODOLOGY AND IMPLEMENTATION

3.1 Experimental design

The experimental work was organised as a sequence of controlled interventions rather than as a search for one large composite model. This distinction matters because the principal question is not merely whether a lower displacement error can be obtained, but which architectural change is responsible for it and whether that change preserves the streaming behaviour of its reference model. The programme in Figure 3.1 separates five comparison families: SEAM state-space integration, SEAM context and attention ablation, SHARP architecture integration, SHARP architectural ablation and the SHARP attention-operator study. Each family has its own matched control. Within that boundary, the data, seed, optimisation schedule, model width and checkpoint rule are fixed; only the intervention named by the hypothesis changes.

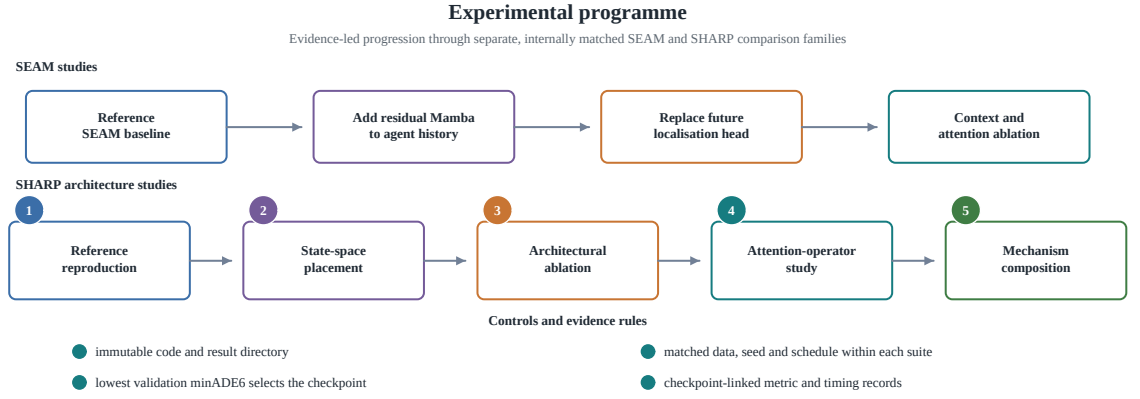


Figure 3.1: Experimental programme used in this dissertation. The SEAM branch contains state-space integration and context and attention ablation. The SHARP branch connects architectural ablation and attention-operator comparisons with reference reproduction, state-space placement and mechanism composition.

Table 3.1 states the experimental factor changed in each family and the scope of the conclusion that it can support. Metric values and elapsed training time are treated as distinct outcomes. Both are compared only within the relevant row of the table; a value from one family is not used to rank a model governed by another training protocol.

Each family follows the selected SEAM implementation or SHARP publication and supplement, with separate ablation controls. Chapter 4 retains evaluation identities, distinguishing minimum-minADE6 selection from full vectors obtained at other epochs.

Table 3.1: Matched-control design of the five experiment families.

Family	Control	Factor	Inference
SEAM SSM	SEAM reference baseline	Agent-history addition or future-head replacement	Effect of state-space placement
SEAM C+A	SEAM ablation baseline	Endpoint uncertainty, scene geometry or QKNorm	Effect of each isolated context or attention mechanism
SHARP INT	SHARP reference reproduction	Residual Mamba placement; composition of context and attention mechanisms	Integration effects in separate matched comparisons
SHARP ABL	SHARP ablation baseline	One local mechanism in each variant	Architectural candidate selection under common controls
SHARP ATT	SHARP MHA control	Attention-score normalisation and cross-head mixing	Effect of each attention operator

Table 3.1: SEAM SSM denotes SEAM state-space integration; SEAM C+A denotes SEAM context and attention ablation; SHARP INT denotes SHARP architecture integration; SHARP ABL denotes SHARP architectural ablation; and SHARP ATT denotes the SHARP attention-operator study. Control is the reference configuration; factor is the deliberately changed mechanism; and inference defines the supported conclusion.

Source, resolved configuration, package inventory, logs and checkpoints identify each experiment. Continuation restores configuration and optimiser state; architectural, schedule or batch changes define a new protocol. Each distributed process, or rank, executes a microbatch. Gradient accumulation combines forward/backward passes before an update; global batch records their total.

Fixed seeds control pseudo-random initialisation and sampling, not identical weights across differently parameterised networks. Warm-up raises the learning rate before decay; AdamW adapts updates through gradient statistics with separate weight decay; clipping limits the global gradient norm. These controls remain fixed while the specified representation or operation changes.

3.2 Dataset and evaluation protocol

3.2.1 Argoverse 2 representation

The main experiments use the Argoverse 2 (AV2) motion-forecasting dataset [8]. The released SHARP preprocessing converts each scenario into tensors containing agent positions, headings, velocities, observation masks, categorical attributes, lane samples and lane attributes. All SHARP comparisons in this work use the same processed training and validation directories. The validation split is used for model selection and for the metric vectors reported in this dissertation; no

claim is made that repeated validation-set inspection is equivalent to a held-out challenge-server test.

At an update time s , the focal agent defines a local coordinate frame. If its global position and heading are $\mathbf{o}_s$ and θ_s , a global point $\mathbf{p}$ is transformed as

$$\bar{\mathbf{p}} = (\mathbf{p} - \mathbf{o}_s)\mathbf{R}(\theta_s), \quad \mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (3.1)$$

This target-centred frame removes absolute translation and rotates the scene into the focal agent’s orientation. Agents outside the configured 150-metre radius, invalid at the update time, or disconnected from the retained lane geometry are excluded by the released data path. Missing observations remain explicitly masked; they are not filled with plausible motion and then treated as measurements.

3.2.2 Metrics and checkpoint selection

The six-mode metrics minADE6, minFDE6, MR and Brier-minFDE6 are complemented by single-mode minADE1 and minFDE1 to expose ranking behaviour. Chapter 2 defines these lower-is-better measures. Every suite selects the lowest validation minADE6, but each complete vector comes from one evaluated model state, never independent column minima. SEAM’s selection and final-epoch vectors retain separate epoch identities; SHARP composition independently evaluates both selected checkpoints.

MinADE6 is designated as the principal comparison metric because it corresponds directly to the six-hypothesis output and, through Equation 2.7, measures the closest hypothesis over the complete forecast horizon. It is therefore more informative about full-trajectory coverage than the endpoint-only minFDE6 and more sensitive to incremental displacement changes than the thresholded MR. Fixing this criterion before the ablations also prevents a variant from being selected retrospectively according to whichever metric happens to favour it. The designation is deliberately narrow: minADE6 is an oracle best-of-six measure and does not establish confidence calibration, first-ranked accuracy, endpoint reliability or computational efficiency. Every result is consequently interpreted with minADE1, minFDE1, minFDE6, MR, Brier-minFDE6 and training time, and an isolated minADE6 reduction is not treated as overall dominance when those complementary measures deteriorate.

3.3 Reference-protocol alignment

3.3.1 SEAM reference specification

The SEAM state-space integration study uses the single-agent training specification distributed with the selected reference implementation and checkpoint [9]. Table 3.2 records the direct alignment between that reference protocol and the state-space integration protocol. The local baseline and both Mamba variants inherit the same data, seed, optimisation schedule, regularisation and checkpoint-selection rule.

Execution is adapted from one GPU with batch 32 to two GPUs with microbatch 8 and two-step accumulation, retaining FP32 and optimisation batch 32. The separate three-GPU SEAM ablation retains model width, optimiser, learning-rate limits, regularisation and seed, but defines a different shared budget and batch.

Table 3.2: Alignment of the selected SEAM reference-implementation protocol and the state-space integration protocol.

Setting	SEAM reference	This study
Epochs	80	80
Warm-up	0.167 (13 epochs)	0.167 (13 epochs)
Optimiser	AdamW	AdamW
Peak LR	10^{-3}	10^{-3}
Minimum LR	10^{-5}	10^{-5}
Weight decay	10^{-2}	10^{-2}
Grad. clip	Global norm 5	Global norm 5
Global batch	32	32
Execution	One GPU, batch 32	Two GPUs, batch 8, accum. 2
Precision	FP32	FP32
Augmentation	None	None
Checkpoint	Validation minADE6	Validation minADE6

Table 3.2: SEAM reference denotes the selected single-agent reference-implementation protocol; this study reports the corresponding state-space integration configuration used by the baseline and both Mamba variants. LR denotes learning rate; Grad. clip denotes gradient clipping; accum. denotes gradient-accumulation steps; and Checkpoint denotes the validation criterion used for model selection.

3.3.2 SHARP reference specification

The SHARP baseline is derived from the authors’ single-agent implementation, retained as an immutable experiment copy before local compatibility adaptations were applied [10]. Each variant uses a separate copy of that reference implementation, preventing subsequent development from silently changing an earlier result.

The SHARP publication and supplementary material specify a global batch of 32, 80 AV2 epochs, 13 warm-up epochs, AdamW, a peak learning rate of 10^{-4} , cosine decay to 10^{-5} , weight decay 10^{-2} , gradient clipping at global norm 5, no augmentation and no cross-dataset training

[10, 47]. Table 3.3 records the direct alignment between that selected reference protocol and the architecture integration protocol. Every SHARP architecture integration baseline and variant in the corresponding study uses this 80-epoch specification.

Table 3.3: Alignment of the selected SHARP reference protocol and the architecture integration protocol.

Setting	SHARP reference	This study
Epochs	80	80
Warm-up	13 epochs	13 epochs
Optimiser	AdamW	AdamW
Peak LR	10^{-4}	10^{-4}
Minimum LR	10^{-5}	10^{-5}
Weight decay	10^{-2}	10^{-2}
Grad. clip	Global norm 5	Global norm 5
Global batch	32	32
Hardware	One RTX 8000	Four RTX 2080 Ti GPUs
Augmentation	None	None
Checkpoint	Validation minADE6	Validation minADE6

Table 3.3: SHARP reference denotes the selected publication protocol; this study reports the corresponding setting used for architecture integration. LR denotes learning rate; Grad. clip denotes gradient clipping; and Checkpoint denotes the validation criterion used for model selection.

3.4 Reference SEAM architecture

3.4.1 Sliding-window input and dual-region encoding

The SEAM baseline instantiates the unmodified `Seam` configuration from the selected implementation [9]. Although the source reuses software infrastructure developed for RealMotion [7], no RealMotion checkpoint, prediction output or fusion branch is included. The experimental baseline is therefore SEAM rather than an ensemble. AV2 history is processed through three model invocations at accumulated times of three, four and five seconds. Each invocation receives a 30-sample, three-second history, and the final invocation shares the standard AV2 prediction origin.

Agent states contain planar position and velocity differences, validity and a temporal index. Projection to $D = 128$, four eight-head temporal blocks and masked maximum pooling yield one token per valid agent. Lane tokens encode centreline features and validity; type embeddings distinguish lanes and four agent classes. A two-layer network embeds token centres and headings as $(x, y, \cos \theta, \sin \theta)$ before four scene blocks form the agent-centric context.

For update $t > 1$, previous mode k supplies an endpoint and terminal heading. Current scene centres are transformed into that endpoint-centred frame, and a fixed 30 m radius selects the corresponding target set in Equation 2.29. Two target-context blocks encode each compressed, mode-specific set. The target region is reset independently for every mode, so a lane or road user may

be relevant to one hypothesised destination and absent from another. This is the principal architectural distinction between generic context streaming and SEAM’s endpoint-aware modelling.

The decoder contains three stages. At each stage, six learned mode queries cross-attend to the complete current scene and then to their associated target regions. Two independent output heads map the final query to an 80-step coordinate sequence and one confidence logit. Only the first 60 future samples define the standard AV2 evaluation horizon; the additional 20 samples provide longer-horizon supervision during earlier streaming invocations. Context referencing and trajectory relay transform retained features and predictions between focal coordinate frames as described by Equation 2.32.

3.4.2 Attention-site audit

SEAM contains the same total number of attention calls as the audited SHARP model, but the semantics of those calls differ. Table 3.4 enumerates the instantiated sites. The count is used to define QKNorm precisely: the controlled QKNorm variant changes every row rather than replacing only the most visible scene encoder.

Table 3.4: Attention sites in the audited SEAM implementation.

Subsystem	Blocks	Attention input
Temporal agent	4	Thirty ordered observations belonging to one agent
Scene	4	Current pooled agent and lane tokens
Context reference	2	Current tokens and pose-aligned previous scene context
Trajectory relay	2	Current mode features and aligned previous mode features
Target context	2	Tokens in each previous-endpoint neighbourhood
Scene decoder	3	Mode queries against complete current-scene context
Target decoder	3	Each mode query against its associated target set
Total	20	All score sites changed in the SEAM QKNorm variant

Table 3.4: Blocks gives the number of instantiated attention blocks; attention input identifies the representation processed at each site. Context reference and Trajectory relay denote cross-window update modules; Scene decoder and Target decoder denote the two cross-attention calls in each decoder stage.

Geometry affects the four scene blocks with comparable agent/lane centres and headings; uncertainty affects confidence-conditioned target retrieval. Both retain temporal encoding, streamed context, trajectory relay and decoder topology.

3.4.3 Objective and state-space integration protocol

The released winner-takes-all objective is retained. Smooth L1 regression supervises the mode with the lowest average displacement, cross-entropy supervises its confidence and a dense auxiliary head predicts non-focal-agent futures, giving Equation 2.33. All SEAM variants preserve the six-mode output, masking rules, endpoint-coordinate transforms and auxiliary non-focal-agent

head. The future-head Mamba intervention replaces only the focal trajectory localisation MLP; it does not replace the auxiliary head.

The baseline and both Mamba variants share the SEAM integration specification and seed 2333. Context and attention ablation uses a separate resource profile and control.

3.5 SEAM state-space integration study

Figure 3.2 shows the baseline and the two deliberately different Mamba interventions. They do not add and replace a block at the same location; each tests a distinct ordered sequence.

The two sites test recurrence over one agent’s observations and one mode’s future positions, respectively, without imposing an arbitrary scene-token order.

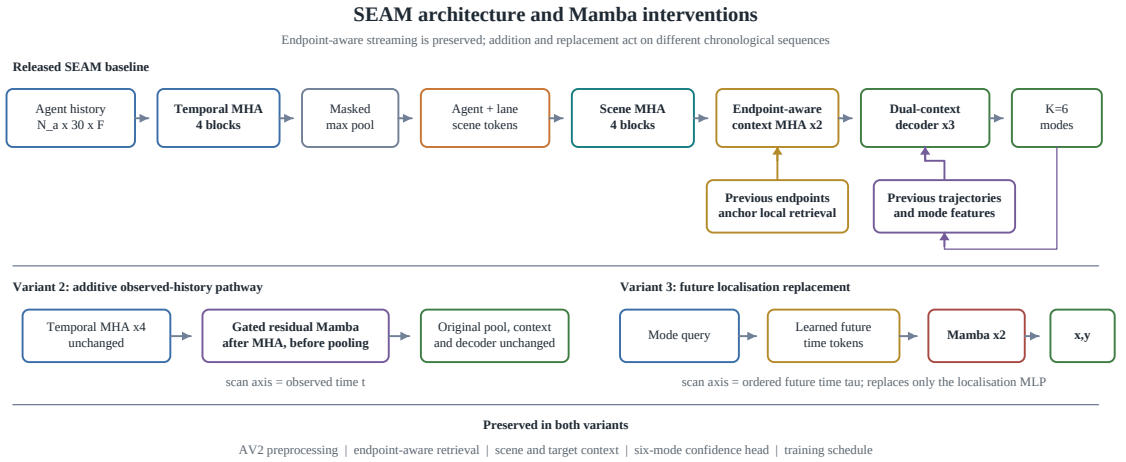


Figure 3.2: Detailed SEAM baseline and modification locations. The agent-history addition refines the ordered observation sequence after temporal MHA and before pooling. The separate future-head experiment leaves the encoder unchanged and replaces only the localisation MLP with time embeddings and two Mamba blocks.

The *agent-history addition* places one gated residual Mamba block after all four temporal MHA blocks and before temporal maximum pooling. It therefore complements rather than removes the released temporal encoder. The block uses $D = 128$, state dimension 16, convolution width 4 and expansion 2. With a trainable scalar a initialised to -2 , its update is

$$\mathbf{X}' = \mathbf{X} + \sigma(a) \text{Mamba}(\text{LN}(\mathbf{X})), \quad (3.2)$$

subject to the original observation mask. Since $\sigma(-2) \approx 0.119$, training starts with a modest recurrent correction rather than an abrupt replacement of the established temporal features.

The sigmoid σ bounds the gate between zero and one. Layer normalisation (LN) normalises each token’s channels before learned scaling and shifting, without batch statistics. Expansion 2

doubles width D internally; state dimension 16 sets per-channel memory; convolution width 4 causally mixes four neighbouring positions.

The *future-head replacement* changes only the decoder localisation MLP. Each mode query is repeated over 80 future positions and combined with learned time embeddings. Two residual Mamba blocks, layer normalisation and a linear $D \rightarrow 2$ projection generate coordinates. This tests explicitly ordered future state against a flat MLP, which already couples its outputs but has no recurrent ordering. The encoder, endpoint context, scene cross-attention, confidence prediction and trajectory relay remain unchanged. Both variants use fused selective scan on the same execution stack.

The history block zeroes invalid inputs and masks residual outputs without compaction. The future head processes each mode’s query-and-time embeddings without future ground truth, connecting positions within rather than across hypotheses.

3.6 SEAM context and attention ablation study

The SEAM ablation compares the baseline with uncertainty-aware target context, relative-geometry bias and QKNorm. Each variant contains only its named intervention, isolating three mechanisms separately from state-space integration.

Figure 3.3 locates the three changes relative to the reference path. The diagram distinguishes retrieval support, relation-aware scoring and attention-score normalisation so that “modified SEAM attention” does not conceal three different hypotheses.

Table 3.5: Shared controls for the SEAM context and attention ablation study.

Setting	Value
Dataset	AV2, common read-only processed training and validation tensors
Architecture	$D = 128$, eight heads, six modes and the original endpoint-aware stream
Epochs	20 with warm-up ratio 0.167
Optimiser	AdamW; learning rate 10^{-3} to 10^{-5}
Regularisation	Weight decay 10^{-2} ; gradient clipping at global norm 5
Global batch	48: eight per rank, three ranks and two-step accumulation
Precision / seed	FP32 / 2333
Checkpoint	Lowest validation minADE6; full-vector epoch recorded separately

Table 3.5: Setting identifies the common data, model, optimisation or selection property; value gives the resolved configuration shared by all four models. Global batch is the number of scenarios contributing to one optimiser update.

Baseline SEAM

The control retains the reference modules, a fixed 30 m target radius, unit feature gain and standard scaled dot-product MHA. It shares the ablation’s 20-epoch schedule with the three variants.

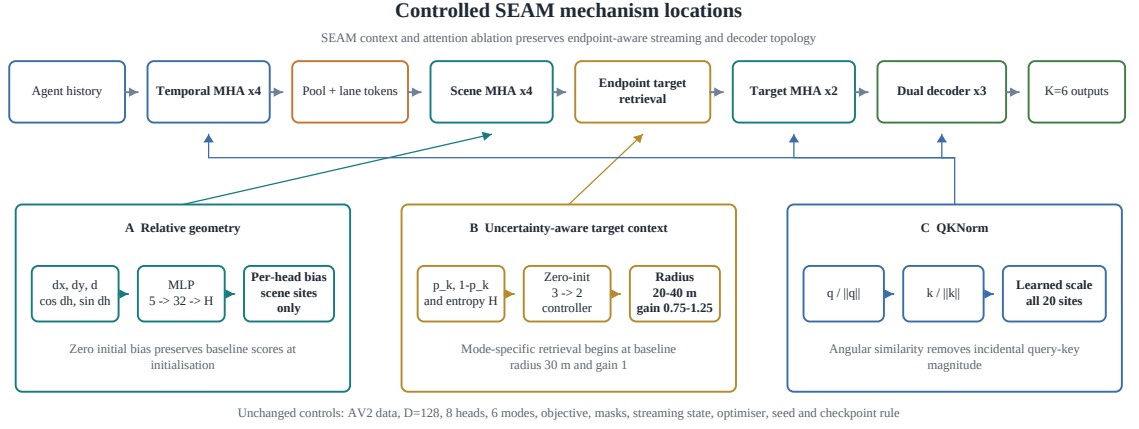


Figure 3.3: Controlled SEAM intervention locations. Uncertainty modulates endpoint-centred target context through a radius-and-gain controller; geometry biases only current-scene self-attention; and QKNorm changes the score rule at all 20 attention sites while preserving the SEAM topology.

Uncertainty-aware target context

The previous confidence logits are converted to probabilities p_k and normalised entropy $H(\mathbf{p})$ using Equation 3.12. A zero-initialised linear controller acts separately on every mode:

$$\begin{bmatrix} u_k^{(r)} \\ u_k^{(g)} \end{bmatrix} = \mathbf{W}_u \begin{bmatrix} p_k & 1 - p_k & H(\mathbf{p}) \end{bmatrix}^\top + \mathbf{b}_u, \quad r_k = 30 + 10 \tanh u_k^{(r)}, \quad \gamma_k = 1 + 0.25 \tanh u_k^{(g)}. \quad (3.3)$$

The radius is therefore bounded between 20 and 40 m and the target-feature gain between 0.75 and 1.25. Zero initialisation gives $r_k = 30$ m and $\gamma_k = 1$ at the start of training, exactly matching the control. The hypothesis is that a diffuse or low-confidence mode should inspect a broader region, whereas a concentrated mode should avoid irrelevant distant tokens. The gain permits the model to reduce the influence of noisy target context without deleting the endpoint-aware branch.

The controller maps three inputs to two scalars, each with a bias. Hyperbolic tangent $\tanh$ bounds them between -1 and 1 ; the constants convert these bounds into a radius in metres and a dimensionless feature gain.

The Boolean distance mask provides no ordinary membership-selection gradient, so the zero-initialised radius row retains the 30 m boundary. Differentiable feature multiplication trains the gain. The intervention therefore tests confidence-conditioned context strength, not learned neighbourhood expansion.

Relative-geometry scene bias

For current-scene tokens i and j , the implementation forms the same five physical features as Equation 3.14. A two-layer $5 \rightarrow 32 \rightarrow H$ network produces one bias per head,

$$\alpha_{ijh} = \text{softmax}_j \left(\frac{\mathbf{q}_{ih}^\top \mathbf{k}_{jh}}{\sqrt{d_h}} + b_h(\mathbf{r}_{ij}) \right), \quad (3.4)$$

where invalid keys remain masked. The final layer of b_h is zero-initialised, so the initial scores equal baseline MHA. Geometry is confined to the four scene blocks because those tokens possess comparable centres and headings; it is not imposed on temporal samples, abstract mode queries or compressed streamed states. The hypothesis is that direct access to range, bearing and heading compatibility reduces the burden on content embeddings when identifying agent–agent and agent–lane relations.

Here the offsets are key minus query: $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$ and $\Delta \theta_{ij} = \theta_j - \theta_i$. The distance term is symmetric, whereas the signed offsets and sine term distinguish opposite relations. The output width H denotes eight attention heads, not the entropy function $H(\mathbf{p})$. Zeroing the final bias layer after model-wide initialisation preserves the stated baseline score at the first forward pass.

QKNorm attention

QKNorm replaces the score kernel at all 20 sites in Table 3.4. Queries and keys are L2-normalised per head, then multiplied by a learned positive scale as defined in Equation 2.19. Each scale starts at $\sqrt{d_h}$ and is capped at 100. Values, masks, residual paths, feed-forward layers and topology are unchanged. This broad substitution tests whether norm-independent angular similarity stabilises SEAM’s heterogeneous attention calls, from temporal encoding to target-specific decoding. It is intentionally broader than relative geometry, which alters only scene interaction. Query–key normalisation changes the score even at initialisation: the initial temperature does not recover the token-dependent norms used by ordinary MHA.

The shared ablation schedule uses three warm-up epochs and global batch 48: three ranks, microbatch eight and two-step accumulation.

All four audited variants share the schedule. Baseline, uncertainty and QKNorm select epoch 19; geometry selects epoch 18. Chapter 4 keeps those scores separate from the complete epoch-19 vectors, particularly geometry’s later secondary metrics.

3.7 Reference SHARP architecture

3.7.1 SHARP stream construction

The standard AV2 scenario contains five seconds of observed motion sampled at 10 Hz. SHARP does not repeatedly encode the full accumulating history. Its released AV2 streaming configuration uses a ten-sample short history and update points

$$\mathcal{S} = \{10, 20, 30, 40, 50\}. \quad (3.5)$$

For update $s \in \mathcal{S}$, the input window is

$$\mathbf{X}^{(s)} = \{\mathbf{x}_{s-9}, \dots, \mathbf{x}_s\}, \quad (3.6)$$

so each invocation encodes one second of fresh observations while memory and prior trajectories provide continuity. The data configuration allocates up to 100 future samples from the earliest origin; the available suffix shortens as the origin advances. Final-origin evaluation uses the standard six-second AV2 horizon.

3.7.2 Agent and lane encoding

SHARP projects per-timestep motion features and categorical attributes into $D = 128$ dimensions. Position differences encode displacement between valid consecutive samples; velocity differences and agent type provide additional motion and semantic information. Four pre-normalised temporal self-attention blocks, each with eight heads and a feed-forward sublayer, process the ten-step history of every agent. A masked maximum over time produces one token per agent. Lane polylines are embedded separately from sampled geometry, lane attributes and intersection information.

The pooled agent tokens and lane tokens are concatenated into a scene sequence. Four scene-attention blocks permit direct agent–agent, agent–lane and lane–lane information exchange. The attention operation is the multi-head form of Equation 2.17; the released model uses a feed-forward expansion ratio of four, GELU activations and stochastic depth up to 0.2. The sequence order in this scene tensor is an implementation ordering, not a chronological axis. This distinction later governs where a recurrent state-space model is physically meaningful.

3.7.3 Streaming context and decoder

The key SHARP contribution is retained in every ablation. After each short window, an instance-aware context module matches agent identifiers to the preceding update, transforms the stored features into the current focal frame and updates each matched instance separately. Agents that

enter the scene have no invented past state; agents that disappear are excluded by the observation and identity masks. Previous trajectories and mode features are also transformed and relayed to the next invocation. Figure 3.4 summarises this dataflow.

The diagram separates three sequence axes. One agent’s temporal tensor, $T \times D$, follows observation time. The pooled scene tensor, $(N_a + N_l) \times D$, stores N_a agents and N_l lane tokens in an implementation-defined order. Target context instead groups a neighbourhood for each of $K = 6$ hypotheses. Their common width $D = 128$ does not make attention and Mamba interchangeable: each axis represents different information.

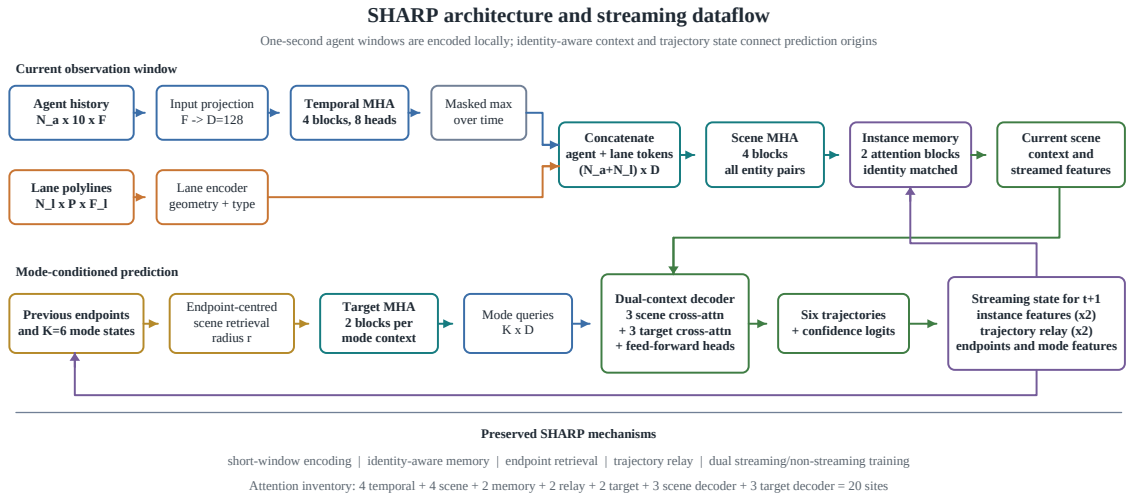


Figure 3.4: Detailed SHARP architecture used as the reference model. Solid arrows show the current-window path; the lower feedback path shows identity-aware context and trajectory relay from the previous prediction origin. Numbers beside attention stages give the instantiated block counts.

Table 3.6 makes the twenty attention sites explicit. The count is obtained from the instantiated module lists rather than from a high-level block diagram. This map is important for the QKNorm ablation because replacing only a subset would produce a different hypothesis from replacing the attention score consistently throughout SHARP.

For each of six trajectory modes, the previous endpoint defines a local target neighbourhood. Scene tokens within the target radius pass through two target-context attention blocks and refine the corresponding mode query. The decoder then uses scene and target cross-attention to predict future coordinates and confidence logits. This endpoint-conditioned retrieval is related to the endpoint-aware design developed in SEAM, while SHARP extends it with short-window instance context and trajectory relay [9, 10]. Dual training retains both streaming and non-streaming paths with shared parameters; none of the local ablations removes this training structure.

Table 3.6: Attention sites in the audited SHARP implementation.

Subsystem	Blocks	Attention input
Temporal agent	4	Valid observations belonging to the same agent
Scene	4	Pooled agent and lane tokens in the current window
Instance memory	2	Current agent state and identity-matched streamed context
Trajectory relay	2	Current mode features and transformed previous prediction state
Target context	2	Tokens in each previous-endpoint neighbourhood
Scene cross-attn.	3	Mode queries against current scene context
Target cross-attn.	3	Mode queries against endpoint-conditioned context
Total	20	All score sites changed in the attention ablation

Table 3.6: *Blocks* is the number of attention blocks in each subsystem; *attention input* identifies the representation processed at that site. *Temporal agent*, *Scene*, *Instance memory* and *Target context* denote their corresponding encoders or update modules; *Scene cross-attn.* and *Target cross-attn.* denote decoder scene and target cross-attention, respectively.

3.7.4 Forecasting objective

The released training loss retains SHARP’s multimodal regression, mode classification and auxiliary non-target regression. In compact form,

$$\mathcal{L}_{\text{SHARP}} = \lambda_{\text{reg}} \mathcal{L}_{\text{reg}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}, \quad (3.7)$$

where the regression term uses the best-matching trajectory mode, the classification term trains its confidence and the auxiliary term supervises other valid agents. The scalar coefficients λ weight their relative contributions. Unless a variant explicitly adds a named auxiliary loss, the objective and its released weights are unchanged.

3.8 SHARP optimisation and execution

Table 3.7 lists the common controls for the SHARP mechanism composition study. The four RTX 2080 Ti implementation uses distributed data parallel (DDP) training with eight scenarios per rank. Gradients are reduced across four ranks, preserving global batch 32 without linear learning-rate scaling. SyncBatchNorm synchronises the mini-batch moments used by batch normalisation across ranks; without it, each replica would estimate those moments from only eight scenarios [48]. This is a hardware adaptation rather than bitwise reproduction of the single RTX 8000 run: data ordering, collective arithmetic, CUDA kernels and library versions can still change the optimisation path.

Table 3.7: Controlled settings for the SHARP mechanism composition study.

Setting	Value
Dataset	AV2, common SHARP-processed training and validation splits
Stream	Ten samples per window; split points 10, 20, 30, 40 and 50
Epoch schedule	80 epochs / 13 warm-up epochs
Optimiser	AdamW
LR schedule	10^{-4} / 10^{-5} with cosine decay
Regularisation	Weight decay 10^{-2} / gradient clipping at global norm 5
Global batch	32 (8 per rank on four GPUs)
Precision / seed	FP32 / 2333
Representation	$D = 128$, eight attention heads, six output modes
Data augmentation	None; no external training data
Checkpoint	Lowest validation minADE6; complete metric vector from that checkpoint

Table 3.7: Setting identifies the controlled training or model property; value gives its resolved configuration for both models in the final comparison. Epoch schedule comprises total and warm-up epochs; LR schedule comprises peak and minimum learning rates; Regularisation comprises weight decay and gradient clipping; Data augmentation also records external training data; and Checkpoint denotes the model-selection rule.

The global optimisation batch is

$$B_{\text{global}} = N_{\text{GPU}} B_{\text{rank}} = 4 \times 8 = 32. \quad (3.8)$$

After the thirteen-epoch linear warm-up, the learning rate follows the cosine schedule

$$\eta(e) = \eta_{\min} + \frac{\eta_{\max} - \eta_{\min}}{2} \left[1 + \cos \left(\pi \frac{e - E_w}{E - E_w} \right) \right], \quad E_w \leq e \leq E, \quad (3.9)$$

where $E_w = 13$, $E = 80$, $\eta_{\max} = 10^{-4}$ and $\eta_{\min} = 10^{-5}$. Holding the global batch and this schedule fixed is essential: changing either would alter the optimisation trajectory at the same time as the architecture and would prevent a causal interpretation of the ablation.

The attention-operator study uses batch 24 and its own MHA control, separate from batch-32 integration. Accumulation preserves examples per optimiser update, not the batch-normalisation moments of a larger forward pass. Per-rank batch and accumulation therefore specify execution as well as global batch.

Batch-normalisation moments are the activation mean and variance within a forward pass. Synchronising these across devices differs from accumulating gradients across passes; layer normalisation instead uses each token’s channels.

3.9 SHARP architectural ablation study

The SHARP architectural ablation study compares ten models, each trained for 20 epochs. Every run uses the same AV2 tensors, seed 2333, global batch 32, four GPUs, AdamW, weight decay 10^{-2} , clipping at 5, peak and minimum learning rates of 10^{-4} and 10^{-5} , SyncBatchNorm, $D = 128$, eight heads and six output modes. The original 13-epoch warm-up is deliberately retained. Consequently, only seven epochs remain after the learning-rate peak; the screen ranks early-training behaviour and cannot establish full convergence. Each intervention in Table 3.8 changes one mechanism relative to test 1, except that the final test deliberately replaces the complete temporal stack.

Table 3.8: Controlled interventions in the SHARP architectural ablation study.

Test	Intervention	Location	Rationale
1	Baseline	No architectural change	Establish the within-suite reference
2	Memory gate	Instance-memory residual update	Reject stale streamed context when it conflicts with the current token
3	Consistency	Additional loss between adjacent forecasts, weight 0.05	Reduce unnecessary forecast changes between successive windows
4	Temp. pooling	Replaces masked temporal maximum	Select the most informative valid observation rather than the largest value per channel
5	Uncertainty	Target-neighbour radius and target-feature gate	Inspect wider endpoint context for uncertain modes
6	Geometry bias	Added to all four scene-attention logits	Supply distance and heading relations directly to each head
7	Kinematic stem	Before temporal-agent encoding	Expose changes in displacement and speed-difference features
8	Endpoint refine.	Decoder output with auxiliary loss weight 0.2	Correct coarse endpoints and improve final displacement
9	Lane graph	Lane tokens before scene interaction	Exchange messages between geometrically neighbouring lane segments
10	Mamba replace.	Replaces four temporal MHA blocks	Test selective recurrence as the entire agent-history encoder

Table 3.8: Test gives the sequence number; intervention is the isolated change; location identifies where it acts in SHARP; and rationale states the expected mechanism of improvement. Memory gate denotes confidence-gated memory; Consistency denotes cross-window consistency; Temp. pooling denotes learned temporal pooling; Uncertainty denotes uncertainty-aware target context; Geometry bias denotes relative-geometry attention bias; Kinematic stem denotes the kinematic motion stem; Endpoint refine. denotes endpoint refinement; Lane graph denotes the lane-topology graph; and Mamba replace. denotes temporal Mamba replacement.

3.9.1 Test-by-test definitions and hypotheses

Shared notation and components recur across variants. $\odot$ denotes component-wise multiplication; LayerScale applies a learned per-channel residual scale, often initialised to 0.01. Dropout suppresses activations and stochastic depth bypasses residual branches during training, regularising the active computation. GELU smoothly attenuates negative inputs; ReLU sets them to zero.

Xavier initialisation sets weight variance from input and output widths; model-wide initialisation can override module defaults.

Test 1 – Baseline SHARP

The first test is the unmodified SHARP control [10]. It retains the input projections, four temporal-agent MHA blocks, masked temporal maximum, four scene-attention blocks, identity-aware context streaming, trajectory relay, endpoint-conditioned target-context retrieval and multimodal decoder. No auxiliary loss, additional feature or new trainable branch is introduced. Its purpose is not to improve accuracy directly, but to establish the metric vector, optimisation trajectory and computational cost against which tests 2–10 are interpreted. A separate baseline is required for this ablation because its 20 training epochs differ from the 80 used for SHARP architecture integration.

Test 2 – Confidence-gated memory

This test modifies only the instance-memory residual update. It receives the current token $\mathbf{z}$, the standard SHARP memory update $\tilde{\mathbf{z}}$ and their difference. A learned element-wise gate computes

$$\mathbf{z}' = \mathbf{z} + \sigma(f([\mathbf{z}, \tilde{\mathbf{z}}, \tilde{\mathbf{z}} - \mathbf{z}])) \odot (\tilde{\mathbf{z}} - \mathbf{z}). \quad (3.10)$$

The constructor’s zero weights and bias 2 are overwritten by SHARP-wide Xavier weights and zero biases. The gate is therefore input-dependent initially, with channels that learn to retain or suppress memory updates. Context may remain useful for a behaviourally consistent identity but become stale after a manoeuvre or imperfect association. Conditioning on current features, memory and their disagreement is hypothesised to preserve continuity while suppressing contradictory history, without removing instance-aware streaming.

Test 3 – Cross-window consistency

The third test retains the forward architecture and adds a Smooth L1 penalty between successive confidence-weighted trajectories after alignment to the current frame. Its coefficient, 0.05, leaves the original objective dominant. Confidence weighting avoids hard correspondence between modes whose ordering can change across windows. Penalising unsupported jumps between one-second updates is hypothesised to reduce temporal jitter and improve continuity. The penalty remains soft: a newly observed manoeuvre can require substantial revision, and excessive consistency would preserve an obsolete forecast.

Test 4 – Learned temporal pooling

This test replaces only the masked per-channel temporal maximum. For valid temporal features $\mathbf{h}_t$,

$$a_t = \frac{\exp(f(\text{LN}(\mathbf{h}_t)))}{\sum_{u \in \mathcal{V}} \exp(f(\text{LN}(\mathbf{h}_u)))}, \quad \mathbf{h}_{\text{pool}} = \sum_{t \in \mathcal{V}} a_t \mathbf{h}_t, \quad (3.11)$$

where f is a $D \rightarrow D/2 \rightarrow 1$ network and $\mathcal{V}$ contains valid timesteps. Unlike channel-wise maxima drawn from different observations, the learned distribution assigns each timestep one importance weight. It is hypothesised to emphasise turns, acceleration and interaction responses while suppressing noisy or uninformative samples.

Test 5 – Uncertainty-aware target context

The fifth test changes how endpoint-conditioned context is retrieved. It begins from the previous mode probabilities p_k . Normalised entropy

$$H(\mathbf{p}) = -\frac{\sum_{k=1}^K p_k \log(p_k + \epsilon)}{\log K} \quad (3.12)$$

is approximately zero for a concentrated distribution and approximately one for uniform confidence. Here $\log$ is the natural logarithm and the small positive ϵ prevents taking the logarithm of zero. The implementation parameterises a minimum radius and span, initialised to 24 m and 16 m. For mode k ,

$$r_k = \text{clip}(r_{\min}, 15, 35) + \text{clip}(s, 1, 25)(1 - p_k). \quad (3.13)$$

A low-confidence mode consequently searches a wider endpoint neighbourhood, whereas a confident mode uses a compact region. A sigmoid gate conditioned on $(p_k, H(\mathbf{p}))$ also scales the retrieved target feature; its module-level zero weights and bias 2 are superseded by the same model-wide Xavier/zero initialisation as the memory gate. The hypothesis is that ambiguous modes require broader evidence to discover alternative lanes or interactions, while confident modes benefit from excluding distant distractors. This couples predictive uncertainty to context extent rather than merely reporting uncertainty after decoding [23, 22].

Hard thresholding supplies no retrieval-membership gradient to radius parameters. SHARP nevertheless varies radius with previous mode probability through $(1 - p_k)$, while the forecasting loss trains the continuous feature gate. Confidence-dependent retrieval is therefore distinct from learning its boundary through token selection.

Here $r_{\min}$ sets the compact neighbourhood, s its additional span and $\text{clip}(x, a, b)$ restricts x to $[a, b]$. Hard-mask membership is a binary decision, not a smooth function of radius like continuous feature weighting.

Test 6 – Relative-geometry attention bias

This test augments the logits of all four scene-attention blocks without changing their value vectors or masks. For scene tokens i and j , relative centres are divided by 30 m and represented by

$$\mathbf{r}_{ij} = [\Delta x_{ij}/30, \Delta y_{ij}/30, \|\Delta \mathbf{p}_{ij}\|_2/30, \cos \Delta \theta_{ij}, \sin \Delta \theta_{ij}]. \quad (3.14)$$

A $5 \rightarrow 32 \rightarrow H$ GELU network predicts one additive bias per attention head. The module constructor zeroes its final layer, but SHARP’s subsequent global Xavier initialisation overwrites that setting. The implemented bias is therefore learned from a nonzero initial mapping rather than an exactly baseline-preserving first forward pass. Content-only attention must infer distance, bearing and heading compatibility indirectly from token embeddings. Supplying those physical relations directly could improve sample efficiency and make interactions between agents and lanes easier to discriminate. The design adapts relation-aware attention [11] to continuous road geometry while preserving SHARP’s scene-token representation.

In the SHARP adapter, signed offsets use query minus key, $\Delta \mathbf{p}_{ij} = \mathbf{p}_i - \mathbf{p}_j$, with the same direction for heading differences. This is the reverse of the SEAM adapter’s signed convention; each remains internally consistent, and the two networks are trained separately. The 30 m scale makes the positional inputs dimensionless without changing the neighbourhood mask.

Test 7 – Kinematic motion stem

The seventh test inserts a residual feature stem before temporal-agent encoding. Finite differences operate on SHARP’s already processed planar-displacement and speed-difference channels. The five added components are the first and second temporal differences of planar displacement and the first difference of the speed-difference channel, masked at invalid output timesteps. They are discrete feature changes, not physical velocity or acceleration estimates in SI units. Five resulting scalar components are projected by a $5 \rightarrow D \rightarrow D$ network and added to the ordinary motion embedding through dropout 0.2 and LayerScale 0.01. The small initial residual protects the original representation while allowing the branch to grow if useful. The hypothesis is that these higher-order changes expose braking, acceleration and turning tendency more directly than the original embedding alone, helping the temporal encoder distinguish geometrically similar histories

with different dynamics. The corresponding risk is that differentiation amplifies localisation noise and duplicates information that temporal attention can already infer [6].

Test 8 – Endpoint-refinement decoder

This test retains the multimodal decoder but adds a residual correction after its coarse trajectory prediction. Each coarse endpoint is embedded and combined with the corresponding mode query; lightweight heads then predict a scaled trajectory correction and a confidence correction. LayerScale 0.01 limits the initial perturbation, while an auxiliary regression loss with weight 0.2 supervises the coarse output so that the new head cannot rely entirely on repairing a poor first estimate. Endpoint-conditioned forecasting methods demonstrate that destination quality strongly constrains the subsequent path [33, 9, 10]. The hypothesis is therefore that a dedicated local correction can reduce terminal displacement and sharpen endpoint-conditioned context while preserving the decoder’s six-mode structure.

Test 9 – Lane-topology graph

The ninth test modifies lane tokens immediately before dense scene interaction. Each lane segment selects at most eight neighbouring lane segments within 40 m. Distance determines neighbour selection; an edge network receives the two normalised offset components and cosine and sine of the relative heading, uses these four features to weight messages and applies a normalised residual update with LayerScale 0.01. No external topology annotation is introduced; connectivity is constructed from geometry already present in the processed scene. The hypothesis is that plausible motion follows locally connected lane structures, so an explicit sparse map message can expose route continuation and turn compatibility before agent–map attention. This reduces the burden on content-only dense attention to rediscover lane adjacency and follows the structured-map principle demonstrated by lane-graph forecasting models [4].

Test 10 – Temporal Mamba replacement

The final test is the strongest intervention. It removes all four temporal-agent MHA blocks and substitutes four unidirectional Mamba blocks with state dimension 16, convolution width 4 and expansion factor 2. The input projection, observation mask, masked temporal maximum, scene interaction, instance memory, target context, trajectory relay and decoder remain unchanged. The hypothesis is that each agent history has a stable chronological order, allowing a selective state-space recurrence to retain informative motion state and suppress irrelevant observations with linear sequence scaling [15]. Unlike the residual Mamba tests, this configuration asks whether recurrence can replace direct pairwise temporal comparison entirely. It consequently distinguishes the benefit

of a state-space operator from the safer possibility of using Mamba only as a small complementary path.

3.10 SHARP attention-operator study

The attention-operator study compares MHA, QKNorm, Talking-Heads and their combination over 80 epochs. Each substitution covers the twenty audited sites in the temporal, scene, memory, relay, target and decoder pathways. Non-attention layers, masks, losses, data and seed remain unchanged.

QKNorm applies L2 normalisation to every query and key vector before their dot product and learns one temperature per head, as formalised in Equation 2.19 [13]. The local scale starts at $\sqrt{d_h}$ and its exponentiated value is capped at 100 for numerical safety. Talking-Heads inserts learned linear mixing across heads immediately before and after softmax, following Equation 2.20 [14]. Both mixing matrices are initialised to identity. Identity head mixing preserves the ordinary attention operation initially; QKNorm does not, because normalising queries and keys removes their original norms. The two interventions therefore have distinct initial behaviours as well as distinct score mechanisms.

Boolean and additive masks are converted to a common representation without changing permitted interactions. The study uses three GPUs, SyncBatchNorm and global batch 24. Ordinarily each rank processes eight scenarios; the combined operator also uses microbatch four with two-step accumulation, preserving optimisation batch but changing per-forward normalisation. The selected operator is subsequently tested within the batch-32 composition study rather than ranked against its baseline directly.

3.11 SHARP state-space placement

Mamba provides input-dependent selective state updates with linear sequence complexity [15]. The relevant methodological question is not simply whether to add a Mamba block, but which tensor dimension has a defensible sequence meaning. Figure 3.5 distinguishes the three SHARP placements tested in this work.

An *addition* supplements the baseline path with a scaled residual; a *replacement* assigns an existing operator’s function to a new block. A small nonzero residual is conservative, not an unchanged mapping. SHARP composition combines similarity scaling, geometric scoring and target retrieval, excluding sequential propagation.

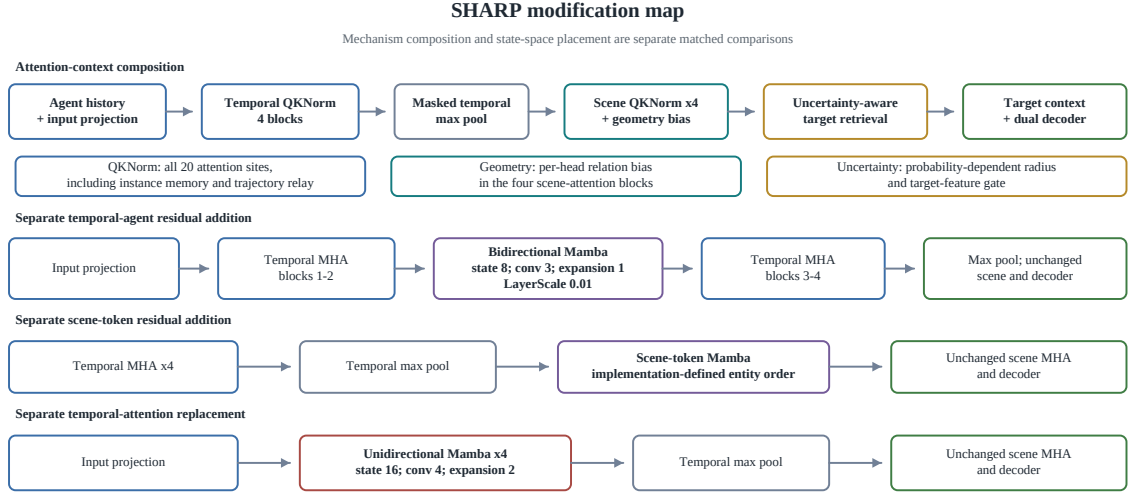


Figure 3.5: Detailed SHARP intervention map. *QKNorm* changes the score at all twenty attention sites; relative geometry acts only in the four scene blocks; uncertainty controls target-neighbour retrieval; residual temporal Mamba is inserted between temporal blocks 2 and 3; scene-token Mamba is an earlier post-pooling addition; and the replacement variant removes the four temporal MHA blocks.

3.11.1 Scene-token residual addition

The scene-token experiment retains temporal attention and inserts fused Mamba after pooling, before scene processing. It scans assembled agent and lane tokens to test selective refinement at lower asymptotic cost than another all-to-all block. Because adjacency follows token assembly rather than time or road geometry, the scan imposes a directional prior on an implementation order. This provides the contrast with chronological agent-history recurrence.

3.11.2 Residual temporal-agent addition

The second long experiment removes the scene-token Mamba entirely and inserts one bidirectional residual Mamba module between temporal-agent attention blocks 2 and 3. Each sequence now represents one agent's valid chronological observations before temporal pooling. Let $\mathbf{X} \in \mathbb{R}^{T \times D}$ be the compacted valid history. Two independent selective state-space modules process forward and reversed order:

$$\mathbf{U}_{\rightarrow} = \text{Mamba}_{\rightarrow}(\text{LN}(\mathbf{X})), \quad (3.15)$$

$$\mathbf{U}_{\leftarrow} = \text{Reverse}(\text{Mamba}_{\leftarrow}(\text{Reverse}(\text{LN}(\mathbf{X})))) . \quad (3.16)$$

A learned per-channel gate $\mathbf{g} = \sigma(\alpha)$ combines both directions, and the block returns

$$\mathbf{X}' = \mathbf{X} + \gamma \odot \text{Dropout}(\mathbf{g} \odot \mathbf{U}_{\rightarrow} + (1 - \mathbf{g}) \odot \mathbf{U}_{\leftarrow}) . \quad (3.17)$$

Zero gate logits α initially weight both directions equally. The configuration uses $D = 128$, state dimension 8, convolution width 3, expansion 1 and dropout 0.1, with LayerScale $\gamma = 0.01$ keeping the initial mapping close to SHARP. Only compacted valid timesteps are reversed; padding is restored afterwards. Groups of 128 agents bound memory without dividing the temporal axis. Fused CUDA selective scan and equivalent framework causal convolution keep the recurrence GPU-resident.

3.11.3 Complete temporal-attention replacement

The replacement variant removes all four temporal-agent MHA blocks and substitutes four unidirectional Mamba blocks with state dimension 16, convolution width 4 and expansion 2. The original input projection and masked temporal maximum remain. This is a stronger causal intervention than Equation 3.17: it tests whether selective recurrence can perform all within-agent temporal modelling. Scene attention, instance memory, target context, trajectory relay and the decoder remain SHARP components, so the defining short-window streaming mechanism is preserved even though the local temporal operator changes.

3.12 SHARP mechanism composition study

The two-model SHARP composition study tests whether the selected mechanisms retain their benefit under the article-aligned integration protocol:

1. **Official SHARP baseline:** The pinned single-agent architecture, article schedule and compatibility changes that preserve model behaviour.
2. **QKNorm + uncertainty + geometry:** QKNorm replaces every MHA constructor; Equation 3.13 controls endpoint-neighbour retrieval; Equation 3.14 supplies additive biases to the four scene-attention blocks.

Both models share data, seed, optimiser, schedule, global batch and checkpoint criterion. Their checkpoint-backed contrast in Chapter 4 tests the package directly against an unmodified baseline, not an effect inferred from preceding component studies.

Resolved configurations differ only in output directories; architecture audits identify the interventions. Baseline and composition contain 4,635,611 and 4,636,232 trainable parameters, respectively, with no Mamba modules. The minADE6 rule selects epochs 78 and 79, both evaluated independently on one GPU with batch 32. The primary metric, five secondary measures and training duration retain their distinct roles.

3.13 Evaluation and evidence controls

The scientific variants are separated from the shared execution layer. Behaviour-preserving compatibility changes are applied identically to all members of a controlled suite, recorded in the experiment manifest and excluded from the list of learnable architectural interventions. This keeps the causal comparison focused on the named attention, context, geometry or Mamba change. Recording source, configuration, environment and evaluation artefacts follows established reproducibility guidance for machine-learning experiments [19].

The shared compatibility layer preserves the intended linear algebra and masking semantics across the controlled variants. The inverse of the orthonormal rotation matrix in Equation 3.1 is evaluated by its equivalent transpose; index tensors are colocated with their operands; attention masks use a common representation; and metric aggregation receives the true validation-batch cardinality. Optimiser construction assigns every trainable parameter to exactly one AdamW group. None of these controls creates a learnable pathway unique to a scientific variant.

Pre-training checks verify source, configuration and optimiser coverage. A real AV2 batch exercises predictions, loss, gradients and an update; communication and bounded four-GPU tests then verify the intended batch construction.

Per-epoch checkpoints retain the model and optimiser state, while minimum-minADE6 checkpoints provide the model-selection record. Exact experiment and result roots associate each configuration with its checkpoint and validation artefacts. Figure 3.6 summarises the resulting evidence path from pinned source to validated artefact.

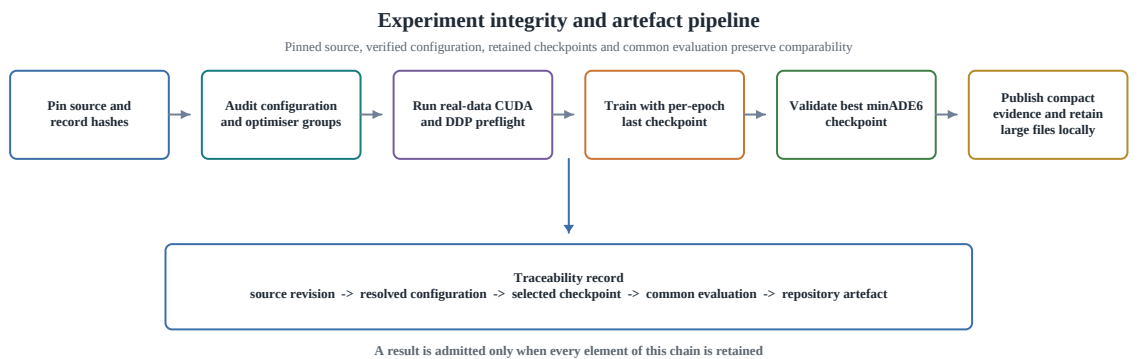


Figure 3.6: Integrity pipeline used for the long-running suites. Source, configuration, checkpoints, evaluation and publication remain traceable for every scientific variant.

Independent one-GPU validation avoids replicated samples from distributed padding. Retained artefacts include metric vectors, checkpoint inventories and SVG figures with vector-PDF derivatives; PNGs are previews. Model-specific outputs reside under `Studies/`, curated com-

parisons under `Results/` and large checkpoints and raw logs in experiment storage. Credentials are excluded.

3.14 Chapter summary

Each intervention changes a defined representation while retaining the reference streaming structure. The implementation distinguishes trainable modulation from hard retrieval and model-wide initialisation from module defaults. These distinctions and the family-specific controls govern the accuracy and timing interpretation in Chapter 4.

4 RESULTS AND DISCUSSION

4.1 Evaluation evidence and comparison rules

This chapter evaluates accuracy and training duration within the five matched study families, retaining each model’s protocol and evaluation identity.

All reported forecasting metrics are lower-is-better. For a metric m , the absolute and relative differences between a variant and its matched control are

$$\Delta m = m_{\text{variant}} - m_{\text{control}}, \quad \Delta m\% = 100 \frac{m_{\text{variant}} - m_{\text{control}}}{m_{\text{control}}}. \quad (4.1)$$

A negative value therefore denotes an improvement. Percentage changes are used to communicate scale rather than statistical significance. The principal studies use one seed, so a difference of a few thousandths of a metre cannot be interpreted as a population-level effect without repeated runs. A stronger inferential design would repeat each matched configuration across seeds and construct paired scenario-level bootstrap intervals for every metric delta. The present evidence consequently supports configuration-specific conclusions and candidate selection, rather than a claim of universal model ordering [19, 18].

Best checkpoint minADE6 identifies the lowest monitored validation score; a *metric vector* contains all six measures from one model state. Different checkpoints are reported separately: an epoch-66 selection cannot replace minADE6 in an epoch-79 vector.

A paired bootstrap resamples the same scenarios for both models to estimate sampling uncertainty in their metric difference; repeated seeds instead measure training variation.

SEAM integration and context ablation precede the SHARP studies. Shared hypotheses connect them; differing schedules and batches preclude a common numerical ranking.

4.1.1 How to read a six-metric result

The six metrics distinguish path coverage, endpoint coverage and confidence ranking: a better endpoint need not imply a better path, nor an improved oracle trajectory a better highest-confidence prediction.

For this reason, a configuration is said to *dominate* a control only when it is no worse on every reported metric and strictly better on at least one. Some variants satisfy this condition in their retained evaluation, while others improve only a subset of the metrics or none. A mixed vector is

a trade-off, not proof that the configuration belongs to the Pareto frontier of all alternatives. Confidence calibration is especially relevant because neural-network probabilities can be overconfident even when ranking accuracy is high [23]. b-minFDE6 partly exposes that problem, but it is not a complete proper scoring-rule analysis; proper scores are designed so that the true predictive distribution minimises expected loss [27].

The absolute scale must also be interpreted in context. A reduction of 0.003 m in a dataset-level mean is three millimetres, but it is averaged over many trajectories and timesteps and does not imply that every prediction moved three millimetres. Conversely, the small mean does not prove irrelevance: a consistent paired improvement could be systematic. The present single-seed summaries cannot separate that possibility from optimisation variance. Benchmark studies show that initialisation, sampling and hyperparameter choices can change apparent rankings, so repeated seeds are required before millimetre-scale differences are treated as robust [18].

4.2 SEAM state-space integration study

4.2.1 Baseline reproduction

SEAM’s endpoint-aware architecture provides the first setting for examining selective recurrence. The selected reference checkpoint and the local baseline are compared in Table 4.1. The local run follows the same training schedule distributed with the reference implementation, and both results use the unmodified SEAM architecture [9].

The local baseline is close to the reference vector: final-epoch minADE6 is 0.001314 m higher, or 0.20%, while minFDE6 is 0.023190 m higher, or 1.86%. The retained checkpoint selector gives minADE6 0.662859 at epoch 77. The complete vector in Table 4.1 instead comes from the final epoch-79 model, evaluated through the two-GPU training pipeline. The score difference therefore reflects distinct model states, not a single-GPU reevaluation of the selected checkpoint. These records establish a local reference control in the expected forecasting regime without asserting numerical identity with the reference checkpoint or equivalence between distributed and single-device evaluation.

Table 4.1: Selected SEAM reference checkpoint and final-epoch distributed evaluation of the local SEAM baseline on AV2.

Source	MR	b-minFDE6	minADE1	minADE6	minFDE1	minFDE6
Reference	0.153000	1.848000	1.598000	0.663000	3.962000	1.249000
Baseline	0.159462	1.882245	1.600877	0.664314	3.964651	1.272190
Δ absolute	+0.006462	+0.034245	+0.002877	+0.001314	+0.002651	+0.023190

Table 4.1: Source distinguishes the selected SEAM reference from its local reproduction. Reference denotes the selected reference checkpoint; Baseline denotes the local model’s final epoch 79 evaluated on two GPUs; and Δ absolute is Baseline minus Reference. MR denotes miss rate; b-minFDE6 denotes Brier-adjusted minimum final displacement error over six modes; minADEK and minFDEK denote minimum average and final displacement error over K modes, respectively. Displacement errors are measured in metres, and lower values are better.

4.2.2 Agent-history addition and future-head replacement

Table 4.2 records the state-space integration study. All three final epoch-79 models use the same two-GPU consolidated evaluation procedure, so their six-metric vectors form a direct within-SEAM comparison. Figure 4.1 plots the separately retained checkpoint-selection minADE6 values. The two Mamba interventions remain architecturally distinct, as specified in Figure 3.2: one adds a residual history path, whereas the other replaces the localisation head.

Agent-history Mamba gives a mixed vector: MR, b-minFDE6 and minFDE6 decrease by 0.005322, 0.009376 and 0.010767, while minADE1, minADE6 and minFDE1 increase by 0.013335, 0.000538 and 0.011993. Selected minADE6 is also 0.001955 m (0.29%) higher than baseline, so endpoint gains do not extend to the primary criterion.

Table 4.2: SEAM state-space integration results. The selected-checkpoint six-mode average displacement error and the common consolidated six-metric evaluation are reported separately.

(a) Common six-metric evaluation						
Variant	MR	b-minFDE6	minADE1	minADE6	minFDE1	minFDE6
Baseline	0.159462	1.882245	1.600877	0.664314	3.964651	1.272190
History Mamba	0.154141	1.872869	1.614212	0.664852	3.976645	1.261423
Head Mamba	0.145149	1.864566	1.583175	0.648480	3.931687	1.239389

(b) Checkpoint-selection metric		
Variant	Best minADE6	Best epoch
Baseline	0.662859	77
History Mamba	0.664814	77
Head Mamba	0.648480	79

Table 4.2: Baseline denotes the local SEAM baseline; History Mamba denotes agent-history residual Mamba; and Head Mamba denotes future-head Mamba replacement. Panel (a) reports two-GPU validation of the final epoch-79 models; Best minADE6 and Best epoch in panel (b) identify checkpoint selection. MR denotes miss rate; b-minFDE6 denotes Brier-adjusted minimum final displacement error over six modes; minADEK and minFDEK denote minimum average and final displacement error over K modes, respectively. Displacement errors are measured in metres, and lower values are better.

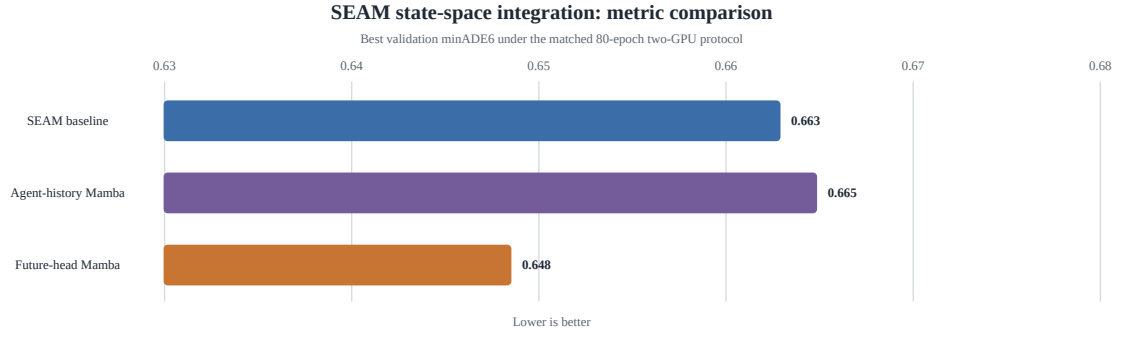


Figure 4.1: Best retained minADE6 for the SEAM baseline, agent-history Mamba addition and future-head Mamba replacement. All bars use the matched 80-epoch two-GPU protocol.

Future-head replacement alone improves every metric in this comparison. Selected minADE6 reaches 0.648480, below baseline by 0.014380 m (2.17%) and history addition by 0.016334 m (2.46%). In the common vector, MR falls by 0.014313 (8.98%) and minFDE6 by 0.032801 m (2.58%); b-minFDE6, minADE1, vector minADE6 and minFDE1 also improve. The benefit spans path accuracy and endpoint coverage.

4.2.3 Interpretation of the recurrent scan location

History addition's MR reduction of 0.005322 (3.34%) places at least one endpoint within the miss threshold in more scenarios; minFDE6 falls by 0.010767 m (0.85%). Higher single-mode errors nevertheless indicate no improvement to the top-ranked trajectory. Aggregate metrics do not distinguish changed trajectory shape from changed confidence ordering.

The +0.000538 m minADE6 change indicates little aggregate path change despite better endpoint coverage. Per-horizon ADE could test whether late-horizon mode separation offsets intermediate errors.

Future-head replacement uses two recurrent blocks over 80 mode-conditioned future positions before coordinate projection. The original MLP couples outputs; Mamba adds temporal ordering. The favourable vector supports this formulation, not decoder replacement universally.

4.2.4 Training-time comparison

Table 4.3 reports the complete SEAM timing comparison independently of Tables 4.1 and 4.2. Figure 4.2 plots the same durations in hours. Each duration spans model initialisation, 80 training epochs and the final explicit validation under the same two-GPU execution protocol.

History addition adds 4 h 27 min (6.3%) to baseline; future-head replacement adds 2 h 52 min (4.1%), taking 1 h 35 min less than history addition. Final training epochs take 51 min 48 s, 55 min 06 s and 54 min 39 s for baseline, history addition and replacement, with validation passes of

68 s, 72 s and 69 s. Both duration measures preserve this ordering; neither isolates selective-scan cost or inference latency.

Table 4.3: Wall-clock comparison for the SEAM state-space integration study. Every row uses two GPUs, microbatch 8 and gradient accumulation 2, giving global batch 32.

Variant	Wall time	min/epoch	Δ time (%)
Baseline	70 h 31 min	52.9	0.0%
History Mamba	74 h 58 min	56.2	+6.3%
Head Mamba	73 h 23 min	55.0	+4.1%

Table 4.3: Baseline denotes the local SEAM baseline; History Mamba denotes agent-history residual Mamba; and Head Mamba denotes future-head Mamba replacement. Wall time denotes total elapsed training time; min/epoch denotes the mean minutes per epoch; Δ time denotes the wall-time difference from the stated control; and Δ time (%) denotes the corresponding relative change. The stated control is Baseline; wall time is shown in hours and minutes.

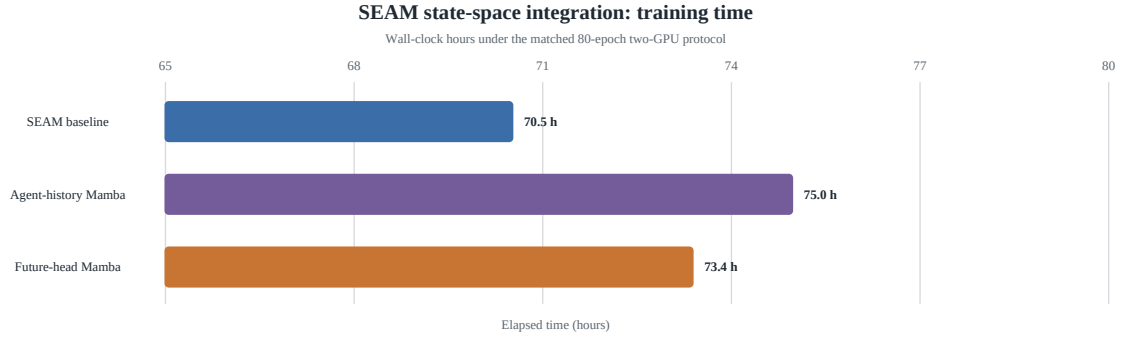


Figure 4.2: Wall-clock hours for the SEAM state-space integration study. All bars use the matched 80-epoch two-GPU protocol.

4.3 SEAM context and attention ablation study

4.3.1 Four-model results

The second SEAM family transfers three mechanisms to locations determined by SEAM’s own representations rather than copying their SHARP placements mechanically. The baseline retains fixed-radius endpoint retrieval and standard scaled dot-product attention. The uncertainty variant applies confidence-conditioned target-feature gain while its hard-selection radius remains at the initial boundary; the geometry variant biases only current-scene self-attention; and the QKNorm variant changes the score rule at every audited attention site. All four models use the common controls in Table 3.5.

All four models use the same 20-epoch schedule. Baseline, uncertainty and QKNorm select epoch 19, and their final validation vectors agree with their selected minADE6. Geometry selects epoch 18 at 0.717261; its complete final-epoch vector instead contains minADE6 0.718735. Table 4.4 separates checkpoint selection from final-epoch evaluation, while Figure 4.3 plots only the selection values. In particular, geometry’s epoch-18 score is not substituted into its epoch-19 vector.

Table 4.4: Checkpoint selection and final-epoch validation for the SEAM context and attention ablation study.

(a) Checkpoint-selection evidence						
Variant	Best minADE6			Best epoch		
Baseline	0.727668			19		
Uncertainty	0.708755			19		
Geometry	0.717261			18		
QKNorm	0.703845			19		

(b) Final-epoch validation vectors						
Variant	MR	b-minFDE6	minADE1	minADE6	minFDE1	minFDE6
Baseline	0.194958	2.025440	1.751487	0.727668	4.372325	1.418612
Uncertainty	0.187595	1.995327	1.723248	0.708755	4.325522	1.389651
Geometry	0.187955	1.997381	1.713196	0.718735	4.282954	1.394853
QKNorm	0.181393	1.992296	1.687949	0.703845	4.244209	1.375525

Table 4.4: Baseline denotes the controlled SEAM baseline; Uncertainty denotes uncertainty-aware target context; Geometry denotes relative-geometry attention bias; and QKNorm denotes Query–Key Normalisation at all SEAM attention sites. Epochs are zero-indexed. All full vectors use the same three-GPU distributed validation at final epoch 19. Geometry’s selected score is from epoch 18, not the vector in panel (b); the other three selected epochs are 19. MR denotes miss rate; b-minFDE6 denotes Brier-adjusted minimum final displacement error over six modes; minADEK and minFDEK denote minimum average and final displacement error over K modes, respectively. Displacement errors are measured in metres, and lower values are better.

These effect sizes use the common ablation control and distributed validation sampler, not the differently trained SEAM integration baseline.

4.3.2 Pre-specified metric contrasts

For metric m , the three controlled effects are defined only against the SEAM context and attention ablation baseline B :

$$\Delta_U m = m_U - m_B, \quad \Delta_G m = m_G - m_B, \quad \Delta_Q m = m_Q - m_B, \quad (4.2)$$

where U , G and Q denote uncertainty, geometry and QKNorm. Negative contrasts are favourable. These independent one-factor tests include neither mechanism interactions nor Mamba.

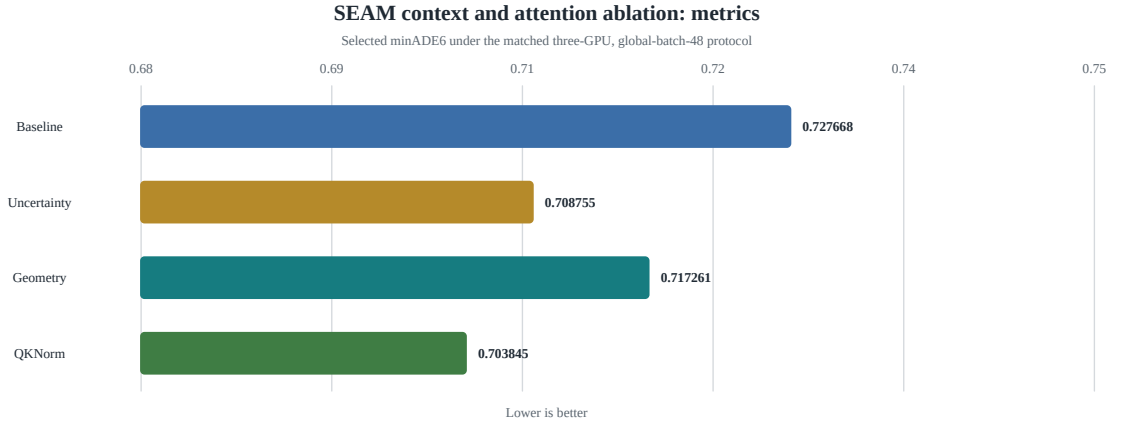


Figure 4.3: Selected minADE6 for the SEAM context and attention ablation study. Relative to the common baseline, uncertainty, geometry and QKNorm reduce error by 2.60%, 1.43% and 3.27%, respectively. Geometry selects epoch 18; the other models select epoch 19.

Uncertainty-aware context lowers minADE6 from 0.727668 to 0.708755, a reduction of 0.018913 m (2.60%). MR falls from 0.194958 to 0.187595 by 0.007363: 0.7363 percentage points or 3.78% relative. Together, these decreases indicate better six-mode paths and fewer scenes beyond the endpoint miss threshold.

Uncertainty also reduces b-minFDE6 by 0.030113 (1.49%), minFDE6 by 0.028961 m (2.04%), minADE1 by 0.028239 m (1.61%) and minFDE1 by 0.046803 m (1.07%). The benefit extends to the top-ranked path and confidence-sensitive endpoint error, without establishing calibration.

Relative geometry lowers selected minADE6 from 0.727668 to 0.717261, an absolute reduction of 0.010407 m (1.43%). Its final-epoch minADE6 is 0.718735, still 0.008933 m (1.23%) below baseline despite a 0.001474 m increase from the selected checkpoint. Final-epoch MR falls from 0.194958 to 0.187955, a reduction of 0.007003, or 0.7003 percentage points (3.59%). minFDE6 decreases by 0.023759 m (1.67%) to 1.394853, and b-minFDE6 decreases by 0.028059 (1.39%) to 1.997381. The single-mode reductions are 0.038292 m (2.19%) for minADE1 and 0.089372 m (2.04%) for minFDE1. Geometry therefore improves both the top-ranked prediction and six-mode coverage, although its smaller oracle displacement gain does not exceed uncertainty's.

QKNorm gives the lowest value in every final-epoch metric column. minADE6 decreases from 0.727668 to 0.703845, an absolute reduction of 0.023823 m (3.27%); the selected and final values coincide at epoch 19. MR decreases from 0.194958 to 0.181393, a reduction of 0.013565, or 1.3565 percentage points (6.96%). minFDE6 falls by 0.043087 m (3.04%) to 1.375525, while

b-minFDE6 falls by 0.033144 (1.64%) to 1.992296. The highest-confidence trajectory improves as well: minADE1 decreases by 0.063538 m (3.63%) to 1.687949, and minFDE1 decreases by 0.128117 m (2.93%) to 4.244209. The agreement between oracle, top-ranked and miss measures is stronger evidence of a broadly useful representation change than an isolated improvement in one selected metric, although repeated seeds are needed to assess its stability.

All three interventions improve the final-epoch vector, but their ordering depends on the metric. Geometry gives lower single-mode errors than uncertainty; uncertainty gives lower oracle path and endpoint errors. Preferred-trajectory quality and six-mode coverage therefore distinguish two mechanisms that both improve baseline.

4.3.3 Mechanism-specific interpretation

The uncertainty controller acts directly on SEAM’s endpoint-conditioned stream. A previous mode with diffuse probability mass or high entropy provides weaker evidence about the spatial support of its next target set. Equation 3.3 permits a bounded adjustment of retrieval radius and target-feature gain while starting at the baseline values. The measured reductions in minADE6, MR and endpoint errors support the implemented confidence-conditioned context modulation. The differentiable gain controller regulates the contribution of endpoint evidence to the decoder; the Boolean radius mask does not provide a data gradient for learning a wider neighbourhood. This interpretation follows SEAM’s endpoint-aware design [9]; predictive uncertainty provides a broader motivation for adapting how uncertain evidence is used [22]. The controller contains both radius and gain outputs, but the zero-initialised radius branch retains its baseline boundary under the recorded hard-selection implementation. A test of genuinely learned retrieval extent would require a differentiable selection mechanism; the present result should not be interpreted as that test.

The geometry variant addresses SEAM’s current-scene interaction rather than its endpoint retrieval. Content projections can encode relative position indirectly, but they must learn the transformation from separate token features. Equation 3.4 supplies displacement, distance and heading compatibility directly to each of the four scene-attention blocks. The restriction to scene blocks is central: agent and lane centres have a common physical interpretation there, whereas temporal samples and abstract mode queries do not. The observed ADE and FDE reductions support explicit physical relations as a useful complement to learned token content, consistent with relation-aware attention [11]. Lower single-mode errors than uncertainty suggest more useful interaction evidence for the preferred trajectory, but do not establish which road situations benefit. Curved roads,

junctions and closely interacting agents are plausible diagnostic subsets, not measured subgroup findings. The epoch-18 minimum followed by a slightly higher final score also shows why check-point selection and final-epoch reporting must remain distinct.

QKNorm tests a network-wide score-normalisation hypothesis. SEAM’s 20 attention calls span ordered histories, heterogeneous scene entities, retained context, trajectory state, endpoint neighbourhoods and decoder cross-attention. Their query and key norms can therefore encode different incidental scales. Per-head L2 normalisation removes magnitude from similarity and delegates concentration to a learned positive temperature [13]. The reduction across all six metrics is consistent with useful score-scale control in this architecture: attention concentration need not be determined by feature norm alone, and several hypotheses can retain access to informative context. However, neither the learned temperatures nor attention-entropy trajectories were analysed, so reduced softmax saturation is a mechanistic explanation to test rather than a directly observed cause. Replacing all 20 kernels also cannot identify whether temporal, scene, target or decoder attention supplies most of the effect; a site-wise ablation would answer that narrower question.

4.3.4 Training-time comparison

Table 4.5 and Figure 4.4 compare training durations separately from forecasting accuracy. Baseline, uncertainty, geometry and QKNorm require 15 h 36 min 46 s, 15 h 40 min 25 s, 15 h 45 min 02 s and 17 h 55 min 47 s, respectively. Their mean durations are 46.84, 47.02, 47.25 and 53.79 minutes per epoch.

Table 4.5: Training-time comparison for the SEAM context and attention ablation study. All rows use three GPUs, microbatch 8 and two-step accumulation, giving global batch 48.

Variant	Time (h)	min/epoch	Δ time (%)
Baseline	15.613	46.84	0.00%
Uncertainty	15.674	47.02	+0.39%
Geometry	15.751	47.25	+0.88%
QKNorm	17.930	53.79	+14.84%

Table 4.5: Baseline denotes the controlled SEAM baseline; Uncertainty denotes uncertainty-aware target context; Geometry denotes relative-geometry attention bias; and QKNorm denotes Query–Key Normalisation at all SEAM attention sites. Time is the retained training duration in decimal hours; min/epoch is the mean minutes per epoch; and Δ time is the relative change from Baseline.

Uncertainty adds 219 seconds (0.39%) and geometry adds 496 seconds (0.88%) to the matched baseline duration. These differences are small relative to the full training interval; repeated executions would be needed to establish their stability as system costs.

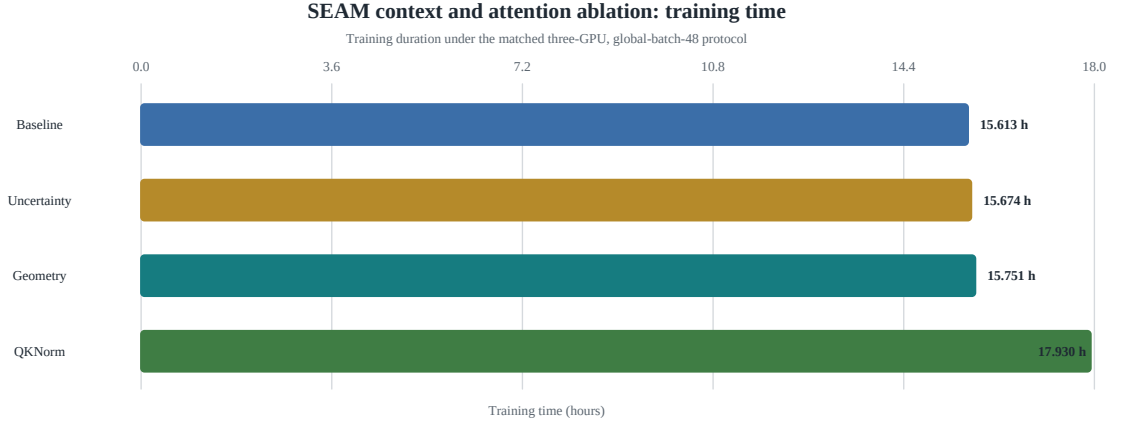


Figure 4.4: Training durations for the SEAM context and attention ablation study. Relative to baseline, uncertainty adds 3 min 39 s (0.39%), geometry adds 8 min 16 s (0.88%) and QKNorm adds 2 h 19 min 01 s (14.84%).

QKNorm adds 8,341 seconds (14.84%), taking 2 h 15 min 22 s longer than uncertainty. Repeated norm reductions and rescaling are plausible contributors, but the end-to-end record does not isolate their cost or measure inference latency.

4.4 SHARP reference reproduction

4.4.1 Argoverse 2 validation

Table 4.6 compares the SHARP article’s AV2 validation vector with the official local baseline. The local model retains the original architecture and follows the article-aligned protocol of 80 epochs, global batch 32, 13 warm-up epochs and checkpoint selection by validation minADE6 [10].

Table 4.6: SHARP article AV2 validation result and official SHARP reproduction.

Source	MR	b-minFDE6	minADE1	minADE6	minFDE1	minFDE6
Article	0.140000	1.822000	1.569000	0.639000	3.850000	1.197000
Baseline	0.155955	1.924560	1.676867	0.679282	4.138639	1.287697
Δ absolute	+0.015955	+0.102560	+0.107867	+0.040282	+0.288639	+0.090697
Δ relative	+11.40%	+5.63%	+6.87%	+6.30%	+7.50%	+7.58%

Table 4.6: Source identifies the reported or reproduced evaluation. Article denotes the SHARP article’s AV2 result; Baseline denotes the official SHARP reproduction selected at epoch 78; Δ absolute is Baseline minus Article; and Δ relative expresses that difference as a percentage of Article. MR denotes miss rate; b-minFDE6 denotes Brier-adjusted minimum final displacement error over six modes; minADEK and minFDEK denote minimum average and final displacement error over K modes, respectively. Displacement errors are measured in metres, and lower values are better.

Local minADE6 exceeds the article value by 0.040282 m (6.30%). MR increases by 11.40%, and displacement and confidence-sensitive measures by 5.63–7.58%. These differently sensitive statistics do not identify a constant coordinate offset. Aligned epochs, global batch and schedule rule out deliberate budget shortening; sample order, preprocessing and numerical execution remain unisolated possibilities. The measured baseline supplies the control for architectural contrasts.

The SHARP article also reports AV2 test-set MR 0.14, b-minFDE6 1.83, minADE6 0.64 and minFDE6 1.19 [10]. Those values are not subtracted from the local validation row because validation and hidden-test measurements are different evidence sources. All subsequent references to the SHARP architecture integration baseline use the local row in Table 4.6.

4.5 SHARP architectural ablation study

4.5.1 Matched architectural results

The ten-model suite compares nine isolated interventions with one SHARP control. Every row uses AV2, seed 2333, 20 epochs, global batch 32, the same 13-epoch warm-up and the same checkpoint rule. Table 4.7 reports the retained metrics, and Figure 4.5 plots checkpoint-selection minADE6. The delta column is the selected value for the variant minus that of the baseline. The historical summary obtains the other metrics from saved variant summaries or matching validation records; it does not establish an independent six-metric reevaluation of every selected checkpoint.

4.5.2 Geometry and confidence-conditioned context

Relative geometry leads the screen at minADE6 0.749679, improving baseline by 0.002660 m (0.35%). It also reduces MR by 0.001800 (0.93%) and minFDE6 by 0.002356 m (0.16%); b-minFDE6, minADE1 and minFDE1 improve too. These small gains extend across the vector rather than only the selected criterion.

The result is consistent with the role of relative geometry in scene reasoning. Standard attention must infer displacement, range and heading relationships indirectly from token embeddings. The added bias presents those pairwise physical relations directly to the attention score, analogously to relation-aware attention in other domains [11]. Its effect is credible because the intervention changes how existing tokens interact rather than adding a parallel predictor or changing the streaming state.

Uncertainty ranks second on minADE6 at 0.749961, improving by 0.002378 m (0.32%). minADE1 falls by 0.023206 m (1.23%), but MR rises by 0.005923 (3.07%); b-minFDE6 and both endpoint errors also worsen. Better average paths therefore coexist with poorer endpoint coverage. Its selection score and distinct retrieval function motivate composition with geometry’s attention-score modification.

These sub-centimetre gains justify candidate selection, not conclusive superiority or proof of the proposed mechanisms.

Table 4.7: SHARP architectural ablation results on AV2.

<i>(a) Checkpoint selection, coverage, confidence and average displacement</i>						
Variant	Epoch	MR	b-minFDE6	minADE1	minADE6	Δ minADE6
Baseline	19	0.192652	2.064007	1.886442	0.752339	+0.000000
Memory	19	0.194413	2.067328	1.876097	0.755349	+0.003010
Consistency	18	0.202497	2.096022	1.872110	0.776207	+0.023868
Pooling	19	0.201897	2.080306	1.891240	0.757637	+0.005298
Uncertainty	19	0.198575	2.074243	1.863236	0.749961	−0.002378
Geometry	19	0.190852	2.051891	1.875754	0.749679	−0.002660
Kinematics	19	0.198175	2.068197	1.942671	0.785164	+0.032825
Endpoint	19	0.196254	2.058200	1.867779	0.755898	+0.003559
Lane graph	19	0.200616	2.065051	1.903412	0.754238	+0.001899
Mamba	19	0.212822	2.141835	1.993448	0.780190	+0.027851
<i>(b) Final displacement</i>						
Variant			minFDE1	minFDE6		
Baseline			4.672195	1.434819		
Memory			4.694828	1.443534		
Consistency			4.785388	1.476766		
Pooling			4.739688	1.463850		
Uncertainty			4.676969	1.447852		
Geometry			4.660576	1.432463		
Kinematics			4.701560	1.448078		
Endpoint			4.641530	1.441044		
Lane graph			4.696756	1.443290		
Mamba			4.973433	1.514158		

Table 4.7: Baseline denotes baseline SHARP; Memory denotes confidence-gated memory; Consistency denotes cross-window consistency; Pooling denotes learned temporal pooling; Uncertainty denotes uncertainty-aware target context; Geometry denotes relative-geometry attention bias; Kinematics denotes the kinematic motion stem; Endpoint denotes endpoint refinement; Lane graph denotes the lane-topology graph; and Mamba denotes temporal Mamba replacement. Best epoch identifies the checkpoint selected by validation minADE6; Δ minADE6 is the selected value minus the architectural ablation baseline value. MR denotes miss rate; b-minFDE6 denotes Brier-adjusted minimum final displacement error over six modes; minADEK and minFDEK denote minimum average and final displacement error over K modes, respectively. Displacement errors are measured in metres, and lower values are better.

4.5.3 Temporal and decoder interventions

The kinematic stem gives the highest minADE6 (0.785164) and second-highest minADE1 (1.942671). Differencing processed motion features may amplify noise or duplicate temporal information, not recover physical velocity and acceleration. These explanations remain untested.

Cross-window consistency also degrades strongly. Its minADE6 is 0.776207 and minFDE6 is 1.476766. Agreement is not automatically correctness: during early training, a consistency penalty can preserve an inaccurate previous prediction and oppose correction by new observations. This risk is especially relevant because 13 of the 20 epochs are warm-up.

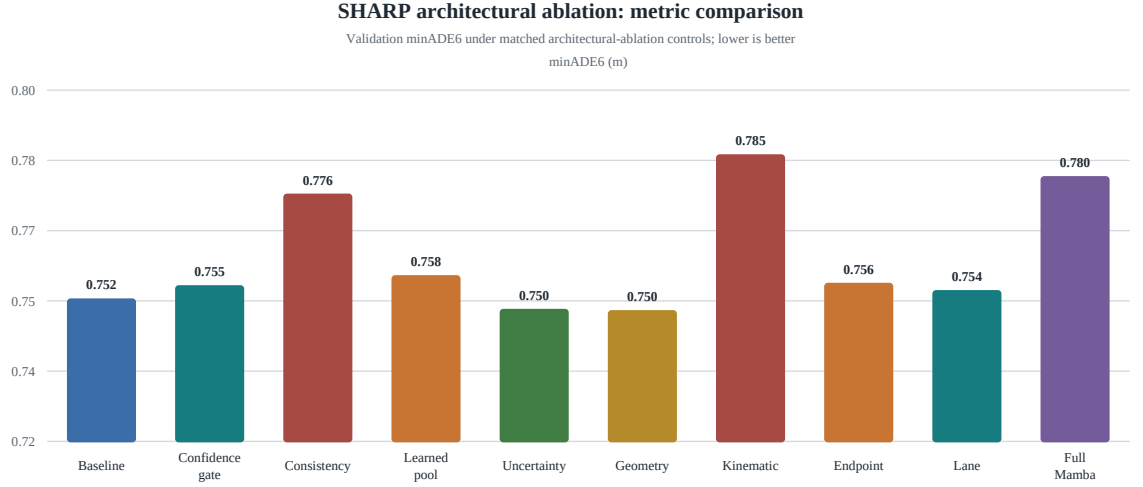


Figure 4.5: Best minADE6 in the SHARP architectural ablation study. Lower is better, and all bars use the same shortened training budget.

Learned pooling, lane topology and memory gating remain close to baseline. Endpoint refinement increases minADE6 by 0.003559 but lowers minFDE1 from 4.672195 to 4.641530. Its correction may improve the preferred endpoint without improving six-mode path quality or diversity; minFDE1 alone would therefore suggest a different conclusion.

Replacing all four temporal-attention blocks raises minADE6 from 0.752339 to 0.780190, an increase of 0.027851 m (3.70%). MR rises by 0.020170 (10.47%) and minFDE6 by 0.079339 m (5.53%). This configuration has the second-highest minADE6 and the highest MR, minADE1, minFDE1 and minFDE6 in the suite. The conclusion is specific to replacing the short-window attention encoder under these controls, not to selective state-space models generally.

With 65% of epochs devoted to warm-up and validation-based selection introducing optimism, the screen identifies early-adaptation candidates rather than converged rankings. Geometry and uncertainty advance on that basis.

4.5.4 Result-space structure across the ten variants

Among information-selection changes, geometry improves the full vector, uncertainty both ADE values and memory gating only minADE1. Explicit physical relations give the most consistent benefit.

Pooling, endpoint refinement and lane topology remain within 0.0053 m of baseline minADE6. Refinement benefits the preferred endpoint, not the primary path criterion. Lane messaging may duplicate globally accessible relations, an untested explanation.

Consistency, kinematics and temporal replacement change the temporal learning problem more substantially and give the largest primary-metric increases. Their outcomes motivate closer attention to how new temporal constraints interact with an existing short-window encoder, rather than assuming that additional structure is beneficial.

Geometry (0.749679) and the kinematic stem (0.785164) span 0.035485 m in minADE6, or 4.72% of baseline. The screen distinguishes promising changes from candidates requiring refinement; geometry and uncertainty advance on selection scores and complementary roles, not capacity alone.

4.5.5 Training-time comparison

Table 4.8 separates computational cost from the metric comparison in Table 4.7. It reports wall time, average minutes per epoch and absolute and relative differences from the matched baseline; Figure 4.6 plots the same complete durations in hours.

Table 4.8: Retained wall-clock comparison for the SHARP architectural ablation study. All configurations use 20 epochs, global batch 32 and the same four-GPU execution platform.

Variant	Wall time	min/epoch	Δ time	Δ time (%)
Baseline	35 h 13 min	105.7	0 h 00 min	0.0%
Memory	33 h 10 min	99.5	−2 h 04 min	−5.9%
Consistency	28 h 47 min	86.4	−6 h 26 min	−18.3%
Pooling	34 h 32 min	103.6	−0 h 41 min	−1.9%
Uncertainty	30 h 29 min	91.5	−4 h 44 min	−13.4%
Geometry	38 h 26 min	115.3	+3 h 12 min	+9.1%
Kinematics	37 h 33 min	112.7	+2 h 20 min	+6.6%
Endpoint	37 h 53 min	113.6	+2 h 39 min	+7.5%
Lane graph	37 h 45 min	113.3	+2 h 32 min	+7.2%
Mamba	46 h 09 min	138.4	+10 h 55 min	+31.0%

Table 4.8: Baseline denotes baseline SHARP; Memory denotes confidence-gated memory; Consistency denotes cross-window consistency; Pooling denotes learned temporal pooling; Uncertainty denotes uncertainty-aware target context; Geometry denotes relative-geometry attention bias; Kinematics denotes the kinematic motion stem; Endpoint denotes endpoint refinement; Lane graph denotes the lane-topology graph; and Mamba denotes temporal Mamba replacement. Wall time denotes total elapsed training time; min/epoch denotes the mean minutes per epoch; Δ time denotes the wall-time difference from the stated control; and Δ time (%) denotes the corresponding relative change. The stated control is Baseline; wall-time and Δ time values are shown in hours and minutes.

Relative geometry takes 38 h 26 min, 9.1% longer than the 35 h 13 min baseline, while uncertainty takes 30 h 29 min, 13.4% less. Consistency has the shortest retained duration at 28 h 47 min; temporal-Mamba replacement has the longest at 46 h 09 min, 31.0% above baseline. These end-to-end records characterise the tested execution paths without attributing the duration of an entire experiment to one added operator.

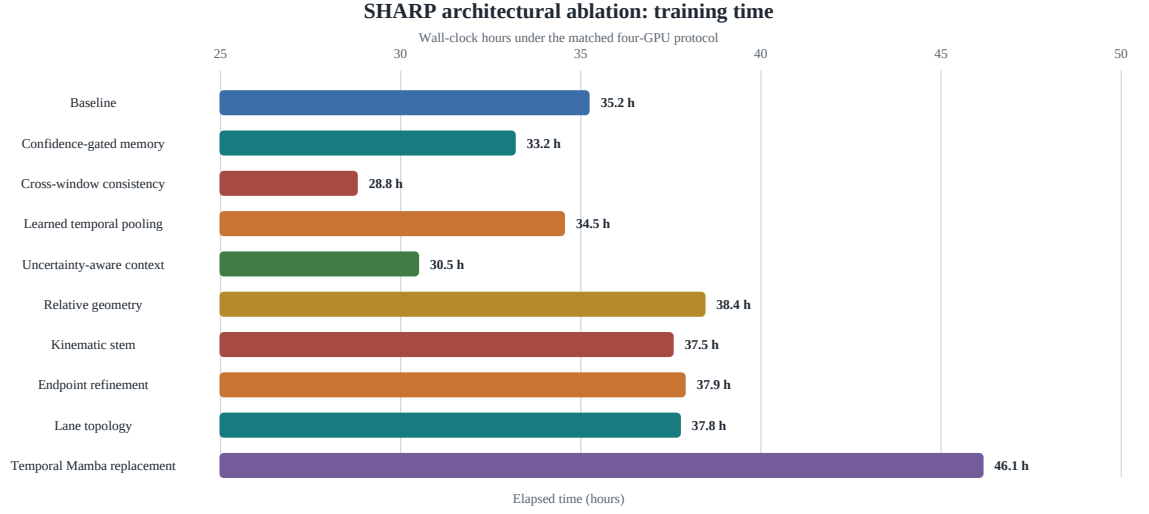


Figure 4.6: Wall-clock hours for the SHARP architectural ablation study. Every bar uses the matched four-GPU, 20-epoch protocol.

4.6 SHARP attention-operator study

4.6.1 Controlled attention results

The attention-operator suite isolates the attention mechanism under an 80-epoch, global-batch-24 protocol. The same operator is substituted at all 20 configured attention sites identified in Chapter 3. Table 4.9 separates best-checkpoint minADE6 from the latest complete vector, while Figure 4.7 plots only the checkpoint-selection values so that the baseline’s epoch-66 selection is not confused with its epoch-79 metrics.

Table 4.9: SHARP attention-operator evidence on AV2. The selected-checkpoint and common-evaluation six-mode average displacement errors may come from different epochs.

(a) Checkpoint selection and common-evaluation average displacement				
Attention	Best epoch	Best minADE6	Eval. epoch	Eval. minADE6
MHA	66	0.673460	79	0.686853
QKNorm	65	0.669777	65	0.669777
TH	71	0.674991	71	0.674991
QK+TH	71	0.674492	71	0.674492

(b) Remaining common-evaluation metrics					
Attention	MR	b-minFDE6	minADE1	minFDE1	minFDE6
MHA	0.153231	1.927635	1.671288	4.063601	1.287932
QKNorm	0.154471	1.899642	1.671904	4.119903	1.266240
TH	0.156510	1.927414	1.675181	4.130968	1.285642
QK+TH	0.152064	1.931623	1.663308	4.090390	1.291059

Table 4.9: MHA denotes standard multi-head attention; TH denotes Talking-Heads; and QK+TH denotes QKNorm combined with Talking-Heads. Best epoch and Best minADE6 identify checkpoint selection. Eval. epoch identifies the model supplying the retained evaluation: final epoch 79 for MHA and the selected checkpoints for QKNorm, TH and QK+TH. The MHA vector is therefore not its best-checkpoint vector. MR denotes miss rate; b-minFDE6 denotes Brier-adjusted minimum final displacement error over six modes; minADEK and minFDEK denote minimum average and final displacement error over K modes, respectively. Displacement errors are measured in metres, and lower values are better.

QKNorm improves best minADE6 from 0.673460 to 0.669777. The absolute reduction is 0.003683 m and the relative reduction is 0.55%. Query/key normalisation removes magnitude variation before the dot product, so the learned temperature controls logit scale more directly [13]. This is a plausible benefit in a streaming network that repeatedly combines heterogeneous agent, lane and target tokens.

Talking-Heads attention is 0.001531 m worse than MHA on best minADE6. Mixing logits and attention weights across heads can exchange information that independent heads cannot share directly [14], but that additional flexibility does not improve the selected metric here. The combined QKNorm–Talking-Heads operator is also 0.001032 m worse than MHA on best minADE6. It nevertheless produces the lowest retained MR, 0.152064, and lowest minADE1, 1.663308. This metric trade-off suggests that it may distribute hypotheses to avoid complete misses while slightly increasing the oracle average path error.

The retained QKNorm vector also lowers b-minFDE6 by 0.027993 (1.45%) and minFDE6 by 0.021692 m (1.68%) relative to the retained MHA vector. Conversely, MR is 0.001240 (0.81%) higher and minFDE1 is 0.056302 m (1.39%) higher. QKNorm’s explicit evaluation uses its selected epoch-65 checkpoint, whereas the MHA vector is from epoch 79 rather than its selected epoch 66. These secondary contrasts therefore describe the retained evaluations, not a matched comparison of both best-checkpoint vectors. The common selection criterion favours QKNorm by 0.55% in this run.

The pre-specified minADE6 criterion therefore selects QKNorm without Talking-Heads for subsequent composition with the two context candidates.

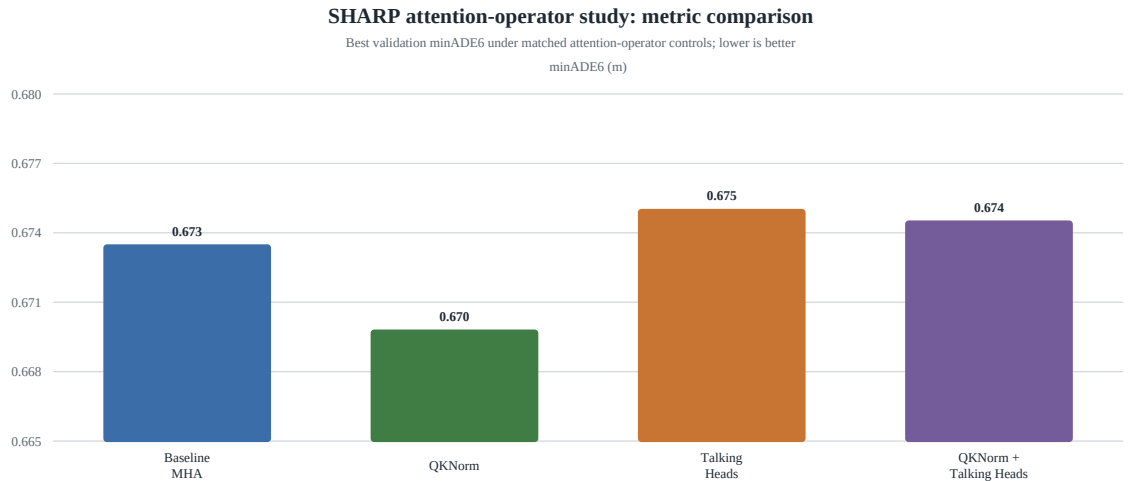


Figure 4.7: Best checkpoint minADE6 for the attention operators. QKNorm is the only operator to improve the matched MHA control on the selection metric.

4.6.2 Score normalisation and cross-head communication

QKNorm constrains similarity geometry and learns concentration; Talking-Heads learns communication between head logits and weights. Their comparison distinguishes score normalisation from additional expressiveness.

Normalisation may help by removing incidental norm differences across SHARP’s attention inputs; head mixing supplies neither geometry nor chronology. Attention-norm and entropy diagnostics would test this explanation.

The combination adds pre- and post-softmax mixing to QKNorm. Lower retained MR and minADE1 coexist with selected minADE6 above QKNorm and MHA, distinguishing endpoint coverage, preferred-trajectory quality and oracle path accuracy rather than establishing calibration. Part of training uses microbatch 4 with two-step accumulation instead of microbatch 8. Both give global batch 24 but different normalisation populations, so the result characterises the combined operator and execution path, not head mixing alone.

4.6.3 Training-time comparison

The four attention operators also have distinct end-to-end costs, reported separately from their forecasting metrics in Table 4.10. The table presents wall time, average minutes per epoch and the relative difference from standard MHA; Figure 4.8 provides the matched wall-clock comparison.

Table 4.10: Retained wall-clock comparison for the SHARP attention-operator study. All rows use three GPUs and global batch 24.

Attention	Wall time	min/epoch	Δ time (%)
MHA	99 h 39 min	74.7	0.0%
QKNorm	118 h 41 min	89.0	+19.1%
TH	118 h 07 min	88.6	+18.5%
QK+TH	199 h 54 min	149.9	+100.6%

Table 4.10: MHA denotes standard multi-head attention; TH denotes Talking-Heads; and QK+TH denotes QKNorm combined with Talking-Heads. Wall time denotes total elapsed training time; min/epoch denotes the mean minutes per epoch; Δ time denotes the wall-time difference from the stated control; and Δ time (%) denotes the corresponding relative change. The stated control is MHA; wall time is shown in hours and minutes.

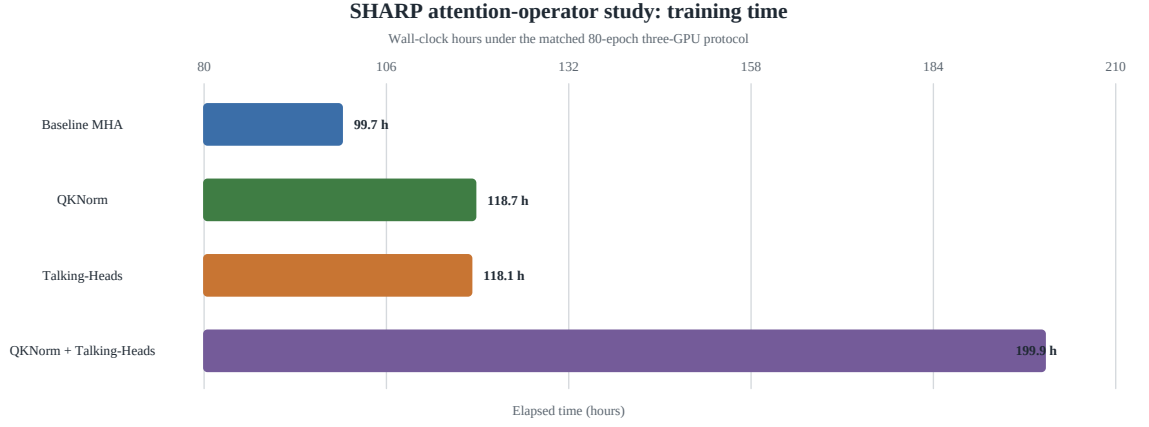


Figure 4.8: Wall-clock hours for the SHARP attention-operator study. Every bar uses the 80-epoch schedule under the matched three-GPU protocol.

QKNorm and Talking-Heads take 118 h 41 min and 118 h 07 min versus MHA’s 99 h 39 min, increases of 19.1% and 18.5%. Their combination takes 199 h 54 min (100.6% more), including its smaller-microbatch accumulation path. These are end-to-end durations, not attention-kernel latencies.

4.7 SHARP architecture integration study

4.7.1 Residual-Mamba placement

The SHARP state-space placement experiments test whether selective state-space recurrence is useful as an auxiliary pathway and whether sequence semantics determine its effectiveness. Table 4.11 separates each retained metric vector from its best checkpoint value, while Figure 4.9 visualises the checkpoint-selection metric. The official SHARP baseline and both Mamba additions use 80 epochs, global batch 32 and four GPUs, making the official baseline the sole SHARP state-space placement comparator.

Temporal Mamba scans history; scene Mamba scans an implementation-defined list. Exchanging lane tokens alters the recurrent path, not the physical scene. Attention can remain permutation equivariant when geometry and masks follow the exchange. This distinction fits the result, but ordering and capacity effects were not isolated.

Table 4.11: SHARP state-space placement comparison on AV2 validation.

(a) Common six-metric evaluation							
Variant	Eval. ckpt	MR	b-minFDE6	minADE1	minADE6	minFDE1	minFDE6
Baseline	78/80	0.155955	1.924560	1.676867	0.679282	4.138639	1.287697
Scene Mamba	78/80	0.157353	1.926419	1.680364	0.680362	4.130003	1.290422
Agent Mamba	79/80	0.157000	1.920000	1.640000	0.676000	4.050000	1.290000

(b) Checkpoint-selection metric	
Configuration	Best minADE6
Baseline	0.679282
Scene Mamba	0.680362
Agent Mamba	0.673736

Table 4.11: Baseline denotes the official SHARP baseline; Scene Mamba denotes scene-token residual Mamba; and Agent Mamba denotes temporal-agent residual Mamba. Eval. ckpt identifies the checkpoint used for the common six-metric evaluation: 78/80 denotes the selected epoch 78 checkpoint, whereas 79/80 denotes the final epoch 79 checkpoint. Best minADE6 is the checkpoint-selection value. MR denotes miss rate; b-minFDE6 denotes Brier-adjusted minimum final displacement error over six modes; minADEK and minFDEK denote minimum average and final displacement error over K modes, respectively. Displacement errors are measured in metres, and lower values are better.

Residual temporal-agent Mamba obtains best minADE6 0.673736, which is 0.006626 m lower than scene-token Mamba. Relative to the scene-token value, this is a 0.97% improvement. The result supports the placement hypothesis: an individual agent’s observation history is chronological, whereas the order of pooled scene tokens is an implementation detail. A selective state-space scan can meaningfully retain or suppress information along time, but it has no equivalent physical interpretation when it traverses an arbitrary list of agents and lane segments [15].

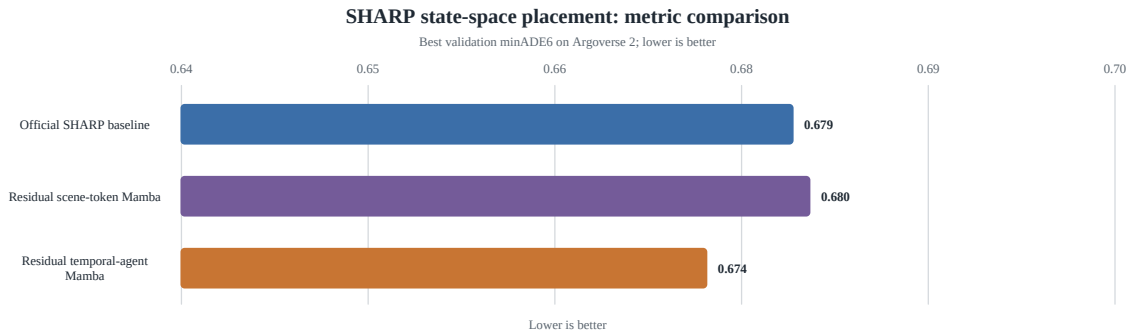


Figure 4.9: Best retained minADE6 for the official SHARP baseline and two residual-Mamba additions. All three use the 80-epoch, global-batch-32 protocol.

Relative to the official baseline, temporal-agent Mamba reduces best minADE6 by 0.005546 m, or 0.82%. In its retained vector, minADE1 falls from 1.676867 m to 1.640000 m, an absolute reduction of 0.036867 m or 2.20%, whereas MR rises by 0.001045, or 0.67%. The remaining metrics are also mixed: b-minFDE6, vector minADE6 and minFDE1 improve, while minFDE6 increases slightly. Temporal-agent Mamba therefore improves the principal selection metric with-

out dominating the baseline across every measure. Scene-token Mamba is 0.001080 m, or 0.16%, worse than the official baseline on best minADE6; its only retained improvement is a 0.008636 m, or 0.21%, reduction in minFDE1. The contrast strengthens the placement conclusion while requiring metric-specific interpretation.

4.7.2 Sequence semantics and residual refinement

The residual pathway between attention blocks 2 and 3 complements direct comparison over ten observations. Valid-prefix compaction excludes padding from both scan directions; LayerScale 0.01 limits the initial correction while the gate and state-space parameters learn. This preserves temporal attention without preserving its exact initial mapping. Initialisation was not isolated as the cause of improvement.

4.7.3 Training-time comparison

Table 4.12 compares wall time and average minutes per epoch within the SHARP state-space placement comparison, and Figure 4.10 plots the corresponding complete wall-clock durations.

Table 4.12: Wall-clock comparison for SHARP state-space placement. All rows use 80 epochs, global batch 32 and the same four-GPU platform.

Configuration	Wall time	min/epoch	Δ time (%)
Baseline	125 h 42 min	94.3	0.0%
Scene Mamba	129 h 17 min	97.0	+2.8%
Agent Mamba	160 h 20 min	120.3	+27.5%

Table 4.12: Baseline denotes the official SHARP baseline; Scene Mamba denotes scene-token residual Mamba; and Agent Mamba denotes temporal-agent residual Mamba. Wall time denotes total elapsed training time; min/epoch denotes the mean minutes per epoch; Δ time denotes the wall-time difference from the stated control; and Δ time (%) denotes the corresponding relative change. Here, the stated control is Baseline; wall time is shown in hours and minutes.

Both additions retain attention and add selective scans. Their end-to-end durations do not measure operator latency, and linear scan complexity does not imply a speed-up on ten-step histories.

Scene-token Mamba requires approximately 3 h 35 min more than the baseline, a 2.8% increase. Temporal-agent Mamba requires approximately 34 h 38 min more than baseline (27.5%) and 31 h 03 min more than scene-token Mamba. The values retain the common training budget and execution platform.

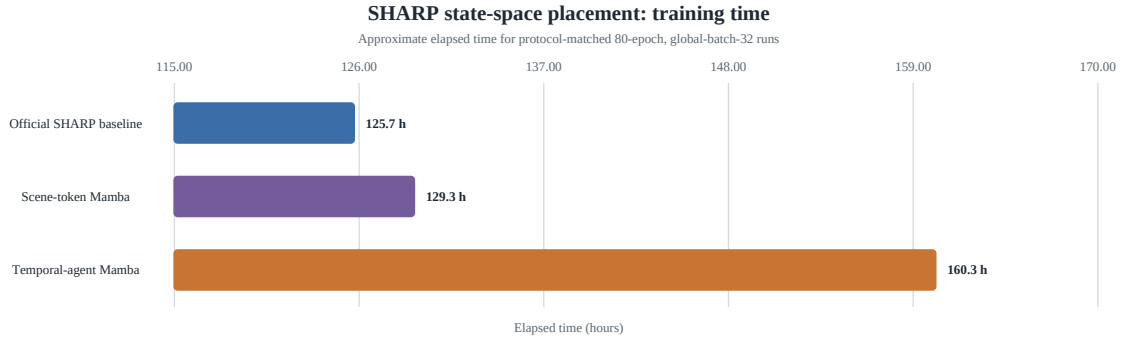


Figure 4.10: Approximate wall-clock hours for SHARP state-space placement. All configurations use 80 epochs, global batch 32 and the same four-GPU platform.

4.7.4 Mechanism composition

The final comparison adds the selected QKNorm, uncertainty and geometry mechanisms to article-aligned SHARP. Both models share AV2 preprocessing, 80 epochs, global batch 32, four GPUs, seed and checkpoint rule.

The baseline selects epoch 78 and the combined model selects epoch 79, with epochs numbered from zero. Each selected checkpoint is evaluated independently on one GPU with batch 32 over the same AV2 validation split. Table 4.13 reports these full-vector evaluations. The minADE6 encoded in each checkpoint filename agrees with its reevaluation at the six-decimal precision shown; no value from a different checkpoint is substituted into either vector.

The combined model does not improve the primary comparison metric. minADE6 increases from 0.679282 to 0.680589 m, an absolute increase of 0.001307 m, or 0.19%. minADE1 also increases, from 1.676867 to 1.677496 m, by 0.000628 m (0.04%). Figure 4.11 visualises the minADE6 contrast. These millimetre-scale differences favour the unmodified baseline in this evaluation; they do not establish a general ranking across random seeds.

Table 4.13: SHARP mechanism composition comparison under the common selected-checkpoint evaluation protocol.

Configuration	MR	b-minFDE6	minADE1	minADE6	minFDE1	minFDE6
Baseline	0.155955	1.924560	1.676867	0.679282	4.138639	1.287697
Q+U+G	0.155515	1.918870	1.677496	0.680589	4.130601	1.285462

Table 4.13: Configuration identifies the model evaluated at its selected checkpoint. Baseline denotes the official SHARP baseline at epoch 78; Q+U+G denotes QKNorm with uncertainty-aware target context and relative-geometry attention bias at epoch 79. Both vectors use one-GPU validation with batch 32. MR denotes miss rate; b-minFDE6 denotes Brier-adjusted minimum final displacement error over six modes; minADEK and minFDEK denote minimum average and final displacement error over K modes, respectively. Displacement errors are measured in metres, and lower values are better.

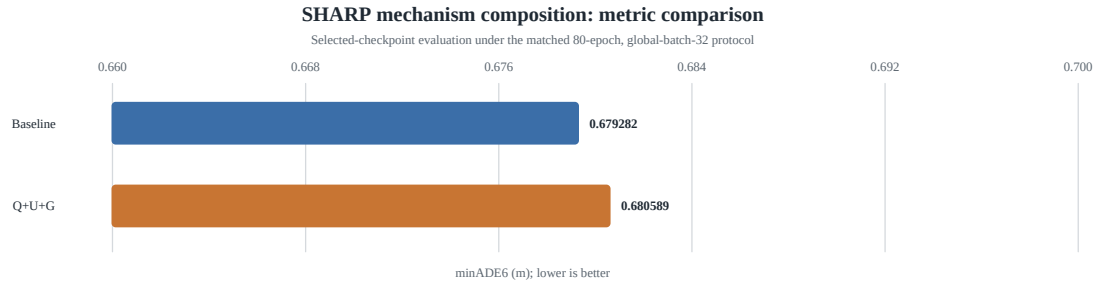


Figure 4.11: Selected-checkpoint minADE6 for the SHARP mechanism composition comparison. *QKNorm*, uncertainty-aware target context and relative-geometry bias together increase error by 0.001307 m (0.19%) relative to the matched baseline. The horizontal scale is restricted to distinguish the small difference.

Four secondary metrics move in the opposite direction. MR decreases from 0.155955 to 0.155515, a reduction of 0.000440 (0.28% relative, or 0.044 percentage points). Brier-adjusted minFDE6 decreases from 1.924560 to 1.918870 by 0.005690 (0.30%). minFDE1 falls from 4.138639 to 4.130601 m, a reduction of 0.008038 m (0.19%), and minFDE6 falls from 1.287697 to 1.285462 m, a reduction of 0.002235 m (0.17%). All differences and percentages are calculated from unrounded evaluation outputs. The outcome is therefore mixed, not a uniform improvement or deterioration. Figure 4.12 isolates the MR contrast.

Better endpoints can coexist with slightly larger intermediate errors because ADE averages the path and FDE measures its endpoint. The aggregates locate neither horizon-specific nor road-specific changes; lower b-minFDE6 also does not establish calibration. Per-horizon, scenario-level and calibration diagnostics would resolve these questions.

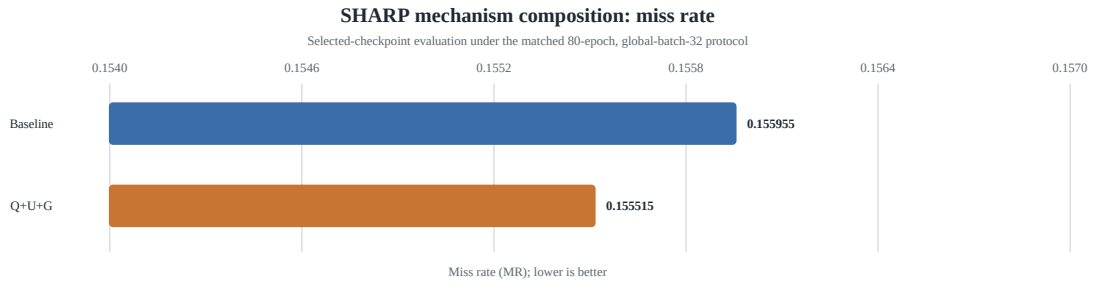


Figure 4.12: Selected-checkpoint miss rate for the SHARP mechanism composition comparison. *Q+U+G* denotes *QKNorm*, uncertainty-aware target context and relative-geometry bias. MR decreases from 0.155955 to 0.155515, a reduction of 0.000440 (0.28% relative, or 0.044 percentage points). The horizontal scale is restricted to distinguish the small difference; lower values are better.

Training time and parameter cost

Table 4.14 reports training time separately from the metric comparison. The baseline takes 125 h 42 min 14 s and Q+U+G takes 151 h 53 min 20 s. The additional duration is 26 h 11 min 06 s,

or 20.83%; the mean rises from 94.28 to 113.92 minutes per epoch. Figure 4.13 plots the same within-protocol timing contrast.

Table 4.14: Training-time comparison for SHARP mechanism composition. Both configurations use the same 80-epoch protocol.

Configuration	Time (h)	min/epoch	Δ time (%)
Baseline	125.704	94.28	0.00%
Q+U+G	151.889	113.92	+20.83%

Table 4.14: Baseline denotes official SHARP; Q+U+G denotes QKNorm, uncertainty-aware target context and relative-geometry attention bias. Time is training duration in decimal hours; min/epoch is the mean minutes per epoch; and Δ time is relative change from Baseline.

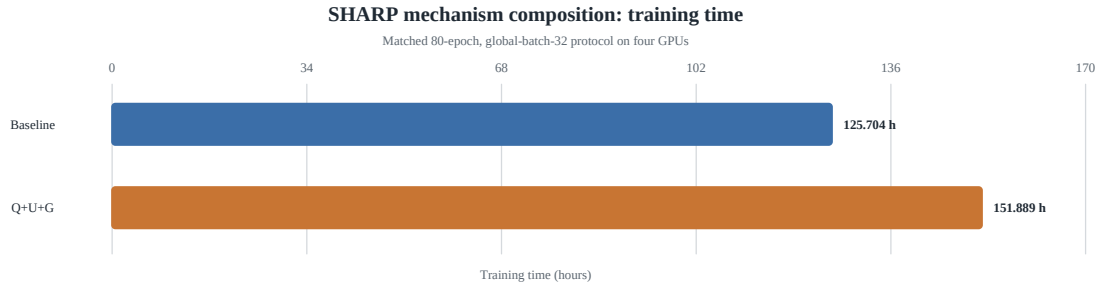


Figure 4.13: Training duration for the SHARP mechanism composition comparison. The combined mechanisms add 26 h 11 min 06 s (20.83%) relative to baseline under the same epoch, batch and hardware controls.

Q+U+G adds 621 parameters (0.0134%), from 4,635,611 to 4,636,232. The larger proportional duration change shows that repeated tensor operations, not parameter count alone, determine cost. The record measures the joint implementation without assigning a fraction to each mechanism or implying equal inference overhead.

Interpretation of the joint intervention

The scientific motivation was complementarity: QKNorm controls the scale of content similarity, geometry exposes physical relations and uncertainty changes the spatial support and strength of target context. These mechanisms nevertheless act on coupled representations. In a scene block, the score combines the learned QKNorm scale with an additive geometry term. Their relative magnitudes jointly determine whether content alignment or physical proximity controls attention. Target-context adaptation then changes which evidence is available downstream. Useful effects in isolation need not remain useful after those distributions and gradients change [13, 11].

The package improves endpoint measures without improving minADE6. Three simultaneous changes and separately controlled prior studies prevent component-level attribution. A factorial study, testing each component and their combinations under one protocol, would separate individual effects from interactions.

4.8 Research-question findings

RQ1 – Mamba placement in SEAM: Future-head replacement improves selected minADE6 and all final-epoch metrics; history addition gives a mixed vector.

RQ2 – Context and attention mechanisms in SEAM: All interventions improve the six-metric vector, led by QKNorm, without establishing learned radius expansion or composition.

RQ3 – Mamba placement in SHARP: Residual history processing improves selected minADE6 with mixed secondary effects; scene addition and temporal replacement do not improve their controls.

RQ4 – Context and attention mechanisms in SHARP: Geometry, uncertainty and QKNorm improve their separate selection criteria. Composition improves four secondary measures but raises minADE6 by 0.19%.

RQ5 – Architecture-dependent mechanism transfer: Context and attention gains recur; Mamba favours SEAM future generation and SHARP history refinement. Transfer is representation-specific, not additive.

4.9 Chapter summary

The findings favour representation-specific intervention: future generation and context conditioning in SEAM; selected context, attention and chronological pathways in SHARP. Composition requires direct confirmation after candidate selection. Chapter 5 develops these design implications.

5 CONCLUSIONS AND FUTURE WORK

5.1 Overall conclusion

This dissertation examined whether compact streaming motion predictors can be improved through representation-matched attention, geometric context and selective state-space processing without removing the mechanisms that make the predictors incremental. SEAM supplies an architecture in which previous endpoints define mode-specific target regions and a dual-context decoder combines global scene evidence with destination-local evidence [9]. SHARP provides a complementary architecture that processes short observation windows and transfers instance-aware context and trajectory state between them [10]. The central methodological condition was that every intervention preserve the defining streaming behaviour of its reference model and be judged only against a control trained under the same experimental protocol.

SEAM benefits from future-head recurrence, which reduces selected minADE6 by 2.17%, and independent confidence, geometry and QKNorm interventions. Each improves all six final-epoch metrics. History recurrence’s mixed outcome reinforces the importance of function-specific placement.

SHARP supplies complementary evidence within its own controls. Relative geometry and uncertainty-aware context improve the architectural ablation’s primary criterion; QKNorm improves the independent attention-operator control; and residual chronological Mamba improves selected minADE6 within the state-space placement comparison. The direct QKNorm–uncertainty–geometry composition, however, improves four secondary measures while increasing minADE6 by 0.19%. The central conclusion is therefore conditional: targeted mechanisms can improve compact streaming predictors, but a favourable placement or isolated effect does not establish the performance of a different placement or combined model. The following answers connect that conclusion to the research questions and measurable objectives established in Chapter 1.

5.2 Answers to the research questions

5.2.1 RQ1 – Mamba placement in SEAM

The SEAM baseline closely matches its selected reference. Its retained checkpoint has minADE6 0.662859, compared with 0.663 at the available reference precision. Final epoch-79 two-GPU validation gives minADE6 0.664314 and small differences across the remaining metrics, establishing the local SEAM control under its own two-GPU protocol [9].

Agent-history addition improves MR, b-minFDE6 and minFDE6, but not either ADE measure or minFDE1: endpoint coverage improves without better average path accuracy.

Future-head replacement improves all six final-epoch metrics, including reductions of 0.015835 m in minADE6 and 8.98% in MR. Selected minADE6 falls from 0.662859 to 0.648480; selection and final-epoch comparisons remain distinct.

RQ1 favours recurrence across mode-specific future positions over history refinement, not addition or replacement inherently. Trajectory smoothness was not measured.

5.2.2 RQ2 – Context and attention mechanisms in SEAM

All interventions improve the six final-epoch metrics. Uncertainty reduces selected minADE6 by 2.60% and MR by 3.78% through trainable feature gain within the fixed 30 m Boolean boundary, not learned neighbourhood expansion.

Geometry reduces selected minADE6 by 1.43%. Its lower single-mode but higher oracle errors than uncertainty distinguish preferred-trajectory quality from six-mode coverage.

QKNorm leads all six metrics, reducing minADE6 by 3.27% and MR by 6.96%. Its all-site substitution preserves masks and pathways without isolating site-specific effects.

RQ2 supports these three mechanisms independently, not their unmeasured combination.

5.2.3 RQ3 – Mamba placement in SHARP

The SHARP reproduction reached minADE6 0.679282 under the publication-aligned 80-epoch schedule, compared with 0.639000 in the article. This gap prevents a claim of numerical replication, but the run establishes the protocol-matched local control required for architectural study. Within the separate SHARP state-space placement comparison, residual processing over chronological agent history lowers selected minADE6 to 0.673736, a 0.82% reduction relative to that control. The retained six-metric vector is mixed and training time increases by 27.5%. Scene-token Mamba reaches 0.680362 and adds 2.8% training time, providing neither a selected-metric nor a cost advantage. The evidence therefore supports chronological placement when minADE6 is prioritised, but it does not support Mamba as an unconditional accuracy or efficiency improvement [15].

The 0.006626 m selection difference favours observations between attention blocks 2 and 3 over pooled scene tokens, although ordering remains coupled to local architecture. Complete temporal-attention replacement does not improve its separate control. RQ3 therefore supports complementary chronological processing, not wholesale substitution.

5.2.4 RQ4 – Context and attention mechanisms in SHARP

Independent architectural effects: Geometry reduces selected minADE6 by 0.35% and improves every retained secondary metric, supporting explicit physical relations in scene-interaction scores.

Uncertainty reduces minADE6 by 0.32% and also improves minADE1, with mixed secondary outcomes. Unlike SEAM, SHARP explicitly varies its retrieval radius with mode probability.

The remaining seven variants do not improve the screening criterion; their mechanisms and outcomes are analysed in Chapter 4. This early-optimisation screen identifies candidates, not a ranking of converged architectures.

Attention-operator selection: QKNorm is the strongest operator under the matched attention-study protocol. Selected minADE6 decreases from 0.673460 for standard MHA to 0.669777, an absolute reduction of 0.003683 m or 0.55%. This supports the hypothesis that normalising queries and keys can control similarity scale more directly than unconstrained dot products when attention is applied repeatedly to heterogeneous temporal, scene and target tokens [13].

Neither Talking-Heads variant improves selected minADE6, so QKNorm alone advances. Secondary comparisons retain the MHA checkpoint difference and combined operator’s altered microbatch construction.

Direct mechanism composition: The combination increases minADE6 from 0.679282 to 0.680589 m (0.19%) and minADE1 by 0.04%, while improving MR, b-minFDE6 and both endpoint errors by 0.17–0.30% under common selected-checkpoint evaluation. RQ4 identifies useful independent candidates but a mixed joint effect. These aggregate scores neither locate horizon-specific changes nor isolate component interactions.

5.2.5 RQ5 – Architecture-dependent mechanism transfer

Uncertainty, geometry and QKNorm improve their respective selection criteria in both architectures. Mamba instead favours SEAM future generation and SHARP residual history processing, supporting representation-specific rather than universal placement.

The publication-aligned SHARP reproduction uses 80 epochs, global batch 32, 13 warm-up epochs and validation-minADE6 checkpoint selection. Its selected checkpoint gives minADE6 0.679282 with MR 0.155955, b-minFDE6 1.924560, minADE1 1.676867, minFDE1 4.138639

and minFDE6 1.287697. The corresponding article values are lower on every reported measure, including minADE6 0.639000 [10].

Matched local controls separate architectural effects from the unexplained 6.30% reproduction gap. Direct composition further shows that favourable independent effects need not be additive.

5.3 Research contributions

5.3.1 A representation-specific intervention map

The first contribution maps implemented and evaluated interventions to chronological histories, scene sets, endpoint-indexed context, mode-specific futures and cross-window memory. These representations require ordered propagation, pairwise interaction or hypothesis-conditioned retrieval. The investigation concerns adaptation, placement and controlled assessment of established operators within two compact streaming architectures.

SEAM tests history and future recurrence alongside context and attention interventions; SHARP adds architectural screening, operator selection, residual placement and direct composition. The map connects hypotheses without pooling protocols.

This addresses the architectural-explanation objective: each intervention has a defined input, operation, retained pathway and control, distinguishing new relations from altered weighting, scoring or propagation rather than additional capacity alone.

5.3.2 Evidence for functional placement and non-additive composition

The second contribution establishes the importance of sequence function alongside order. Both SEAM placements are chronological, yet future generation benefits more than history refinement. SHARP favours residual temporal placement over scene-token scanning. These contrasts support representation-matched design, not chronology as a sufficient condition for improvement.

The context studies extend this principle beyond recurrence. SEAM benefits across all six metrics from each intervention; SHARP's geometry, uncertainty and QKNorm effects are more metric-specific. Similar mechanisms can therefore be useful in both architectures without producing identical outcomes.

Direct SHARP composition establishes the boundary of this evidence: promising components do not guarantee a joint minADE6 gain. Candidate selection and joint confirmation answer different questions, and a measured package effect does not identify its component interactions.

5.3.3 An inspectable record of accuracy and training duration

The third contribution links model definitions and configurations to checkpoints, evaluation vectors and timing artefacts. SHARP composition uses common selected-checkpoint evaluation; historical comparisons retain their documented model states rather than combining selection scores with later metrics.

Training duration is measured within each family rather than inferred from parameter count or asymptotic complexity. SEAM’s history and future-head interventions add 6.3% and 4.1% to their control’s duration; its separate context interventions add 0.39%, 0.88% and 14.84% for uncertainty, geometry and QKNorm. These records accompany, rather than replace, the forecasting criteria.

The independently controlled attention-operator study selects QKNorm on the pre-specified metric. Selected minADE6 decreases from 0.673460 for standard MHA to 0.669777, a 0.55% reduction, while end-to-end wall time increases by 19.1%. Talking-Heads and QKNorm with Talking-Heads do not improve the selection criterion; the combined operator requires 100.6% more wall time than MHA. The result supports query–key scale control rather than unrestricted head mixing under this protocol, but the small single-seed effect remains a candidate for confirmation rather than a universal operator ranking [13].

The remaining SHARP timings likewise use separate controls. In particular, composition adds 20.83% training time despite adding only 621 parameters, demonstrating why parameter count cannot substitute for measured execution cost. Chapter 4 retains all durations and comparisons; none is an inference-latency measurement.

These records meet the reproducible-comparison objective, linking model identity directly to checkpoint reevaluation and scenario-level analysis.

5.4 Study-specific design guidance

Five study-specific principles guide further model development, not universal operator rankings.

1. **Preserve the streaming function:** Retain short-window updates, identity-associated memory, endpoint context and trajectory relay. Lower offline error alone does not establish equivalent streaming behaviour.
2. **Match recurrence to its role:** SEAM favours future generation; SHARP favours residual history processing on selected minADE6. Ordering alone guarantees neither effect; addition and replacement require separate tests.

3. **Separate content, geometry and confidence:** Distinguish similarity scale, physical relations and target-evidence conditioning, including whether gradients train retrieval boundaries.
4. **Confirm combinations directly:** SHARP’s mixed composition result requires a matched control and pre-specified primary metric, not gains inferred from individual components.
5. **Retain the measurement context:** Identify checkpoint, full vector, protocol and within-family duration. Endpoint gains and parameter counts do not replace the primary criterion.

5.5 Principal limitations

5.5.1 Repeatability and candidate selection

Single-seed SHARP and SEAM gains describe retained models, not average effects across initialisations or the likelihood of reproducing their ranking.

The validation set is used both to select checkpoints and to screen architectural candidates, so the selected differences are expected to be more favourable than an untouched confirmatory estimate. Multiple training seeds would quantify optimisation variation, while paired scenario-level bootstrap intervals would quantify uncertainty in the fixed-checkpoint metric difference. Both are required because they answer different statistical questions [18, 19].

The ablations measure early adaptation, not guaranteed convergence. SHARP retains 13 warm-up epochs out of 20; SEAM uses its own shared warm-up. Candidates require longer-schedule confirmation.

5.5.2 Execution and checkpoint comparability

SEAM integration retains optimiser batch 32 through accumulation, not the reference per-forward SyncBatchNorm population. SEAM ablation uses batch 48 on three devices; SHARP reproduction uses four RTX 2080 Ti devices rather than one RTX 8000. Aligned schedules and batches therefore need not reproduce sample order, reductions or normalisation.

Historical selection scores and full vectors sometimes identify different checkpoints, notably SEAM integration and MHA. Common independent evaluation resolves this for SHARP composition only.

5.5.3 Forecasting and deployment scope

The main evaluation is open-loop and oracle-based. minADE and minFDE use the trajectory closest to the ground truth after the future is known. They do not show whether confidence is calibrated, whether hypotheses are diverse without duplication, whether a planner selects a safe

response, or whether predictions remain stable between streaming updates. b-minFDE incorporates confidence but does not replace a complete calibration or proper-scoring analysis [23, 27].

The conclusions concern AV2 forecasting, not closed-loop safety, on-vehicle latency or unmeasured combinations and datasets. Training duration includes validation, loading and communication; DeMo is a literature precedent, not a control.

5.6 Priority directions for future work

5.6.1 Confirm repeatability with common checkpoint evaluations

First, repeat SEAM’s future-head and context interventions and SHARP’s placements and composition against their original controls. At least three seeds, reported individually with mean and standard deviation, would estimate variability; effects small relative to variation would require more.

Paired bootstrap intervals would separate evaluation-sample uncertainty from retraining variation; neighbouring late checkpoints would test whether gains persist beyond a selected minimum.

Common full-vector evaluation should also cover SEAM geometry’s selected epoch-18 checkpoint and the attention-study MHA selected checkpoint before further variants are added.

5.6.2 Isolate mechanism interactions and learned pathways

A factorial SHARP study could evaluate all combinations of QKNorm, uncertainty-aware retrieval and relative geometry under one protocol. Pairwise interaction terms would distinguish complementary mechanisms from redundant effects that alter the same retrieved evidence. This would explain the mixed joint result in Table 4.13 beyond a single package-level contrast, without attributing its primary-metric deterioration to a component that was not independently isolated in that protocol.

The 2^3 design tests eight presence/absence combinations. Relative geometry/content-logit magnitude, attention entropy, retrieved-token counts and feature gains would diagnose interactions. Per-horizon and scenario-level displacement would test redistribution between path and endpoint accuracy.

SEAM needs learned versus fixed gain at radius 30 m, separate geometry features and subsystem-specific QKNorm tests. Learning the radius would require differentiable selection or another justified estimator. These tests would locate individual gains before composition.

5.6.3 Test persistent state and decision-relevant prediction

Mamba studies should next isolate cross-update persistence: vary state carry-over in SEAM’s future head and pair identity-keyed, forward-only state with SHARP’s within-window attention. Neither follows automatically from the current within-sequence findings.

Persistent state needs identity association, lifecycle rules and missing-observation or identity-switch handling. Graph detachment retains values without indefinitely extending gradients; future-position state also needs horizon alignment. These distinguish cross-update memory from recomputed within-window recurrence.

Future evaluation should include negative log-likelihood or another proper scoring rule, confidence calibration, mode diversity, off-road rate, collision rate and temporal consistency between successive streaming predictions. These measures would explain cases where minADE6 improves while MR or minFDE1 worsens and would separate hypothesis coverage from probability quality [27, 23].

Immediate priorities are calibrated confidence and cross-update consistency. Later work can assess planning, latency and memory under fixed execution conditions; transfer beyond AV2 requires matched retraining.

5.7 Concluding statement

The dissertation shows that representation-specific changes can improve motion forecasting while preserving SEAM’s and SHARP’s streaming functions. SEAM benefits from future-head Mamba and independent uncertainty, geometry and QKNorm interventions. SHARP benefits from residual agent-history Mamba on the selected primary metric, while architectural screening and attention-operator comparison identify useful context and scoring mechanisms. These findings connect an operator’s usefulness to the information it processes and its functional location, rather than to architectural substitution alone.

The main contribution is a tested intervention map linking context retrieval, relevance scoring and temporal propagation. Direct SHARP composition shows that endpoint and miss-rate gains can coexist with a small increase in average path error; complementary mechanisms therefore require joint evaluation. Matched controls, retained metric vectors and separate training-duration comparisons make these findings inspectable and provide a basis for focused development of streaming predictors.

BIBLIOGRAPHY

- [1] A. Gupta, J. Johnson, L. Fei-Fei, S. Savarese, and A. Alahi, “Social GAN: Socially acceptable trajectories with generative adversarial networks,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, June 2018, pp. 2255–2264, doi: 10.1109/CVPR.2018.00240. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2018/html/Gupta_Social_GAN_Socially_CVPR_2018_paper.html.
- [2] T. Salzmann, B. Ivanovic, P. Chakravarty, and M. Pavone, “Trajectron++: Dynamically-feasible trajectory forecasting with heterogeneous data,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 683–700, doi: 10.1007/978-3-030-58523-5_40. [Online]. Available: https://www.ecva.net/papers/eccv_2020/papers_ECCV/html/3094_ECCV_2020_paper.php.
- [3] J. Gao *et al.*, “VectorNet: Encoding HD maps and agent dynamics from vectorized representation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, June 2020, pp. 11525–11533, doi: 10.1109/CVPR42600.2020.01154. [Online]. Available: https://openaccess.thecvf.com/content_CVPR_2020/html/Gao_VectorNet_Encoding_HD_Maps_and_Agent_Dynamics_From_Vectorized_Representation_CVPR_2020_paper.html.
- [4] M. Liang *et al.*, “Learning lane graph representations for motion forecasting,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 541–556, doi: 10.1007/978-3-030-58536-5_32. [Online]. Available: https://www.ecva.net/papers/eccv_2020/papers_ECCV/html/3622_ECCV_2020_paper.php.
- [5] S. Arbabi, D. Tavernini, S. Fallah, and R. Bowden, “Planning for autonomous driving via interaction-aware probabilistic action policies,” *IEEE Access*, vol. 10, pp. 81699–81712, 2022, doi: 10.1109/ACCESS.2022.3193492.
- [6] S. Mozaffari, E. Arnold, M. Dianati, and S. Fallah, “Early lane change prediction for automated driving systems using multi-task attention-based convolutional neural networks,” *IEEE Trans. Intell. Veh.*, vol. 7, no. 3, pp. 758–770, Sep. 2022, doi: 10.1109/TIV.2022.3161785.

- [7] N. Song, B. Zhang, X. Zhu, and L. Zhang, “Motion forecasting in continuous driving,” in *Adv. Neural Inf. Process. Syst.*, 2024, vol. 37, pp. 78147–78168, doi: 10.52202/079017-2484. [Online]. Available: https://papers.nips.cc/paper_files/paper/2024/hash/8ec078518dcce6be1324cfd3de11ed24-Abstract-Conference.html.
- [8] B. Wilson *et al.*, “Argoverse 2: Next generation datasets for self-driving perception and forecasting,” in *Proc. Neural Inf. Process. Syst. Track Datasets Benchmarks*, 2021, vol. 1. [Online]. Available: <https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/4734ba6f3de83d861c3176a6273cac6d-Abstract-round2.html>.
- [9] A. Prutsch, D. Schinagl, and H. Possegger, “Streaming real-time trajectory prediction using endpoint-aware modeling,” in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2026, pp. 3005–3014, doi: 10.1109/WACV61042.2026.00294. [Online]. Available: https://openaccess.thecvf.com/content/WACV2026/html/Prutsch_Streaming_Real-Time_Trajectory_Prediction_Using_Endpoint-Aware_Modeling_WACV_2026_paper.html.
- [10] A. Prutsch, C. Fruhwirth-Reisinger, D. Schinagl, and H. Possegger, “SHARP: Short-window streaming for accurate and robust prediction in motion forecasting,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, June 2026, pp. 32103–32112. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2026/html/Prutsch_SHARP_Short-Window_Streaming_for_Accurate_and_Robust_Prediction_in_Motion_CVPR_2026_paper.html.
- [11] P. Shaw, J. Uszkoreit, and A. Vaswani, “Self-attention with relative position representations,” in *Proc. NAACL-HLT, vol. 2 (Short Papers)*, New Orleans, Louisiana, June 2018, pp. 464–468, doi: 10.18653/v1/N18-2074. [Online]. Available: <https://aclanthology.org/N18-2074/>.
- [12] A. Vaswani *et al.*, “Attention is all you need,” in *Adv. Neural Inf. Process. Syst.*, 2017, vol. 30, pp. 5998–6008. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

- [13] A. Henry, P. R. Dachapally, S. S. Pawar, and Y. Chen, “Query-key normalization for transformers,” in *Findings Assoc. Comput. Linguistics: EMNLP 2020*, Online, Nov. 2020, pp. 4246–4253, doi: 10.18653/v1/2020.findings-emnlp.379. [Online]. Available: <https://aclanthology.org/2020.findings-emnlp.379/>.
- [14] N. Shazeer, Z. Lan, Y. Cheng, N. Ding, and L. Hou, “Talking-heads attention,” *arXiv preprint arXiv:2003.02436*, 2020. [Online]. Available: <https://arxiv.org/abs/2003.02436>.
- [15] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” in *Proc. First Conf. Lang. Model. (COLM)*, 2024. [Online]. Available: <https://openreview.net/forum?id=tEYskw1VY2>.
- [16] B. Zhang, N. Song, and L. Zhang, “DeMo: Decoupling motion forecasting into directional intentions and dynamic states,” in *Adv. Neural Inf. Process. Syst.*, 2024, vol. 37, pp. 106582–106606, doi: 10.52202/079017-3383. [Online]. Available: https://papers.nips.cc/paper_files/paper/2024/hash/c0ff9e52e94ae331bc0f2d28be06a9ca-Abstract-Conference.html.
- [17] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. Salakhutdinov, and A. J. Smola, “Deep sets,” in *Adv. Neural Inf. Process. Syst.*, 2017, vol. 30. [Online]. Available: <https://proceedings.neurips.cc/paper/2017/hash/f22e4747da1aa27e363d86d40ff442fe-Abstract.html>.
- [18] X. Bouthillier *et al.*, “Accounting for variance in machine learning benchmarks,” in *Proc. Mach. Learn. Syst.*, 2021, vol. 3, pp. 747–769. [Online]. Available: https://proceedings.mlsys.org/paper_files/paper/2021/hash/0184b0cd3cfb185989f858a1d9f5c1eb-Abstract.html.
- [19] J. Pineau *et al.*, “Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program,” *J. Mach. Learn. Res.*, vol. 22, no. 164, pp. 1–20, 2021. [Online]. Available: <https://www.jmlr.org/papers/v22/20-303.html>.
- [20] Z. Zhou, L. Ye, J. Wang, K. Wu, and K. Lu, “HiVT: Hierarchical vector transformer for multi-agent motion prediction,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, June 2022, pp. 8823–8833, doi: 10.1109/CVPR52688.2022.00862. [Online]. Available: <https://openaccess.thecvf.com/content/CVPR2022/html/>

Zhou_HiVT_Hierarchical_Vector_Transformer_for_Multi-Agent_Motion_Prediction_CVPR_2022_paper.html.

- [21] Z. Zhou, J. Wang, Y.-H. Li, and Y.-K. Huang, “Query-centric trajectory prediction,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, June 2023, pp. 17863–17873, doi: 10.1109/CVPR52729.2023.01713. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2023/html/Zhou_Query-Centric_Trajectory_Prediction_CVPR_2023_paper.html.
- [22] A. Kendall and Y. Gal, “What uncertainties do we need in Bayesian deep learning for computer vision?,” in *Adv. Neural Inf. Process. Syst.*, 2017, vol. 30. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/hash/2650d6089a6d640c5e85b2b88265dc2b-Abstract.html.
- [23] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, 2017, vol. 70, pp. 1321–1330. [Online]. Available: <https://proceedings.mlr.press/v70/guo17a.html>.
- [24] B. Lakshminarayanan, A. Pritzel, and C. Blundell, “Simple and scalable predictive uncertainty estimation using deep ensembles,” in *Adv. Neural Inf. Process. Syst.*, 2017, vol. 30. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/hash/9ef2ed4b7fd2c810847ffa5fa85bce38-Abstract.html.
- [25] Argo AI, “Argoverse forecasting evaluation,” 2020. [Online]. Available: https://github.com/argoverse/argoverse-api/blob/master/argoverse/evaluation/eval_forecasting.py.
- [26] Argo AI, “Argoverse 2 motion forecasting evaluation metrics,” 2022. [Online]. Available: https://github.com/argoverse/av2-api/blob/main/src/av2/datasets/motion_forecasting/eval/metrics.py.
- [27] T. Gneiting and A. E. Raftery, “Strictly proper scoring rules, prediction, and estimation,” *J. Amer. Stat. Assoc.*, vol. 102, no. 477, pp. 359–378, 2007, doi: 10.1198/016214506000001437.
- [28] M.-F. Chang *et al.*, “Argoverse: 3D tracking and forecasting with rich maps,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, June 2019, pp. 8748–8757, doi:

- 10.1109/CVPR.2019.00895. [Online]. Available: https://openaccess.thecvf.com/content_CVPR_2019/html/Chang_Argoverse_3D_Tracking_and_Forecasting_With_Rich_Maps_CVPR_2019_paper.html.
- [29] H. Caesar *et al.*, “nuScenes: A multimodal dataset for autonomous driving,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, June 2020, pp. 11621–11631, doi: 10.1109/CVPR42600.2020.01164. [Online]. Available: https://openaccess.thecvf.com/content_CVPR_2020/html/Caesar_nuScenes_A_Multimodal_Dataset_for_Autonomous_Driving_CVPR_2020_paper.html.
- [30] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, “Social LSTM: Human trajectory prediction in crowded spaces,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, June 2016, pp. 961–971, doi: 10.1109/CVPR.2016.110. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2016/html/Alahi_Social_LSTM_Human_CVPR_2016_paper.html.
- [31] N. Lee, W. Choi, P. Vernaza, C. B. Choy, P. H. S. Torr, and M. Chandraker, “DESIRE: Distant future prediction in dynamic scenes with interacting agents,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, July 2017, pp. 336–345, doi: 10.1109/CVPR.2017.233. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2017/html/Lee_DESIRE_Distant_Future_CVPR_2017_paper.html.
- [32] Y. Chai, B. Sapp, M. Bansal, and D. Anguelov, “MultiPath: Multiple probabilistic anchor trajectory hypotheses for behavior prediction,” in *Proc. Conf. Robot Learn. (CoRL)*, 2020, vol. 100, pp. 86–99. [Online]. Available: <https://proceedings.mlr.press/v100/chai20a.html>.
- [33] H. Zhao *et al.*, “TNT: Target-driven trajectory prediction,” in *Proc. 2020 Conf. Robot Learn. (CoRL)*, 2021, vol. 155, pp. 895–904. [Online]. Available: <https://proceedings.mlr.press/v155/zhao21b.html>.
- [34] J. Gu, C. Sun, and H. Zhao, “DenseTNT: End-to-end trajectory prediction from dense goal sets,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 15303–15312, doi: 10.1109/ICCV48922.2021.01502. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2021/html/Gu_

DenseTNT_End-to-End_Trajectory_Prediction_From_Dense_Goal_Sets_ICCV_2021_paper.html.

- [35] S. Shi, L. Jiang, D. Dai, and B. Schiele, “Motion Transformer with global intention localization and local movement refinement,” in *Adv. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 6531–6543, doi: 10.52202/068431-0473. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2022/hash/2ab47c960bfee4f86dfc362f26ad066a-Abstract-Conference.html.
- [36] N. Nayakanti, R. Al-Rfou, A. Zhou, K. Goel, K. S. Refaat, and B. Sapp, “Wayformer: Motion forecasting via simple & efficient attention networks,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2023, pp. 2980–2987, doi: 10.1109/ICRA48891.2023.10160609. [Online]. Available: <https://waymo.com/research/wayformer/>.
- [37] J. Lee, Y. Lee, J. Kim, A. Kosiorek, S. Choi, and Y. W. Teh, “Set Transformer: A framework for attention-based permutation-invariant neural networks,” in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, 2019, vol. 97, pp. 3744–3753. [Online]. Available: <https://proceedings.mlr.press/v97/lee19d.html>.
- [38] C. Ying *et al.*, “Do transformers really perform badly for graph representation?,” in *Adv. Neural Inf. Process. Syst.*, 2021, vol. 34, pp. 28877–28888. [Online]. Available: <https://proceedings.neurips.cc/paper/2021/hash/f1c1592588411002af340cbaedd6fc33-Abstract.html>.
- [39] J. Ngiam *et al.*, “Scene Transformer: A unified architecture for predicting multiple agent trajectories,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022. [Online]. Available: <https://openreview.net/forum?id=Wm3EA50lHsG>.
- [40] R. Girgis *et al.*, “Latent variable sequential set transformers for joint multi-agent motion prediction,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022. [Online]. Available: https://openreview.net/forum?id=Dup_dDqkZC5.
- [41] A. Gu, K. Goel, and C. Ré, “Efficiently modeling long sequences with structured state spaces,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022. [Online]. Available: <https://openreview.net/forum?id=uYLFoz1v1AC>.

- [42] J. T. H. Smith, A. Warrington, and S. W. Linderman, “Simplified state space layers for sequence modeling,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023. [Online]. Available: <https://openreview.net/forum?id=Ai8Hw3AXqks>.
- [43] T. Dao and A. Gu, “Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality,” in *Proc. 41st Int. Conf. Mach. Learn. (ICML)*, 2024, vol. 235, pp. 10041–10071. [Online]. Available: <https://proceedings.mlr.press/v235/dao24a.html>.
- [44] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, “Vision Mamba: Efficient visual representation learning with bidirectional state space model,” in *Proc. 41st Int. Conf. Mach. Learn. (ICML)*, 2024, vol. 235, pp. 62429–62442. [Online]. Available: <https://proceedings.mlr.press/v235/zhu24f.html>.
- [45] S. Jelassi, D. Brandfonbrener, S. M. Kakade, and E. Malach, “Repeat after me: Transformers are better than state space models at copying,” in *Proc. 41st Int. Conf. Mach. Learn. (ICML)*, 2024, vol. 235, pp. 21502–21521. [Online]. Available: <https://proceedings.mlr.press/v235/jelassi24a.html>.
- [46] Z. Dai, Z. Yang, Y. Yang, J. Carbonell, Q. V. Le, and R. Salakhutdinov, “Transformer-XL: Attentive language models beyond a fixed-length context,” in *Proc. 57th Annu. Meeting Assoc. Comput. Linguistics (ACL)*, July 2019, pp. 2978–2988, doi: 10.18653/v1/P19-1285. [Online]. Available: <https://aclanthology.org/P19-1285/>.
- [47] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019. [Online]. Available: <https://openreview.net/forum?id=Bkg6RiCqY7>.
- [48] S. Ioffe and C. Szegedy, “Batch normalization: Accelerating deep network training by reducing internal covariate shift,” in *Proc. 32nd Int. Conf. Mach. Learn. (ICML)*, 2015, vol. 37, pp. 448–456. [Online]. Available: <https://proceedings.mlr.press/v37/ioffe15.html>.

APPENDIX

A.1 Scientific claims and supporting records

Quantitative claims link to their family, model, resolved configuration, checkpoint selection and evaluation output, exposing the comparison’s data-processing and optimisation conditions.

Figure A.1 summarises this chain. The dissertation presents only the scientific evidence required to inspect the experiments. Detailed operational transcripts are retained with the experimental records but are not reproduced in the report because they do not contribute to the interpretation of the model comparisons.

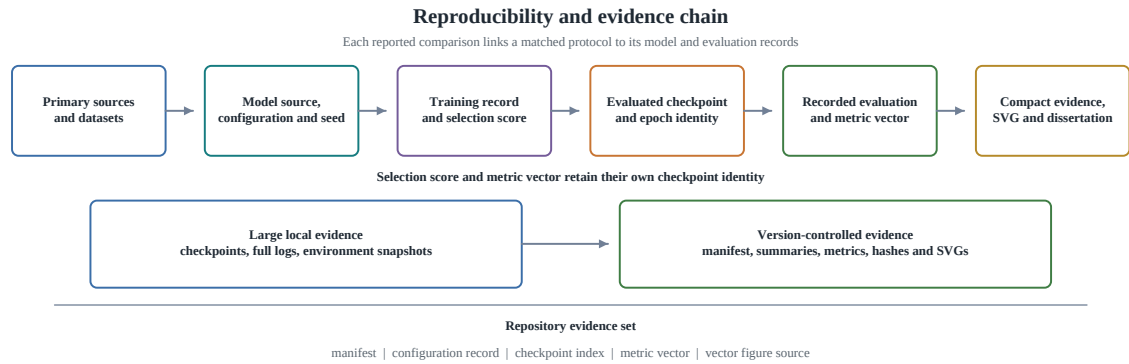


Figure A.1: Traceability chain from a scientific claim to its model, configuration, checkpoint selection, evaluation identity and retained evidence artefact.

A.2 Repository and evidence index

Experimental source, configurations and compact results reside at <https://github.com/madvidd/Salehi-MSc-Thesis>. These records support the sole submitted PDF. AV2 data are not redistributed and remain subject to the official release terms.

The `Studies/SEAM/` and `Studies/SHARP/` directories group source, protocols and run records. Each `README.md` links its study scope, implementation, setup and evidence; `Summary.md` presents the comparison. `SHARP` placement and composition retain separate packages.

The `Results/` index links separate study tables. The repository guide explains conventions, the reproduction guide documents setup and `Tools/` contains inspection utilities.

Three files in the dissertation’s `evidence/` directory provide the principal index:

- `EXPERIMENT_STATUS.md` records the numerical evidence used in the report and distinguishes checkpoint-selection values from separately retained evaluation vectors;
- `RUN_MANIFEST.csv` maps each model variant to its family, dataset, epoch budget, global batch, seed and checkpoint criterion; and
- `ARTIFACT_INDEX.json` supplies the corresponding source-package, configuration and result paths in the experimental repository.

Shared identifiers connect the indexes to study locations; the path migration index maps earlier locations without changing run identifiers or recorded host paths. Executed configurations are distinguished from setup scripts and inventories from weights. GitHub holds compact results, source and configurations; model storage retains large checkpoints and transcripts. Paths and checksums do not themselves provide evaluable weights.

A.3 Reproducible comparison units

The experimental programme contains five comparison families. Each family has its own matched control and fixed protocol, as summarised in Table A.1. Comparisons are made only within a row of this table. In particular, architectural ablation is not ranked against architecture integration, the attention-operator control is not substituted for the SHARP reference reproduction, the two SEAM schedules are not pooled and SEAM variants are not ranked against SHARP variants.

SEAM records distinguish two-device, batch-32 integration from three-device, batch-48 ablation. Both use microbatch eight and two-step accumulation with shared architecture, optimiser, learning-rate limits, regularisation and seed; their epoch and warm-up budgets remain separate.

SEAM ablation records in `evidence/seam20/` preserve full-precision comparisons, durations, configurations and evaluation excerpts. Contrasts distinguish selected minADE6 from final-epoch vectors. The source index separately locates state-space configurations and checkpoint inventories.

SHARP integration records retain the publication-aligned configuration specified in Chapter 3. Composition changes attention, target context and geometry while preserving preprocessing, seed, optimisation, precision and checkpoint selection.

For composition, `evidence/sharp_final/` retains selected-checkpoint metrics, configurations, architecture audits, timings and validation outputs. `Comparison.csv` supplies full-precision measurements; `contrasts.json` records absolute and percentage differences for all

six metrics and training time. The source index and checksums identify imported repository versions. Plots use these records, not rounded report values.

SHARP placement and composition remain distinct comparisons: historical Mamba records do not replace the directly evaluated composition pair.

Table A.1: Matched comparison units and retained evidence.

Family	Control	Interventions	Budget	Evidence
SEAM SSM	SEAM reference baseline	residual agent-history Mamba and future-head Mamba replacement	80 epochs; batch 32	selection records, final-epoch vectors and runtime records
SEAM C+A	SEAM ablation baseline	uncertainty-aware target context, relative-geometry scene bias and all-site QKNorm	20 epochs; batch 48	all four evaluations, timings and checkpoint records
SHARP INT	SHARP reference reproduction	residual Mamba placement; QKNorm, uncertainty-aware target context and relative-geometry bias	80 epochs; batch 32	selection records, metric vectors, runtimes and figures
SHARP ABL	SHARP ablation baseline	nine isolated architectural or objective modifications	20 epochs; batch 32	ten selection records, metric summaries, runtimes and figures
SHARP ATT	MHA control from the same attention suite	QKNorm, Talking-Heads and QKNorm with Talking-Heads	80 epochs; batch 24	four selection records, metric summaries, runtimes and figures

Table A.1: SEAM SSM denotes SEAM state-space integration; SEAM C+A denotes SEAM context and attention ablation; SHARP INT denotes SHARP architecture integration; SHARP ABL denotes SHARP architectural ablation; and SHARP ATT denotes the SHARP attention-operator study. Control is the reference model; interventions are the tested changes; budget gives training epochs and global batch; and evidence lists the retained result types.

A.4 Configuration, architecture and data evidence

Resolved configurations record arguments after inheritance and command-line overrides, including horizons, optimiser schedules and distributed settings. The manifest summarises controls and distinguishes setup records from executed configurations.

Architecture records locate SEAM’s residual history and replacement future-head Mamba, confidence-and-entropy target conditioning, five-feature bias in four scene blocks and QKNorm at 20 sites. They distinguish trainable context gain from non-differentiable radius selection. SHARP records separately identify attention operators, scene and temporal Mamba placements and mechanism composition. Chapter 3 explains each implementation.

Data records identify processed AV2 roots and temporal settings: SEAM’s 30-sample histories and three-, four- and five-second updates and SHARP’s ten-step windows and five update points. Prediction lengths, evaluation horizons and scene radius are retained in the indexed configurations; every within-family variant uses the same prepared splits and temporal construction.

A.5 Checkpoint selection and metric provenance

Let m_e denote validation minADE6 at epoch e , and let $\mathcal{E}$ contain the epoch indices with eligible retained checkpoints. The selection rule gives the index e^* of the checkpoint with the lowest monitored value:

$$e^* = \arg \min_{e \in \mathcal{E}} m_e. \quad (\text{A.1})$$

Epoch indices are zero-based: index 79 denotes the eightieth epoch. The fixed minimum-minADE6 rule separates selection from retrospective secondary-metric interpretation. Each vector must identify its evaluated checkpoint; the selection rule alone does not establish that identity.

A checkpoint filename supplies selected minADE6, not its other metrics. SEAM vectors use final-epoch distributed validation; geometry selects one epoch earlier. MHA uses final-epoch validation while its alternatives use selected checkpoints. SHARP composition evaluates both selections on one device with batch 32. Historical ablation and temporal-Mamba vectors retain recorded precision and provenance, identifying reproducible metric groups.

A.6 Result and runtime traceability

Detailed numerical comparisons are consolidated in Chapter 4; this appendix supplies their source routes rather than repeating the metric tables. Table A.2 links each family to its metric and runtime evidence. This arrangement keeps the scientific interpretation adjacent to the results while providing a direct reproduction route.

Table A.2: Location of metric, runtime and visual evidence for each comparison family.

Family	Metric tables	Time tables	Figures
SEAM SSM	4.1, 4.2	4.3	4.1, 4.2
SEAM C+A	4.4	4.5	4.3, 4.4
SHARP INT	4.6, 4.11, 4.13	4.12, 4.14	4.9, 4.10, 4.11, 4.12, 4.13
SHARP ABL	4.7	4.8	4.5, 4.6
SHARP ATT	4.9	4.10	4.7, 4.8

Table A.2: SEAM SSM denotes SEAM state-space integration; SEAM C+A denotes SEAM context and attention ablation; SHARP INT denotes SHARP architecture integration; SHARP ABL denotes SHARP architectural ablation; and SHARP ATT denotes the SHARP attention-operator study. The remaining columns locate each family’s metric tables, training-time tables and figures.

The Results/Figures/ catalogue contains every numbered figure and the graphical abstract, grouped by architecture, study and supporting evidence. Its index maps figure numbers to editable SVG masters, vector PDFs and PNG previews; a manifest records file checksums. Repository summaries retain numerical inputs, and checkpoint inventories identify the associated model states.

A.7 Reproduction procedure

Inspection starts with the indexed metric, checkpoint and configuration. Retraining and reevaluation additionally require the dataset and model artefacts:

1. identify the experiment family and its matched control in Table A.1;
2. obtain AV2 under its official terms and apply the preprocessing associated with the selected SEAM or SHARP implementation;
3. recreate the recorded software environment from the package inventory;
4. resolve the configuration and verify the dataset, temporal horizon, seed, epoch budget, global batch, optimiser schedule and model-variant fields;
5. train each variant under the fixed family protocol and retain the monitored validation checkpoints;
6. select e^* using Equation A.1;
7. evaluate the checkpoint identified by the reported vector under its recorded validation split, prediction horizon, batch and device settings, retaining the selection score separately when the two records differ; and
8. compare the resulting metric vector and runtime record with the indexed evidence before using the result in a table or figure.

The procedure reproduces a stated within-family comparison, not a ranking across different protocols.

A.8 Reproduction boundaries

GPU architecture, dependency versions, reduction order and sample scheduling can prevent bit-wise equivalence. Equal global batches can also conceal different per-forward normalisation populations. Single-seed results describe the recorded models rather than average effects; matched controls do not eliminate stochastic variation.